ETHIOPIAN AIRLINES GROUP
FLIGHT OPERATIONS

B737 STANDARD OPERATING PROCEDURES MANUAL

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CHAPTER 0 - ADMINISTRATION AND CONTROL OF MANUAL

0.0 FOREWORD

This Standard Operating Procedures (SOP) manual is issued by Flight Operations for guidance in the operation of the B737 aircraft. It is intended to assist the flight crew in operating the aircraft within the limitations of the approved Flight Manual (AFM).

All personnel are expected to exercise sound judgment and consistency in the application of this SOP to the greatest extent possible.

The objective of SOP is to provide a smooth and coordinated flight deck management system that promotes standardization and enhances safety.

Proper use of this manual requires thorough knowledge of airplane systems and company policies. Where a detailed explanation of a given step or policy is required, the detail can be found in the other manuals that supplement this SOP. (Refer to Chapter 0.2)

This SOP is organized in seven chapters.

1. ADMINISTRATION AND CONTROL OF MANUAL
2. GENERAL
3. NORMAL OPERATING PROCEDURES
4. NON-NORMAL OPERATING PROCEDURES
5. NON-NORMAL MANEUVERS
6. ETOPS
7. LOW VISIBILITY OPERATIONS

All pilots shall follow the SOP at all times. The Captain has the authority to deviate from the SOP only when unusual circumstances require him/her to do so in the interest of safety. All applicable personnel shall be briefed whenever deviation from SOP is required.

Any suggestion aimed to improve the contents of this manual and enrich the substance contained in it is highly appreciated and recommended.

Strict adherence by the flight crew to the prescribed procedures in this B737 Standard Operating Procedure Manual will enhance the effort for coordinated performance in the endeavor to establish professionalism.

0.1 INTRODUCTION

The legal basis for Ethiopian to provide and use an Operations Manual is defined in the rules promulgated by the Civil Aviation Authority under ECARAS chapter 8. The content of this Standard Operating Procedures Manual complies with:

- Flight Operations Manual (FOM)
- The applicable regulations of the authority
- Applicable ICAO standards and recommended practices
- The required subparts of the Federal Aviation Regulation (FAR), EASA, etc.
- Additional rules and regulations applied by States within which Ethiopian is authorized to conduct flight operations. and
- The provisions of Ethiopian's Air Operator Certificate (AOC) and Operations Specifications (OPS SPECS) as issued and approved by the Authority

0.2 SUPPLEMENT MANUAL

The B737 SOP is provided for the use and guidance of Flight Operations personnel. The content shall ensure familiarity with applicable laws, regulations, rules, procedures, limitations and/or restrictions pertinent to the performance of duties in areas and conditions where operations are conducted.

The following manuals may supplement this manual as applicable:

- Approved Flight Manual (AFM)
- Flight Crew Operations Manual (FCOM)
- Quick Reference Handbook (QRH)
- Jeppesen Airway Manual
- Flight Crew Training Manual (FCTM)
- Flight Operations Manual (FOM)
- IATA Dangerous Goods Regulation (DGR)
- Emergency Response Guide for A/C Incidents involving Dangerous Goods
- Security Manual
- Dispatch Policy and Procedures Manual
- Flight Operations Training Program Manual (FOTPM)
- Cabin Crew Operations Manual (CCOM)

- Emergency Response Manual
- Route Manual
- Aviation Safety Program Manual
- MEL/CDL
- Airport Analysis/OPT
- PBN Pilot Training Handbook for Ethiopian Airlines Flight Crew
- Flight Data Monitoring Manual
- Ground Handling Manual (For all cargo airplanes)
- North Atlantic Operations and Airspace Manual
- Airport Handling Manual
- PBN Training and Procedure Manual
- Flight Operation Quality Management System Manual
- Flight Operation Safety Management System Manual

0.3 PERSONS AFFECTED

All chapters of this manual, either wholly or partially, are relevant to the flight crew. This manual shall be distributed to all B737 flight crew and Flight/Simulator Instructors/Examiners and will be made available to all in accordance with the distribution list.

This manual is distributed to flight crew via electronic means. Flight Crew shall receive their copy of revision via update of their laptop or tablet and must stay familiar with the revised content. A copy of this manual is also available in the EFB (where applicable) for in-flight use by the flight crew. Flight crew would comply with the revision requirement upon the synchronization of their class I EFB prior to each flight and as a minimum every seven days.

All flight crew must comply with the prescribed methods of operation, detailed procedures and operating limitations contained in the appropriate Approved Flight Manual and Flight Crew Operations Manual.

The Flight Operations Manual has priority over all other fight operations originated documents unless it is otherwise stated in the manual.

This Manual applies to both female and male operations personnel although some references may refer to one gender only. In all cases, reference to one gender shall also be deemed equally applicable to both genders.

Any question about the content or use of this manual shall be directed to Chief pilot B737.

0.4 AMENDMENT AND REVISION

0.4.1 MANUAL CONTROL

Dir. Flying Operations after having received any amendment proposal using the form provided below will present it to the Review Committee.

The Review Committee comprises of Director Flying Operations, Mgr. Training Standards and Facilities and Chief Pilot B737.

The approval process will be in accordance with the directives in the Management Policy Manual (MPM). All amendments will be annotated to show the date of issue (and date of effect if different), the amendment list number and the portion of the text which has been revised as indicated by the vertical margin lines adjacent to the changes.

Each amendment will be accompanied by a revised list of effective pages, with their date of issue and by certificate of receipt/incorporation. An amendment list record will be maintained at the front of each manual.

0.4.2 REVISION TRANSMISSION

Dir. Flying Operations shall:

A. **Transmit revisions to all manual holders**- Acknowledgment of receiving and insertion of revision by all Manual holders shall also be followed up and record kept by Documentation office.

B. **Temporary revision** – Temporary revisions shall be issued pending final revisions per the distribution list to all manual holders.

C. **Emergency revision** – This method is used for revisions that are urgent and need to be immediately circulated pending revision or incorporation in the relevant section of the manual. Emergency revisions shall be transmitted to all manual holders electronically, by fax, or by any other means of fast communication.

D. **Identification and location of revisions** – Temporary revisions printed on yellow papers shall be issued pending final revisions. The temporary revision record shall be filed between the list of effective pages and contents of the affected chapter. Current revisions and copy revisions can be obtained from Documentation office.

0.4.3 RECORD OF REVISION

This manual shall be amended or revised as necessary to ensure that the information contained herein is kept up to date. All such amendments or revisions shall be issued to all personnel that are required to use this manual.

This revision reflects the most current information available to Flight Operations through the subject revision date. The following revision highlights explain changes in this revision. Pages containing revised technical material have revision bars associated with the changed text or illustration. Editorial revisions (for example, spelling corrections) may have revision bars with no associated highlight.

Retain this record in the manual. Upon receipt of revision, insert revised pages in the manual and enter the revision number, revision date, and filed date in the appropriate block.

Revision No.	Revision date	Insertion date	Inserted by	Initial
0	June 2004			
1	April 2006			
2	15 Dec 2009			
3	12 Aug 2011			
3A	21 Nov 2012			
3B	11 DEC 2014			
4	JAN 2016			
5	01-JAN-2018			
6	01-JAN-2020			
7	01-MAR-2022			
8	08-JAN-2024			

0.4.4 REVISION HIGHLIGHTS

This is a complete Revision No. 8 of the B737 SOP. The manual is called B737 Standard Operating Procedures (SOP). This revision cancels the previous version.

0.4.5 REVISION TRANSMITTAL SHEET

Transmittal No.	Chapter/Section	Effective date	Cancel/supersedes (insert old effective date if any)	Date inserted	Entered by (name and initial)

0.4.6 AMENDMENT PROPOSAL FORM

Proposed amendment: _____

Reason for Amendment:

Signature of Proposer: _____

Position _____

Authorized by: _____ Dir. Flying Operations

GUIDELINES ON THE COMPLETION OF THIS FORM

0. Proposed amendments to the SOP will only be considered if submitted on this form.
1. This form shall be completed, signed, and handed to Dir. Flying Operations for comment/approval. Amendments to specific parts shall then be forwarded as follows:
2. The final decision on amendments rests with Director Flying Operations. The company reserves the right to amend any proposal.

0.4.7 DISTRIBUTION

Distribution of the B737 SOP can be accessed through individual class I EFB and shall be available per the distribution list established below.

Manual Cont. No	Issued to
1	Vice President Flight Operations
2	Director Flying Operations
3	Director Flight Training and Standards
4	Mgr. Boeing Fleet Training
5	Chief Pilot B737
6	Mgr. FOPs Quality Management System
7	Mgr. FOPs Safety Management System
8	Simulator Instructors/Examiners
9	Ground Instructors
10	Mgr. FOPs Performance and Technical Support
11	Flight Operations Documentation

0.4.8 ELECTRONIC ACCESS

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0.4.9 LIST OF EFFECTIVE PAGES

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0.5 TERMS AND DEFINITIONS

When the following terms are used in this manual, they have the following meanings.

Shall – The procedure is mandatory and must be carried out.

Should – The procedure is recommended.

May – The procedure is optional.

Note – An operating procedure, technique, etc., is considered essential to be emphasized.

Caution – An operating procedure, technique, etc., may result in damage to personnel or equipment if not carefully carried out.

Warning – An operating procedure, technique, etc., may result in personal injury or loss of life if not carefully followed.

Accelerate Stop Distance Available (ASDA) – The length of the takeoff run available plus the length of stop way if provided.

Aerial Work – Operation in which an aircraft is used in specialized services such as agriculture, construction, photography, surveying, observation and patrol, search and rescue, aerial advertisement etc.

Aerodrome – A defined area on land or water (including any buildings, installations, and equipment) intended to be used either wholly or in part for the arrival, departure, or movement of aircraft.

Aerodrome Operating Minima – The limits of usability of an aerodrome either for takeoff or landing - usually expressed in terms of visibility or runway visual range; decision altitude/height (DA/DH) or minimum descent altitude/height (MDA/H); and cloud conditions.

Aircraft – Any machine that can drive support in the atmosphere from the reactions of the air other than the reactions of the air against the earth's surface.

Aircraft Flight Manual (AFM) – A manual associated with the certificate of airworthiness, containing limitations within which the aircraft is considered to be airworthy, instructions and information necessary to the flight crewmembers for the safe operations of the aircraft. (Flight Crew Operations Manual, Approved Flight Manual, MEL/CDL, etc.)

Air Operator Certificate (AOC) – A certificate authorizing an operator to carryout specified commercial air transport operation.

Authority – Is the Ethiopian or any other Civil Aviation Authority.

Commercial Air Transport Operation – An aircraft operation involving the transport of passengers, cargo or mail for remuneration or hire.

Crewmember – A person assigned by an operator for duty on an aircraft during flight.

Cruising Level – A level maintained during a significant portion of a flight.

Day Off – Periods available for leisure and relaxation free from all duties.

Deadhead Transportation – The period from the time a crewmember reports for the purpose of a positioning flight until the positioning flight is complete.

Dangerous Goods – Articles or substances, which are capable of posing significant risk to health, safety or property when transported by air.

Decision Altitude/Height (DA/DH) – A specified altitude/height in the precision approach at which a missed approach must be initiated if the required visual reference to continue the approach has not been established.

Note: - **1.** Decision altitude (DA) is referenced to mean sea level (MSL) and decision height (DH) is referenced to the threshold elevation.

2. The required visual reference means that the section of the visual aids or the approach area which should have been in view for sufficient time for the pilot to have made an assessment of the aircraft position and rate of change of positions in relation to the desired flight path.

Destination Alternate – An alternate aerodrome to which an aircraft may proceed should it become impossible or inadvisable to land at the aerodrome of intended landing.

Note: - The aerodrome from which an aircraft departs may also be an en-route or destination alternate for that flight.

En-route Alternate – An aerodrome at which an aircraft would be able to land after experiencing an abnormal or emergency condition while en-route.

Flight Crewmember – A licensed pilot crewmember charged with duties essential to the operation of an aircraft during flight.

Flight Duty Period (FDP) – The total time from the moment a flight crewmember commences duty, immediately subsequent to a rest period and prior to making a flight or series of flights, to the moment the flight crewmember is relieved of all duties having completed such flight or series of flights.

Flight Plan – Specified information provided to air traffic services units, relative to an intended flight or portion of a flight.

Flight Recorder – Any type of recorder installed in the aircraft for the purpose of complementing accident/incident investigation.

Flight Simulator - Which provides an accurate representation of the flight deck of a particular aircraft type to the extent that the mechanical, electrical, electronic, etc., aircraft systems control functions, the normal environment of flight crew members, and the performance and flight characteristics of that type of aircraft are realistically simulated.

Flight Time – The total time from the moment an aircraft first moves under its own power for the purpose of taking off until the moment it comes to rest at the end of the flight.

Note: - Flight as here defined is synonymous with the term "block to block" or "chock to chock" time in general usage which is measured from the time the aircraft moves from the loading point until it stops at the unloading point.

Instrument Meteorological Conditions (IMC) – Meteorological condition expressed in terms of visibility, distance from cloud and ceiling less than the minimum specified for visual meteorological condition.

Landing Distance Available (LDA) – The length of the runway, which is declared available and suitable for the ground run of an airplane landing.

Master Minimum Equipment List (MMEL) – A list established for a particular aircraft type by the manufacturer with the approval of the State of manufacturer containing items one or more of which is permitted to be un-serviceable at the commencement of the flight. The MMEL may be associated with special operating conditions, limitations, or procedures.

Maximum Mass – Is maximum certificate takeoff mass.

Minimum Descent Altitude/Height (MDA/H) – A specified altitude/height in a non-precision or circling approach, below which descent must not be made without visual reference.

Minimum Equipment List (MEL) – A list that provides for the operation of aircraft, subject to specified conditions, with particular equipment inoperative and prepared by an operator in conformity with, or more restrictive than, the MMEL established for the aircraft type.

Night Duty – A flight duty period encroaching on any portion of the period between 02:00 and 04:59 hours in the time zone to which the crew is acclimatized.

Obstacle Clearance Altitude/Height (OCA/H) – The lowest altitude/height above the elevation of the relevant runway threshold or above the aerodrome elevation as applicable used in establishing compliance with appropriate obstacle clearance criteria.

Operational Control – The exercise of authority over the initiation, continuation, diversion or termination of a flight in the interest of safety regulatory and efficiency of the flight.

Operational Flight Plan (OFP) – The operator's plan for the safe conduct of the flight based on considerations of aircraft performance, other operating limitations, and relevant expected conditions on the route to be followed and the aerodrome concerned.

Operator - A person, organization or enterprise engaged in or offering to engage in an aircraft operation.

Pilot-In-Command (captain) - The pilot responsible for the operation and safety or the aircraft during flight time.

Pressure-Altitude - An atmospheric pressure expressed in terms of altitude, which corresponds to that pressure in the Standard Atmosphere.

Rating - An authorization entered on or associated with a license and forming a part thereof, stating special conditions, privileges, or limitations pertaining to such license.

Rest Period - Any period of time on the ground or in flight during which the operator relives a flight crewmember of all duties.

Runway Visual Range (RVR) - The range over which the pilot of an aircraft on the centerline of a runway can see the runway surface markings or the lights delineating the runway or identifying its centerline.

Safety Harness - Includes shoulder strap and a seat belt that may be used independently.

State of the Operator - The state in which the operator's principal place of business is located or, if there is no such place of business, the operator's permanent residence.

State of Registry - The State on whose register the aircraft is entered.

Takeoff Alternate – An alternate aerodrome at which an airplane can land should this become necessary shortly after takeoff and it is not possible to use the aerodrome of departure.

Takeoff Run Available (TORA) - The length of runway which is declared available and suitable for the ground run of an airplane taking off. This in most cases corresponds to the physical length of the runway.

Visual Meteorological Conditions (VMC) - Meteorological conditions expressed in terms of visibility, distance from cloud and ceiling, equal to or better than specified minima.

0.6 ABBREVIATION

A/C	Aircraft
A/P	Autopilot
A/T	Autothrottle
AAL	Above Aerodrome Level
ACARS	Aircraft Communications Addressing and Reporting System
ACT	Actual
ADD	Acceptable Deferred Defects
ADF	Automatic Direction Finder
AFE	Above Field Elevation
ALTN	Alternate
AMSL	Above Mean Sea Level
APP	Approach
AR	Authorization Required
ARR	Arrival
ATA	Actual Time of Arrival
ATC	Air Traffic Control
ATD	Actual Time of Departure
ATL	Aircraft Technical Log
ATO	Actual Takeoff Time
ATS	Air Traffic Services
BC	Back Course
BLK	Block
CC	Cabin Crew
CAPT	Captain
CBT	Computer Based Trainer
CFIT	Controlled Flight Into Terrain
Cont	Contingency
CRM	Crew Resource Management
Dist.	Distance
DME	Distance Measuring Equipment
Eng	Engine
Est	Estimated
ECAA	Ethiopian Civil Aviation Authority
EO	Engine Out
ETA	Estimated Time of Arrival
ETAF	Ethiopian Air Force
ETD	Estimated Time of Departure
ETO	Estimated Takeoff
F/D	Flight Director
F/O	First Officer
FAC	Final Approach Course
FIR	Flight Information Region
FL	Flight Level
FLT OPS	Flight Operations

FMA	Flight Mode Annunciation
FMC	Flight Management Computer
FMS	Flight Management System
FOPM	Flight Operations Policy Manual
FPM	Feet per Minute
Freq	Frequency
FT	Feet
FWD	Forward
G/P	Glide Path
G/S	Glideslope
GMT	Greenwich Mean Time
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GPWS	Ground Proximity Warning System
GS	Ground Speed
GTOW	Gross takeoff weight
HAT	Height Above Threshold
HF	High Frequency
HRS	Hours
IAN	Integrated Approach Navigation
ILS	Instrument Landing System
IMC	Instrument Meteorological Conditions
Inst.	Instrument
IOCC	Integrated Operation Control Center
ISFD	Integrated Standby Flight Display
KGS	Kilograms
LAT	Latitude
LAV	Lavatory
LDG	Landing
LNAV	Lateral Navigation
LOC	Localizer
LONG	Longitude
LW	Landing Weight
Max	Maximum
MBS	Millibars
MCD	Main Cargo Door
MCP	Mode Control Panel
MEL	Minimum Equipment List
MGT	Management
MIN	Minimum
MNVR	Maneuver
NDB	Non-Directional Beacon
NM	Nautical Mile
NNC	Non-Normal Checklist
NO.	Number

NOTAM	Notice to Airmen
OFV	Outflow Valve
PA	Passenger
PIC	Pilot-In-Command
PLD	Payload
POB	Persons Onboard
PRM	Persons with Reduced Mobility
Proc	Procedure
PTS	Pilot Training School
PWS	Predictive Windshear System
RA	Radio Altitude | Resolution Advisory
Reg	Registration
RNP	Required Navigation Performance
RTE	Route
RVSM	Reduced Vertical Separation Minimum
RW/RWY	Runway
S	Satisfactory
Sched	Scheduled
SDT	Set-up Departure Time
SIM	Simulator
SVC	Service
T/L	Team Leader
T/O	Takeoff
TA	Traffic Advisory
TAS	True Airspeed
TCAS	Traffic Alert and Collision Avoidance System
TFC	Traffic
TLCS	Team Leader Cabin Services
TO/GA	Takeoff/Go-Around
TRK	Track
TTL	Total
U	Unsatisfactory
UTC	Co-ordinated Universal Time
V/S	Virtical Speed
VFR	Visual Flight Rules
VHF	Very High Frequency
VMC	Visual Meteorological Condition
VNAV	Vertical Navigation
VOR	VHF Omni directional Range
VP	Vice President
VREF	Reference Speed
WT	Weight
XTRA	Extra
X-CHECK	Crosscheck
ZFW	Zero Fuel Weight

CHAPTER 1 - GENERAL

1.1 INTRODUCTION

The primary purpose of this SOP is to identify and describe the standard tasks and duties of the flight crew for each phase. The procedures are to be used in conjunction with Flight Crew Operation Manuals and the Flight Crew Training Manual. The Captain has the overall responsibility for all aspects of each flight. He/she will ensure adherence to the established procedures at all times.

Each crewmember must devote all possible attention to the progress of the flight, aircraft systems, look out and monitoring of other crewmember's actions. All configuration changes shall be accomplished with an announcement of intention, a check on appropriate limitations, and confirmation of completion.

All flight crewmembers in the flight deck shall continuously be aware of current airplane altitude, position, energy state, configuration, and maintain situational awareness appropriate for the phase of flight. Avoid casual and nonessential conversation during critical phases of flight, particularly during taxi, takeoff, approach, and landing. Unnecessary conversation reduces crew efficiency and alertness and is not recommended when below 10,000 feet MSL/FL100. At high altitude airports, adjust this altitude upward as required. (Refer to FOM 2.5(7.1)).

If required all normal and non-normal checklists ought to be read repeatedly from the checklist no matter how few the items are. Under non-normal conditions, the memory items shall be done from memory and only then the checklist has to be read.

This chapter gives:

a) An introduction to the normal procedure's philosophy and assumptions.

b) Step by step normal procedures.

1.2 RESPONSIBILITIES (CAPTAIN'S AND FIRST OFFICER'S)

It shall be in accordance with FOM 1.5.

1.3 GENERAL CONDUCT

Appearance of the crew creates a favorable impression of the airline. It is the Captain's responsibility to ensure that the crews are uniformly dressed in accordance with FOM 1.6(7.3).

All flight crewmembers while coming for training and office visit shall wear a formal or casual business type of attire.

1.4 CREW REPORT TIME

It shall be in accordance with FOM 1.7(7.18).

The captain must make sure that all flying crewmembers are in the aircraft 1 hour before the scheduled or revised time of departure and 45 minutes for deadhead crew and they shall report to the captain.

1.5 CREW RESOURCE MANAGEMENT (CRM)

It is proven that the most effective way to maintain the safety of a flight and resolve abnormal situations is to combine the skills and experience of all available crewmembers in the decision-making process to determine the safest course of action.

Error is inherent in all human beings. Therefore, all available resources and effort must be drawn before executing any decision. The captain shall make sure that everybody involved in the operation is comfortable to suggest, comment or even criticize positively. On all augmented crew flights if a non-normal/emergency situation arises, the relief captain must be called into the flight deck and be advised of the situation as soon as possible. The final decision is always the responsibility of the Pilot in Command. It is imperative that all crewmembers identify and communicate any situation that appears unsafe and out of the ordinary. As all our flights are conducted in a multi-crew environment, it is obviously vital to perform the FORDEC out loud so your colleague is able to follow-up your reasoning.

Following CRM model can be used as a tool for operation.

Facts — Identify the real problem comprehensively or confirm the failure (use every tool at your disposal: EICAS, training, experience); if possible determine the cause. Avoid 'confirmation bias'!

- This is a crucial step as it will determine the rest of the process!
- Retrace all courses of action which led to the present situation.

Options — Objectively consider any possible option to deal with the issue. The emphasis should be set on **any**.

- One option is the best course of action & a good FORDEC should lead you to it.
- When dealing with difficult & complex situations, you may be forced to aim for the least worst outcome!

Risks — Analyze the risk (benefits sometimes included) in accomplishing the viable options considered above

- Weigh the pros and cons carefully as it will necessarily drive the final outcome of your decision making process.

Decide — Decide based on the best option

- Discuss with other crew member(s)!

Execute — Execute required action

- Assign the task to the right people in order to demonstrate synergy in troubleshooting the problem.

Check — Check and evaluate the result of your action

- If the situation changes so as to let a new problem arise, run a new FORDEC.

1.6 CONTROL DISPLAY UNIT (CDU) PROCEDURES

Before taxi, the Captain or First Officer may make CDU entries. The other pilot must verify the entries. When possible, make CDU entries before taxi or when stopped. If CDU entries must be made during taxi, the F/O makes the entries. The Captain must verify the entries before they are executed. In-flight, the PM usually makes the CDU entries. The PF may also make simple CDU entries when the workload allows. The pilot making the entries executes the change only after the other pilot verifies the entries.

During high workload times, for example departure or arrival, CDU entries must be minimized. Comply by using the MCP heading, altitude, and speed control modes. The MCP can be easier to use than entering complex route modifications into the CDU.

1.6.1 CDU DISPLAY SELECTION

Flight crews are encouraged to select a display during any phase of flight that they feel is the most efficient way to get information. The crew may use the following table as a guideline.

PHASE OF FLIGHT	PF	PM
Takeoff	TAKEOFF REF	LEGS
Initial Climb	CLB	LEGS
Climb	CLB	PROGRESS 1
Cruise	CRUISE	PROGRESS 1
Descent	DESCENT	PROGRESS 1
Hold	HOLD	PROGRESS 1
Approach	LEGS	PROGRESS 1
Landing	LEGS	PROGRESS 1

1.7 CALLOUTS

Standard calls enhance crew awareness and co-ordination, and are standardized to remove ambiguity. They are used to achieve this in several ways by providing:

- Situational awareness
- Triggers or cues for action
- Requests for actions
- Decisions of a crewmember
- Detection of an incapacitation (which necessitates positive intervention after callout escalation)
- An unambiguous language

They must be used in all normal and non-normal operations. They shall be assigned to a particular crewmember; however, their execution is the important factor. Therefore, if a call is missed by the responsible crewmember, it shall be made by the other.

This section illustrates the use of standard calls for Ethiopian Airlines Flight Operations Boeing 737 fleet to provide a concise and definitive ruling on standard calls that will be used in all normal and non-normal operations.

Objectives:

- Simplicity.
- Consistency.
- A logical derivation.
- No ambiguity for a multi-cultural operation.
- Cross-fleet consistency (as much as possibly practical)

The broad policy regarding standard calls in flight is that the Pilot Flying (PF) shall make all requests for mode changes, configuration changes, checklists and decision calls, as he / she is primarily responsible for the aircraft flight path. The Pilot Monitoring (PM) shall make all other calls including FMA since they are to assist with the PF's situational awareness.

Both crewmembers should be continuously aware of current airplane altitude, position, energy state, configuration, and maintain situational awareness appropriate for the phase of the flight. One of the basic fundamentals of CRM is that each crewmember must be able to supplement or act as a backup for the other crewmember. Proper adherence to standard callouts is an essential element of a well-managed flight deck. These callouts provide required information about aircraft systems and the participation of the other crewmember. The absence of standard callouts at the appropriate time may indicate a malfunction of an aircraft system or indication or the possibility of incapacitation of the other pilot.

As part of good CRM & TEM, crews shall announce changes on the FMA and verify changes to the thrust mode display when they occur.

Both pilots should check the flight instruments and Flight Mode Annunciations (FMAs) at regular intervals to verify the selections made are correct for the phase of the flight. Both pilots should crosscheck MCP selections with the FMAs to ensure the airplane is responding as expected. Unexpected FMAs should be announced, evaluated, and addressed appropriately.

Any mode selection made on Mode Control Panel (MCP) should be notified by the PF and the PM will announce all FMA changes and armed modes (e.g., LNAV, LOC, GS, etc.). All FMA changes shall be confirmed by the PF. All FMA calls must be crosschecked on the PFD. Callouts are to be based on changes observed on the FMA, not on actions performed on the MCP.

The Pilot Monitoring (PM) makes callouts based on instrument conditions, FMAs, or observations for the appropriate condition. The pilot flying (PF) should verify the condition/location from the flight instruments and acknowledge. If the PM does not make the required call out, the PF shall make it.

The PM should be prepared to call out significant deviations from command airspeed or flight path. Both the PF and PM should call out any abnormal indications of the flight instruments (flags, loss of deviation pointers, etc.).

The PF should acknowledge all GPWS voice callouts except altitude callouts during an approach below 500ft. AFE. No callout is necessary from the PM if the GPWS voice callout has been acknowledged by the PF. The standard callout of "CONTINUE" or "GO-AROUND" at minimums is not considered an altitude callout and should always be made as appropriate. If the automatic electronic voice callout is not heard by the flight crew, the PM should make the callout.

***Note*:** If automatic callouts are not available, the PM may call out radio altitude at 100 feet, 50 feet, and 30 feet (or other values as required) to aid in developing an awareness of eye height at touchdown.

***Note*:** As part of proactive TEM, if a standard callout is not acknowledged, the PM needs to escalate the second callout augmented with an assertive recommendation & if no response, positive intervention (preceded with clear announcement of "I HAVE CONTROL") should be considered when flight parameters are not met or as expected, there is significant deviation from SOP without explanation, if the PF performs an unsafe act (i.e. that may lead to an undesired aircraft state) or there is no response to a challenge (i.e. possible subtle incapacitation)

1.7.1 STANDARD CALLOUTS FOR CLIMB AND DESCENT

	CONDITION/LOCATION	PM	PF
Climb and Descent	Approaching Transition Altitude	"TRANSITION ALTITUDE, SET STANDARD"	"STD SET, PASSING FL __ NOW"
	Approaching Transition Level	"TRANSITION LEVEL, ALTIMETERS SET __"	"QNH__ SET, PASSING ____ FEET NOW"
	10,000 ft. AFE	"10,000 FT. ABOVE"	"EXTERIOR LIGHTS OFF" for climb or "EXTERIOR LIGHTS ON & RECYCLE FASTEN BELTS" for descent.
	1,000 ft. above or below assigned altitude	"1,000 FT. TO LEVEL OFF"	"CHECK"
Descent	10,000 ft. MSL / FL100	"10,000 FEET" or "FLIGHT LEVEL 100"	"CHECK"

1.7.2 STANDARD CALL OUTS FOR ILS APPROACH

CONDITION/LOCATION	PM CALL OUT	PF CALL OUT
First positive inward motion of localizer pointer	"LOCALIZER ALIVE"	"CHECK"
First positive motion of Glideslope pointer	"GLIDESLOPE ALIVE"	"CHECK"
Final approach fix inbound	"OUTER MARKER/FIX, __FEET"	"CHECK"
1000 ft. AFE	"1000 FEET" "STABILIZED" or "UNSTABILIZED"	"CHECK" or "GO AROUND"
500 ft. AFE (check autoland status annunciator, if applicable)	"500 FEET" (F/D Or Single Autopilot Approach)	Radio Altimeter Reading (e.g. "500 Feet"), Autoland Status (e.g. "CMD, LAND 2 ,LAND 3 or NO AUTOLAND", "Roll Out And Flare Armed")
100 ft. above DA(H) (fail passive)	"APPROACHING MINIMUMS"	"CONTINUE"
Individual sequence flasher lights visible	"STROBE LIGHTS"	"CONTINUE"
At AH (fail operational airplanes) - check autoland status annunciator	"ALERT HEIGHT"	"CHECK"
At DA(H) - with individual approach light bars visible	"Approach Lights / red bars…."	"CONTINUE"
At or below DA(H)- Suitable visual reference established, PM calls visual cues	"Runway"	"LANDING"
At DA(H) - Suitable visual reference not established, i.e., PM does not call any visual cues	"NO RUNWAY"	"GO AROUND"
At minimums call out - if no response from PF on second call	"I HAVE CONTROL _____" (State Intention)	"YOU HAVE CONTROL"
Below DA(H)- Suitable visual reference not established, i.e. PM does not call any visual cues	"No Runway"	"Go Around"

1.7.3 STANDARD CALL OUTS FOR NON-ILS APPROACH

CONDITION/LOCATION	PM CALL OUT	PF CALL OUT
First positive inward motion of VOR or LOC course deviation indication	"COURSE/LOCALIZER ALIVE"	"CHECK"
Final approach fix inbound	"VOR/NDB/FIX ___ FEET"	"CHECK"
1000 ft. AFE	"1000 FEET" "STABILIZED" or "UNSTABILIZED"	"CHECK" or "GO AROUND"
100 ft. above DA(H) or MDA(H)	"APPROACHING MINIMUMS"	"CONTINUE"
Individual sequence flasher lights visible	"STROBE LIGHTS"	"CONTINUE"
At DA(H) or MDA(H) - with individual approach light bars visible	"MINIMUMS - APPROACH LIGHTS/ RED BARS" (if installed)	"CONTINUE"
At or below DA(H) or MDA(H)- Suitable visual reference established, i.e. PM calls visual cues	"Runway"	"LANDING"
At DA(H) or MDA(H) - Suitable visual reference not established, i.e., PM does not call any visual cues	"NO RUNWAY"	"GO AROUND"
At minimums call out - if no response from PF on second call	"I HAVE CONTROL _____" (State Intention)	"YOU HAVE CONTROL"
Below DA(H) or MDA(H)- Suitable visual reference established	"Runway"	"Landing"
Below DA(H) or MDA(H)- Suitable visual reference not established, i.e. PM does not call any visual cues	"No Runway"	"Go Around"

1.7.4 STANDARD PHRASEOLOGY

A partial list of recommended words and phrases follows:

Thrust:

- "SET TAKEOFF THRUST"
- "THRUST SET"
- "SET GO AROUND THRUST"
- "SET CONTINUOUS THRUST"
- "SET CLIMB THRUST"
- "SET CRUISE THRUST"
- "THROTTLE HOLD" for THR HLD

Flaps settings:

- "FLAPS UP"
- "FLAPS ONE"
- "FLAPS FIVE"
- "FLAPS FIFTEEN"
- "FLAPS THIRTY"
- "FLAPS FORTY"

Airspeed:

- "SPEED INCREASING"
- "80 KNOTS"
- "V1"
- "ROTATE"
- "SET ___ KNOTS"
- "SET VREF PLUS (additives)"
- "SET FLAPS ___ SPEED"

FMC Entry and Change Commands:

- "CLEAR MESSAGE"
- "SELECT ___ PAGE"
- "SELECT MAX ANGLE / MAX RATE"

To minimize confusion and enhance professionalism use standard terminology at all times.

- ARM(ED) — used for LNAV, VOR/LOC, and G/S
- CHECK — is a command for the other flight crewmember to check an item
- CHECKED — is a response to indicate that an item is checked
- CROSSCHECKED — is a callout verifying information from both flight crew's stations
- CAPTURED — used when an armed mode engages. e.g., "G/S CAPTURED"
- CLEAR — used as a command e.g., "CLEAR LEFT/RIGHT", "CLEAR MESSAGE", …etc.
- ENGAGED — e.g., LNAV engaged
- EXECUTE — request or command to EXECUTE FMC entry change
- GO-AROUND
- I HAVE CONTROL
- LANDING
- ON/OFF — used for some toggle and rotary switches with a position for ON and OFF. e.g., "LANDING LIGHTS ON/OFF", "WIPERS OFF", "ENGINE ANTI-ICE ON/OFF", "CENTER TANK FUEL PUMPS ON/OFF", …etc.
- SELECT — used generally as a command for pushbuttons on MCP, FMC and elsewhere. e.g., "SELECT LNAV", "SELECT PROGRESS PAGE", …etc.
- SET — e.g., "SET QNH ___", "SET___KNOTS", "SET FL___/Altitude ___ ft.", "PARKING BRAKE SET", …etc.
- TUNE/DETUNE — e.g., "TUNE ILS #1/#2", …etc.
- UP/DOWN — e.g., "LANDING GEAR UP/DOWN"
- VERIFY
- YOU HAVE CONTROL

etc.

The following is a guide to deviations which would prompt a standard call:

		PM	PF
Command Speed	+10 KTS/−0 KTS	"SPEED"	"CORRECTING"
Altitude	± 100 feet	"ALTITUDE"	"CORRECTING"
Heading	± 10^0	"HEADING"	"CORRECTING"
Pitch Attitude	Inappropriate pitch	"PITCH"	"CORRECTING"
Roll Attitude	Over 30° Bank	"BANK ANGLE"	"CORRECTING"
Glideslope	½ Dot	"GLIDESLOPE ABOVE OR BELOW"	"CORRECTING"
Localizer	½ Dot	"LOCALIZER L OR R"	"CORRECTING"
Vertical Speed During approach(1)	>1000FPM	"SINK RATE"	"CORRECTING"
Center Line	Significant Deviation from runway centerline	"CENTER LINE"	"CORRECTING"

(1) When depicted on the approach plate, a higher than normal up to 1300 FPM descent rate may be used provided crew and the particular aerodrome approach combinations have been approved. Limitation will not apply during diversion.

1.8 NORMAL PROCEDURES' PHILOSOPHY AND ASSUMPTIONS

Normal procedures verify for each phase of flight that the airplane condition is satisfactory and the flight deck configuration is correct.

Normal procedures are done on each flight. Refer to the FCOM Supplementary Procedures chapter for procedures that are done as needed, for example, No engine bleed takeoff and landing.

Normal procedures are used by a trained flight crew and assume:

a) All systems operate normally,

b) The full use of all automated features (LNAV, VNAV, autoland, autopilot, autothrottle, etc.). This does not preclude the possibility of manual flight for pilot proficiency where allowed.

c) Coordination with the ground crew before:

- Hydraulic system pressurization (1)(2),
- Flight control surface movement,
- Airplane movement

Normal procedures do not include steps for flight deck lighting and crew comfort items.

The Flight deck Check procedures are performed from memory by each pilot according to his/her area of responsibility. All procedural information is task-oriented. The scan flow sequence may be changed at the discretion of the Captain to suit operational requirements provided all preflight requirements are completed prior to engine start. The Captain retains final authority for all actions directed and performed.

(1) For time efficiency it is acceptable to ask for clearance to pressurize at any appropriate time.
(2) Do not overlook the need to coordinate with ground crew & obtain clearance to pressurize HYD systems so as to operate the MCD **[AYP only]**

1.9 CONFIGURATION CHECK

It is the crewmember's responsibility to verify correct system response. Before engine start, use system lights or indications to verify each system's conditions or configuration. After engine start, the master caution system alerts the crew to warnings or cautions away from the normal field of view.

If there is an incorrect configuration or response:

a) Verify that the system controls are set correctly.

b) Check the respective circuit breaker as needed. Maintenance must first determine that it is safe to reset a tripped circuit breaker on the ground. For more information refer to FOM 3.7.

c) Check the respective system light as needed.

Before engine start, use individual system lights to verify the system status. If an individual system light indicates an improper condition:

a) Check the Minimum Equipment List (MEL) to decide if the condition has a dispatch effect.

b) Decide if maintenance is needed.

Note: - For use of MEL Refer FOM 2.4

If during or after engine start, a red warning or amber caution light illuminates:

- Do the respective non-normal checklist (NNC)
- On the ground, check the MEL as required. Refer to FOM 2.4

If during recall, an amber caution light illuminates and then extinguishes after a master caution reset:

- Check the MEL as required.

1.10 CREW DUTY AND DUTY SHARE POLICY

The captain shall be the PF when taking off and landing under the following situations:

- At category C airports
- During low visibility operations
- During Overweight Landing
- Adverse weather conditions (contaminated runway, strong crosswind, tailwinds, etc.)
- Emergency or abnormal situations

Preflight and post flight crew duties are divided between the Captain and First Officer. Phase of flight duties are divided between the Pilot Flying (PF) and the Pilot Monitoring (PM).

To increase the experience of first officers, captains are encouraged to designate them as PF when conditions are suitable.

Duty share policy shall be per FOM 1.10.

Each crewmember is responsible for moving the controls and switches in their area of responsibility. The Captain may direct actions outside of the crewmember's area of responsibility.

The general PF phase of flight responsibilities are:

- Flight path and airspeed control
- Airplane configuration
- Navigation
- Communications during non-normal situations

The general PM phase of flight responsibilities are:

- Checklist reading
- Communications
- Tasks asked for by the PF
- Monitoring of taxiing, flight path, airspeed, airplane configuration and navigation
- Callouts

Normal procedures show who does a step by crew position (CAPT, F/O, PF, or PM):

The mode control panel is the PF's responsibility. When flying manually, the PF directs the PM to make the changes on the mode control panel.

The captain has the final authority for all tasks directed and done.

Crewmembers must remain at their assigned duty stations with seat belts fastened unless their absence is necessary for the performance of their duties in connection with the operation of the aircraft or their absence is essential in connection with their physiological needs. Refer to FOM 2.5 (7.16) for more information.

Whenever the PF has to leave his/her seat, he/she will hand over the aircraft by saying "YOU HAVE CONTROL" by stating the flight mode annunciations (FMC SPD, LNAV, VNAV PTH, etc.) and PM shall acknowledge by saying "I HAVE CONTROL." The PF returning to his seat will take over control by saying "I HAVE CONTROL."

In-flight, minimum required flight crew must not depart far from the flight deck as they cannot return quickly enough if unexpected turbulence is encountered or an emergency situation arises.

In all flights conducted with multiple or double crew, at least one of the non-operating crew must occupy a seat in the flight deck as an additional safety measure for takeoff and landing.

1.11 GENERAL USE OF CHECKLIST

1.11.1 NORMAL CHECKLIST OPERATION

Normal checklists are used after doing all respective procedural items (i.e., do and read method) to verify that certain critical procedural steps have been accomplished.

Normal checklist items are accomplished by the crewmembers using the CHALLENGE/RESPONSE methods except the AFTER LANDING procedures, which are silent.

The pilot-in-command shall ensure that the flight crew utilizes checklists and complies with operating procedures.

If airplane configuration does not agree with the needed configuration:

- Stop the checklist
- Complete the respective procedure step(s)
- Continue the checklist

If it becomes apparent that an entire procedure was not done:

- Stop the checklist
- Complete the entire procedure
- Do the checklist from the start

If possible, checklists must be done before or after high workload times. If the checklist is interrupted for a short time, continue the checklist with the next step. If a pilot is not sure where the checklist was stopped or the interruption was long, do the checklist from the start.

After completion of each checklist, the pilot reading the checklist calls, " _____ CHECKLIST COMPLETE."

1.11.2 NON-NORMAL CHECKLIST OPERATION

Non-normal checklists are provided to cope with or resolve non-normal situations on the ground or in flight. A checklist can have both memory and reference items.

As much of the condition statement to verify that the correct checklist has been selected and as much of the objective statement (if applicable) as needed to understand the expected result of doing the checklist should be read. Some checklists have additional information to provide data the crew may wish to consider. Additional information is not required to be read.

When non-normal situation occurs, at the direction of PF, both crewmembers do all memory items in their areas of responsibility without delay.

The PF calls for the checklist when:

- Flight path is under control
- Airplane is not in critical phase of flight
- ATC advised as applicable.
- All memory items are completed

During in flight non-normal situation, verbal confirmation is required for the actuation of critical controls by saying "CONFIRM" before moving the controls:

- An engine thrust lever
- An engine start lever
- An engine, APU or cargo fire switch
- A generator drive disconnect switch
- An IRS mode selector switch, when only one IRS is failed
- A Flight control switch.

"CONFIRM" is added to the checklist item when both crewmembers must verbally agree before action is taken. "This does not apply to the Loss of Thrust on Both Engines checklist."

The captain may direct reference checklists item to be done by memory if no hazard is created by such action, or if the situation does not allow reference to the checklist.

After completion of each non–normal checklist, the pilot monitoring states "___ CHECKLIST COMPLETE."

1.12 GENERAL BRIEFING

The purpose of briefing is for the pilot flying to inform the pilot monitoring for the planned course of action for both normal and non-normal situations during flight phase and to identify operational risks. Briefing shall be accomplished at the gate before each departure as soon as practical so as not to interfere with the final takeoff preparation and during the last portion of the cruise (150NM to 50NM before the expected top of descent).

The PF normally conducts departure or arrival briefing before engine start and top of descent respectively. After every briefing, the PF shall confirm that it is understood by asking the PM "ANY QUESTION OR SUGGESTIONS?"

Jump seat/observer briefing shall be conducted as stated on FOM 2.5 (7.13) at least on:

a) Handling of seat and seat harness

b) Emergency exits

c) Position, handling, and time of use of the oxygen masks

d) Position and use of the life vests (if required)

e) Behavior in emergency cases

f) Not to touch any controls, switches, instruments, or circuit breakers

g) Sterile Flight Deck policy.

Ethiopian Airlines standard briefing flow guide is used to accomplish a comprehensive briefing with Threat and Error Management concept.

Note: - Irrespective of who performs the briefing the decision to reject or to continue the takeoff shall always be the sole responsibility of the Captain.

1.12.1 DISPATCH BRIEFING

All flights which originate out of Addis Ababa are to be dispatched by Ethiopian flight dispatcher. Ethiopian personnel or the company's designated handling agent performs all outstation dispatches. Collecting all the necessary documents pertaining to the flight shall be the responsibility of the flight crew.

All available payloads will be determined at the dispatch office. Any change in the environmental conditions affecting the takeoff weight shall be consulted with the duty dispatcher and same information will be relayed to load control through the dispatch office.

The onboard performance tool (OPT) is used, when available, on all takeoff and landing performance calculations. Best optimization of performance with the given atmospheric and physical conditions is achieved through selection of optimum on the OPT whenever practical. The maximum available Assumed temperature should also be used for takeoff as much as practical. The OPT gives the pilots the advantage of single-point calculation.

The alternate forward C.G selection for takeoff on the OPT (as installed) should be to select a value immediately forward of the actual takeoff C.G value from the load/trim sheet, and if the actual C.G is less than the minimum value that can be selected in the opt, use FWD limit.

Both crewmembers are required to perform independent Takeoff/Landing performance calculations and crosscheck the resulting calculated data; preferably use "Compare Calculation" feature of OPT (found on Actions tab) & verify there is no mismatch. Any discrepancy shall be resolved prior to Takeoff/Landing.

The importance of crosschecking performance calculation between operating crewmembers is essential for the safe operation of the flight. Wind component should not be used to determine takeoff performance. However, for a tail wind, 150% of the tail wind component shall be used because of its impact on takeoff performance.

***Note*: -** As the 150% calculation is automatically considered by the OPT, flight crew need to put only the reported wind.

Crew must check and note the engine out acceleration height used for actual performance calculation during their preflight briefing.

1.12.2 JOINT BRIEFING

For flights originating out of home base (HAAB), flight crew must report at the assigned briefing room to update dispatch briefings & conduct a joint briefing as per FOM 2.4 (7.4). At out stations where there is a crew change, the captain shall do the joint briefing.

1.13 AUTOPILOT FLIGHT DIRECTOR SYSTEM (AFDS) PROCEDURES

The crew must always monitor:

- Airplane course
- Vertical path
- Speed

It is strongly recommended that the autopilot be engaged as early as possible (but not below 400 ft. AGL) when operating in busy terminal areas to allow the PM to assist more with monitoring duties.

When selecting a value on the MCP, verify that the respective value changes on the flight instruments as applicable. The crew must verify manually selected or automatic AFDS changes. Use the FMA to verify mode changes for the:

- Autopilot
- Flight director
- Autothrottle

During LNAV and VNAV operations, verify all changes to the airplane's:

- Course
- Vertical path
- Thrust
- Speed

1.13.1 AUTOMATIC FLIGHT

Autoflight systems can enhance operational capability, improve safety, and reduce workload. Automatic approach and landing, Category III operations, and fuel-efficient flight profiles are examples of some of the enhanced operational capabilities provided by autoflight systems. Maximum and minimum speed protections are among the features that can improve safety. LNAV, VNAV, and instrument approaches using VNAV are some of the reduced workload features. Varied levels of automation are available. The pilot decides what level of automation to use to achieve these goals by selecting the level that provides the best increase in safety and reduced workload.

When the autopilot is in use, the PF makes AFDS mode selections making the PM aware of MCP configuration change by calling out the change made. The PM may select new AFDS mode selections at the PF's discretion. Both pilots must monitor AFDS mode annunciations and the current FMC flight plan.

PM will:

- Make CDU changes and execute after confirmation with PF
- Make all gear and flaps selection when called for by PF
- Selects and identifies NAV AIDS as required

Both pilots shall monitor the autoflight system by:

- Cross-checking mode control panel status
- observing the result of any mode changes
- Supervising the resulting guidance and aircraft response.

When automatic systems do not perform as expected, the pilot should reduce the level of automation until proper control of path and performance is achieved.

Early intervention prevents unsatisfactory aircraft performance or a degraded flight path. Reducing the level of automation as far as manual flight may be necessary to ensure proper control of the aircraft is maintained. The pilot should attempt to restore higher levels of automation only after aircraft control is assured.

1.13.2 RECOMMENDED PITCH AND ROLL MODES

If the LEGS page and map display reflect the proper sequence and altitudes, LNAV and VNAV are recommended. If LNAV is not used, use HDG SEL or VOR/LOC. When VNAV is not used, the following modes are recommended:

- LVLCHG has logic to allow shallow climbs and descents for small altitude changes. There is no need to use V/S mode for passenger comfort.
- If unplanned speed or altitude restrictions are imposed during the arrival, the continued use of VNAV may induce an excessive workload. If this occurs, use LVLCHG or V/S as appropriate.
- If VNAV is not used during Non-ILS approaches beyond the IAF, V/S is the alternate recommended mode.

1.13.3 MANUAL FLIGHT

The normal company operating procedure is to fully utilize the automatic system, which must be closely monitored. However, manual flying following flight director commands below 10,000 ft. AAL and visual approach in good weather low traffic areas may also be used to maintain proficiency.

Whenever the PF intends to disengage the autopilot and/or autothrottle in order to fly the aircraft manually, the appropriate call “DISENGAGING AUTOPILOT and/or AUTOTHROTTLE” shall be made before doing so. The PM shall acknowledge.

The PM will make all AFDS mode selections and FMC/CDU changes at the request of the PF and must obtain confirmation from the PF before executing any CDU changes. Heading, altitude, and speed selections associated with ATC clearance or speed selections associated with flap position changes may be made without specific direction. However, these selections should be announced, such as “HEADING 170 SET.” The PF must be aware such changes are being made. This enhances overall safety by requiring that both pilots are aware of all selections, while still allowing one pilot to concentrate on flight path control. PM will make all gear and flaps selections when called for by PF.

Ensure the proper Flight Director modes are selected for the desired maneuver. If the flight director commands are not to be followed, the flight director should be selected off.

1.13.4 AUTOTHROTTLE USE

Autothrottle use is recommended during takeoff and climb in either automatic or manual flight. During all other phases of flight, autothrottle use is recommended only when the autopilot is engaged in CMD.

1.14 DISPLAY POLICY

1.14.1 WEATHER RADAR AND TERRAIN DISPLAY POLICY

Whenever the possibility exists for adverse weather and terrain/obstacles near the intended flight path, one pilot should monitor the weather radar display and the other pilot should monitor the terrain display. The use of the terrain display during night or IMC operations, on departure and approach when in proximity to terrain/obstacles, and at all times in non-radar environments is recommended.

Range selection for the purpose of weather/terrain display must be reasonable in order to determine possible threats for takeoff and landing (Optimal range would be 10 to 40NM).

Note: It may be useful to show the terrain display at other times to enhance terrain / situational awareness.

1.14.2. VERTICAL SITUATION DISPLAY

The Vertical Situation Display (VSD) (as installed) helps prevent controlled flight into terrain, and approach and landing accidents. It is a supplementary display, intended to improve situational awareness. Together with the lateral MAP, it creates a clear graphical picture of the airplane's horizontal and vertical position. In addition, it complements other safety features such as the GPWS. The VSD can be used during all phases of flight, but the main benefit is achieved during initial climb, descent, and approach. The VSD is not intended for use as a primary reference, or as a precise terrain following tool.

During departure, the VSD allows crews to recognize possible terrain conflicts more readily, before a GPWS alert is generated. This may be particularly useful if the airplane is held at low altitude for a prolonged time.

During climb and descent, the VSD allows crews to check the vertical flight profile and monitor the vertical flight path vector. This leads to earlier identification of altitude constraints that may not be met.

VSD use is encouraged as much as possible during all approaches because it assists in establishing the correct glide path. If an approach procedure contains one or more step down fixes, the crew can determine that the FMC path and the airplane current flight path angle will comply with the correct path and clear all step down fixes at or above the published altitude. Dedicated decision gates at 1,000 ft. and 500 ft. help the crew achieve a stabilized approach.

During an instrument approach using V/S, the crew can use the dashed vertical speed line to establish and monitor the vertical path. This leads to earlier recognition of an unstable approach or an inappropriate rate of descent.

For visual approaches without a published vertical path (GP angle), a 3° reference vector is displayed. Crews can adjust the flight path angle to overlay the 3° reference line to maintain a stable approach.

To improve speed stability control, crews can use the range-to-target speed symbol (green dot) to show where excess speed will be dissipated along the vertical flight path vector. If excess speed is not an issue, the symbol does not appear on the display.

1.15 MAINTENANCE STATUS (AIRCRAFT ACCEPTANCE)

The captain and the first officer shall check maintenance logbook, cabin logbook, daily/transit checks, and deferred items. Ensure all required documents e.g., airworthiness, insurance, etc. are up to date. The airworthiness of the A/C for acceptance of the flight should be in accordance with the requirement of all regulatory requirements, the AFM, and MEL/CDL. Refer to FOM 2.4(7.8).

Whenever departing with an unserviceable APU, the flight crews are encouraged to coordinate with IOCC to check the availability of ground power and air start unit at the intended airport(s) of destination.

The captain and first officer are responsible for giving early warning through company radio (either HF or VHF) or ACARS to maintenance on the status of the aircraft that adversely affect the next dispatch. At all subsequent stops, they are responsible to update the ATL entries with known or suspected defects reflecting the present status of the aircraft in a legible and no erasable manner. In case errors were committed, corrections should be made in a legible and identifiable manner and initialed.

Ethiopian is required to make a report to ECAA whenever defects affecting airworthiness conditions are identified. This is normally processed in the company safety management system. However, pilots must with due diligence report all those items listed on FOM 3.12. All aircraft defects should be recorded using the Fault Reporting Manual. Repeated remarks must be noted as such.

1.16 NON-ROUTINE REPORTS

Captains are required to report non-routine events such as delays, services, near misses, bird strikes, etc. using dedicated e-forms that are provided in the Logipad application. Maintenance status that affects the next dispatch shall be reported through company radio (HF, VHF) or ACARS the moment any problem is detected before landing (early warning system). It shall also be entered in the maintenance logbook.

1.17 RECORDS TO BE MAINTAINED

Except during cruise, paperwork is kept to a minimum to enhance safe and efficient flight operation. At the end of each flight, first officers are obliged to put back all Jeppesen charts in their appropriate binders and Jeppesen briefcases (as applicable). The captain must ensure its implementation. All flight paper works and required documents must be placed in the dispatch folder. When the flight terminates at Addis Ababa, the F/O has to hand over the dispatch folder to Flight Dispatch. On transit stations, both pilots shall check the flight folder for messages or any special instruction concerning the flight.

1.18 EVENTS REQUIRING MAINTENACE INSPECTION

During ground or flight operations, events may occur which require a maintenance inspection after the flight. Use the following guidance to determine what events require a maintenance inspection:

- Hard landing (Suspected) - specify if the landing was hard on the nose gear only, hard on the main gear only, or hard on both main and nose gear. Specify if the landing was a hard bounced landing.
- Overweight landing
- High drag / high side load event
- Severe turbulence
- Over speed-flap/slat, MMO/VMO, landing gear, landing gear tires
- Rejected takeoff
- High energy stop
- Lightning/ hail strike
- Bird strikes or bird ingestion
- Tail strike
- Spills of hazardous corrosives
- Spills of radioactive materials
- Extreme dust

- Any event that the pilot feels a maintenance inspection could be needed. An example of such event is an overly aggressive pitch up during a TCAS event or a Terrain Avoidance maneuver that could cause structural damage.

  Note 1: - If in doubt, the best course is to report it.

  Note 2: - For more information refer to FOM 3.12(7.6)

1.19 AIRCRAFT LIBRARY CONTENTS

Licenses/Certificates, NAV-KIT Content and Flight Folder / Package Contents shall be in accordance with FOM 1.11(4.6-4.8) and FOM 2.4(7.2.1). On transit stations both pilots shall check the flight folder for messages or any special instruction concerning the flight.

1.20 VIDEO SURVEILLANCE

The video surveillance display is to be used by the crew to identify individuals requesting flight deck entry prior to unlocking the flight deck door.

1.21 ELECTRONIC FLIGHT BAG (EFB)

In order to provide all personnel with the relevant information in the most convenient way and in a timely manner, Ethiopian preferably utilizes electronic data processing via e-mail, aircraft EFB (class II/III), and compatible own device (class I EFB) Tablet or laptop.

All flight crew must ensure that they are aware of relevant information for the flight(s) by synchronizing (updating) their personal class 1 EFB before departure in ADD. Each flight crew has to carry his/her class I EFB on board the aircraft.

For those aircraft not equipped with class II/III EFB's, one Jeppesen airway manual flight bag is a standard requirement. It is the flight crew's responsibility to carry current company-supplied or approved class I EFB. Refer to FOM 2.5 (7.24)

Note: - Crew must avoid fixation on the display or distraction from primary crew duties while using any EFB application during flight.

Normally EFBs can be used during all phases of flight under normal and non-normal situations. However, to avoid destructions pilots shall minimize the manipulation of EFBs below 10,000 feet AGL or during critical phases of flight.

Under some non-normal conditions like EFB failure, smoke in the flight deck, or any other situation that prevents or limits its usage, pilots can revert to any other resources provided by the company in hard or soft copy. (e.g., Class I EFB, QRH. etc.)

While in flight, crewmembers must not become preoccupied with the EFB system at the same time. Workload should be shared between flight crewmembers to ensure ease of use and continued monitoring of other flight crew functions and flight progress.

1.22 CLEAN COCKPIT PROCEDURE

Objects not stored in their dedicated area in the cockpit may fall & cause hazards such as damage to an equipment or inadvertent operation of controls or pushbuttons. Flight crew shall put and store all objects in their dedicated area in the cockpit.

- Cups in the cup holders.
- Books and paper, if any, in the dedicated book compartment or stowage.
- Water bottles with caps should be securely secured in the bottle holders.
- Meal trays should be collected as soon as possible.
- Personal devices should be properly secured in the stowage area.

Keep the cockpit clean at all times and remove any trash and used items out of the cockpit at the end of the flight.

Caution: Flight & Cabin crew must carefully handle liquids in the cockpit area to prevent spillage.

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CHAPTER 2 – NORMAL OPERATING PROCEDURES

Normal operating procedures' philosophy is discussed in the previous chapter (refer 1.8 NORMAL PROCEDURE'S PHILOSOPHY AND ASSUMPTIONS). If a procedure does not apply to all B737 variants, **[NG]**/**[737-8]**/**[CRGO]**/[**AYL]...**etc. on the procedure title and/or at the beginning of the procedure line will indicate the specific airplane type/s the procedure or the specific procedure line will apply to.

2.1 PREFLIGHT

CAPTAIN	FIRST OFFICER
Inspect Aircraft Technical Logbook (ATL) including cabin log, documents and ensure that the daily/transit check is done and valid. (refer to **ATTACHMENT D & E** for a detailed description of ATL, ADD, MEL/CDL)	
Perform electrical power up, as per the FCOM Supplementary Procedures [(2)]	
Perform preliminary preflight[(3)] and exterior inspection, as per the FCOM Amplified procedures.	
Conduct cabin crew briefing in accordance with FOM 2.4(7.4).[(4)]	
Obtain latest T/O data from ATIS or ATC	
Perform preflight procedure captain and first officer and CDU preflight respectively[(5)]	
At home base monitor operations VHF frequency 131.5 while on ground. Always set at least one HF radio to company frequency (5536, 8927, 11256, 17931) and perform SELCAL check for departure out of ADD.	
Authorize passengers boarding in consultation with the ramp coordinator and cabin crew.	After fueling is completed: Fasten Belts sign..... ON[(6)(7)]
Conduct takeoff briefing.[(8)]	
Call "PREFLIGHT CHECKLIST."	Do the PREFLIGHT checklist.
Sign aircraft technical log book (ATL)	Calculate fuel uplift on ATL
After papers arrive, both the captain and first officer shall check ZFW and C.G values, make proper CDU entry and crosscheck FMC T/O weight with load sheet T/O weight then sign. Complete Takeoff performance calculation and after crosschecking each other.[(9)(10)(11)(12)] Do the CDU Preflight Procedure - Performance Data steps.	
N1 bugs ..... Check Verify that the N1 reference bugs are correct. IAS bugs ..... Set	

MCP .....Set AUTOTHROTTLE ARM switch ..... ARM IAS/MACH selector ..... Set V2 Arm LNAV as needed Arm VNAV (13) Initial heading ..... Set Initial altitude ..... Set Stabilizer Trim ..... Set	MCP ..... Check
	Obtain ATC clearance (where applicable)

ATC clearance shall be recorded on the OFP by the F/O. The Captain will confirm the clearance with the F/O & crosscheck the following:

- ✓ RTE pages show ATC cleared airway(s), SID/ Transition routing if applicable.
- ✓ Expected or cleared initial altitude or flight level restriction displayed on PFD.
- ✓ SSR code entered in the transponder.

When all passengers are onboard and after obtaining the status of the cabin, the Captain authorizes the Team Leader Cabin Services to close all entry doors. (14)

Close and Lock Flight Deck Door. (15)

Verify that the LOCK FAIL light is extinguished.

Flight deck windows .....Closed and locked

Time permitting, welcome passengers per the PA guideline in the attachment section.

(1) The flight crew shall verify that relevant information in the OFP is consistent with the filled ats flight plan and with the data programmed into the navigation syatem.

(2) This procedure is accomplished to permit safe application of electrical power.

(3) Full alignment of the IRS must be made for all flights at the start of the flight, at transit stops with more than an hour of ground time and whenever deemed necessary.

(4) Flight Crew shall conduct joint briefing per SOP section 1.12.2. Up on flight origination or whenever a crew change occurs, the captain will conduct a verbal briefing with all the cabin crews present. However, extenuating circumstances such as preflight duties, passenger boarding, rescheduling, etc. may make it impractical to brief the entire cabin crew complement. Regardless of time constraints, the captain must at least brief the Team Leader Cabin Services & the Senior Cabin Crew.

(5) Select: Ignition R, Transponder (ATC) 1 and Weather Radar L (as applicable) on outbound flights from ADD, and Ignition L, Transponder (ATC) 2 and Weather Radar R (as applicable) on inbound flights to ADD. ALT 1 should be selected if CMD A will be used and ALT 2 should be selected if CMD B will be used.

(6) Confirm fueling completed before turning on the fasten belts sign. This can be accomplished by visually checking if the fueling hose has been disconnected or when receiving the fuel sheet.

(7) All Ethiopian flights are non-smoking. Therefore, "NO SMOKING" sign shall be ON at all times. The pilot in command shall ensure that all crew members and passengers are prohibited from smoking onboard or within the vicinity of the aircraft.

(8) The captain will initiate the departure briefing by determining who shall be the PF. The takeoff briefing shall be conducted using the takeoff briefing flow. Briefing shall be accomplished at the gate before each departure as early as practical so as not to interfere with the final takeoff preparation. Pay special attention for aircraft variant within the same type. E.g. short field performance (SFP) or non-SFP. ***DO NOT OVERLOOK THE IMPORTANCE OF BREIFING****, as it has been identified as the sole contributing factor in multiple root cause analysis of easily preventable deviations & incursions.*

(9) Captain enters the actual ZFW in the FMC from the load sheet and the First officer check the load sheet and confirms the entry and advice the Captain if there is any discrepancy Use GTOW from the FMC for T/O speed calculation in the OPT and both pilots shall confirm each entry into the OPT. the captain will make the V speeds entry on the FMC.

(10) The use of "Compare Calculation" feature of the OPT is recommended.

(11) Calculate the GTOW, V1, VR and V2 (Fixed derate and or assumed temperature method as applicable). Both pilots shall confirm OPT tail number and crosscheck results. Do not shut down laptops until after takeoff, this will keep the data in case of minor changes required for takeoff analysis. Acceleration height, takeoff speeds, and special engine out procedure must be noted prior to takeoff. On takeoff REF page 2/2 of the FMC enter the engine out acceleration height from the OPT and enter the intended thrust reduction altitude and acceleration height applicable for the intended type of departure (e.g. noise abatement). Set engine out acceleration altitude on the minimums selector with minimums reference selector on BARO. Set engine out escape route radial on both course selectors.

(12) When applicable use reduced takeoff thrust.

(13) If FMC and FCC versions don't allow it, do not attempt to arm VNAV on the ground before takeoff. Select VNAV after flaps and slats retraction is complete.

(14) Plan to close doors at least 10 minutes before scheduled departure time.

(15) On Cargo configuration Aircraft, a rigid cargo barrier door is used for access between the main deck cargo compartment & the supernumerary area. The barrier helps to prevent smoke from entering the supernumerary area if there is smoke in the main cargo compartment. The door is to remain closed during Taxi, Takeoff & Landing.

(16) The Flight Crew shall follow the procedure below for the purpose of safety and security check of an A/C that he/she is going to operate (for the first flight of the crew, crew change, when the A/C has been left unattended, etc.).

- *Flight crew members shall perform an interior preflight inspection of flight deck systems and emergency equipment for their availability, accessibility and serviceability prior to the first flight of the flight crew on an aircraft during duty period and also after an aircraft has been left unattended by a flight crew for any period of time. The procedure is accomplished as per the FCOM VOLUME-1 preliminary preflight procedure section.*
- *When no cabin crew is used, and if there is supernumerary in the cabin or supernumerary compartment, the flight crew shall perform a preflight inspection of emergency systems and equipment in the cabin and supernumerary compartment for their availability, accessibility and serviceability when a new flight crew has assumed control of the A/C and when the A/C has been left unattended by the flight crew or cabin*

crew. The procedure is accomplished as per the Emergency Equipment layout/EEL. The flight crew are responsible for ensuring all emergency equipment are available, serviceable, proper location, readily accessible, and securely stowed.

END OF PREFLIGHT

2.1.1 NOISE ABATEMENT

General

Noise abatement procedures are secondary to obstacle clearance requirements, air traffic control requirements, and any aircraft emergency. These procedures should only be accomplished where required. Nothing in these procedures shall prevent the pilot-in-command from exercising authority for the safe operation of the airplane.

Noise abatement departure

Comply with the specific local airport procedure by either NADP1, NADP 2 or as described in the airport briefing information.

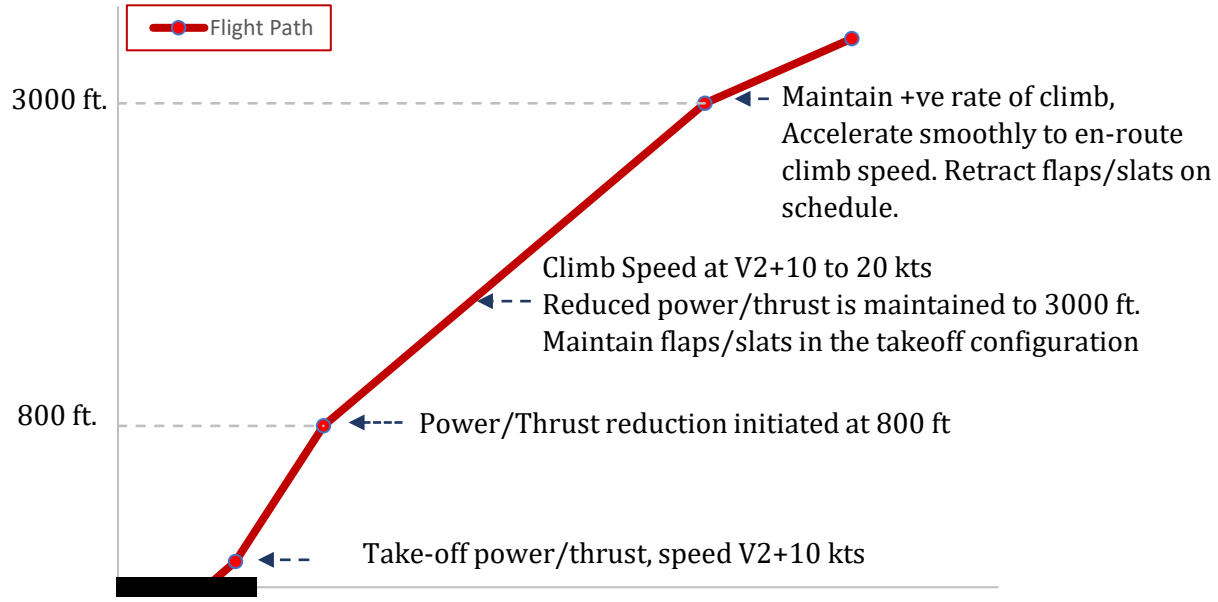
NADP 1

NADP 1 provides noise reduction for noise-sensitive areas in close proximity to the departure end of the runway *(see Figure below).*

To comply with NADP 1 departure:

- on Takeoff REF page 2/2
    - Enter thrust reduction altitude 800ft and acceleration height 3000ft.

NADP 1



NADP 2

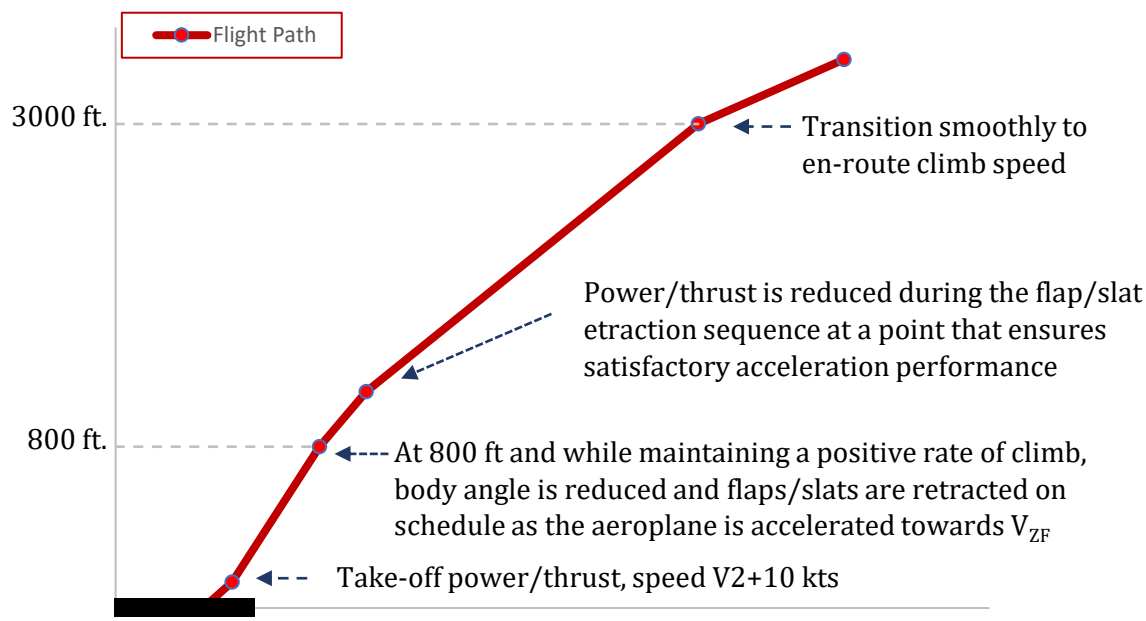
NADP 2 provides noise reduction to areas more distant from the runway end (see Figure below).

To comply with NADP 2 departure:

- Takeoff REF page 2/2
  - Enter an acceleration height 800 ft and enter thrust reduction altitude according to the diagram below.
- On VNAV CLB page insert flaps up speed up to 3000 ft above field elevation

Note: Thrust reduction altitude and acceleration height on Takeoff REF page 2/2 are heights above field elevation.

NADP 2



Note: Always comply with the noise abatement procedure stated in the airport briefing chart of the departure airport.

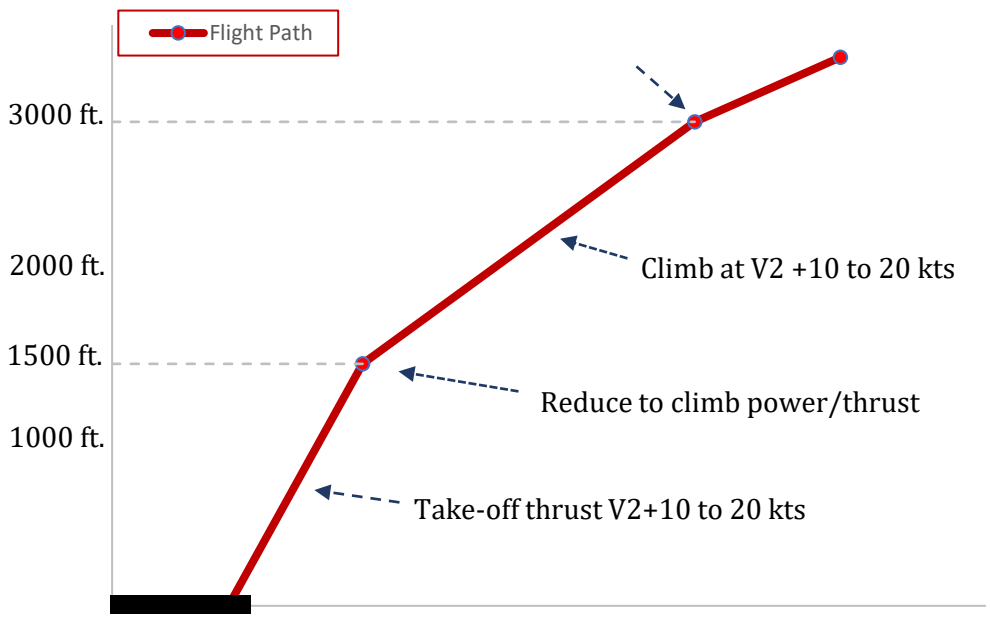
SUPERCEDED NOISE ABATEMENT PROCEDURES

Note: Many locations continue to prescribe the former Noise Abatement Departure Procedures A and B. Though no longer part of the ICAO PANS-OPS Doc. 8168, they have been reproduced in the following pages as supplementary information.

Noise Abatement Departure Procedure A (NADP A)

- Take-off to 450 m (1500 ft) above aerodrome elevation:
  - ✓ Take-off power
  - ✓ Take-off flap
  - ✓ Climb at V2 + 20 to 40 km/h (V2 + 10 to 20 kt) (or as limited by body angle).
- At 450 m (1500 ft):
  - ✓ Reduce thrust to not less than climb power/thrust.
- At 450 m (1500 ft) to 900 m (3000 ft):
  - ✓ Climb at V2 + 20 to 40 km/h (V2 + 10 to 20 kt).
- At 900 m (3000 ft):
  - ✓ Accelerate smoothly to enroute climb speed with flap retraction on schedule.

Noise Abatement Take-Off Climb - Procedure A

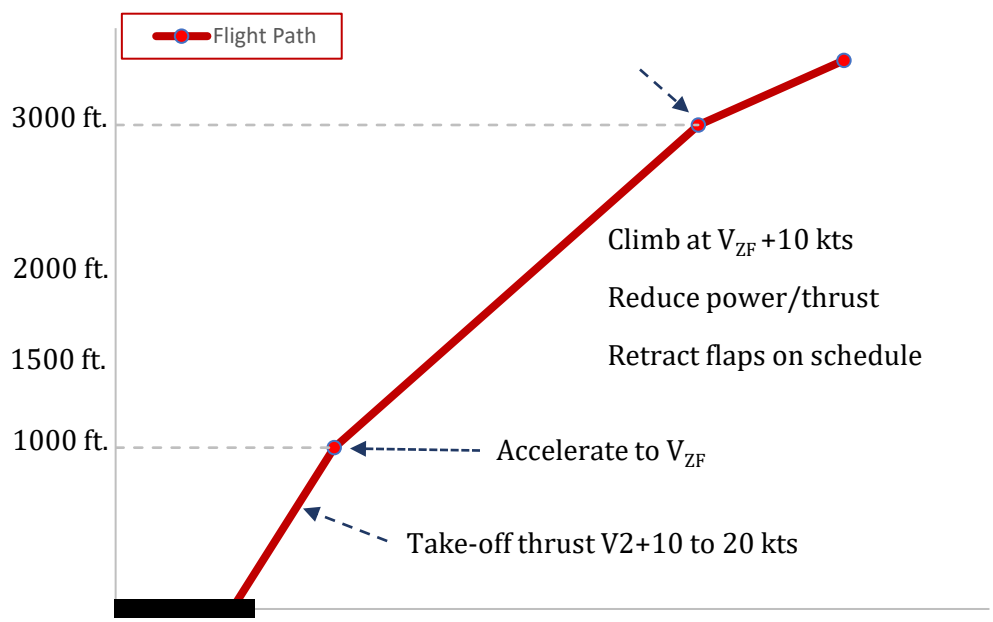


Note: For purposes of these procedures the heights given in metres and feet, and speeds given in kilometers/hour and knots are considered to be operationally acceptable equivalents.

Noise Abatement Departure Procedure B (NADP B)

- Take-off to 300 m (1000 ft) above aerodrome elevation:
  - ✓ Take-off power/thrust
  - ✓ Take-off flap
  - ✓ Climb at V2 + 20 to 40 km/h (V2 + 10 to 20 kt).
- At 300 m (1000 ft):
  - ✓ Maintaining a positive rate of climb, accelerate to zero flap minimum safe manoeuvring speed (V_{ZF}) retracting flap on schedule;
- thereafter, reduce thrust consistent with the following:
  - a) For high by-pass ration engines reduce to normal climb power/thrust;
  - b) For low by-pass ratio engines, reduce power/thrust to below normal climb thrust but not less than that necessary to maintain the final take-off engine-out climb gradient; and
  - c) For aeroplanes with slow flap retracting reduce power/thrust at an intermediate flap setting;
- thereafter, from 300 m (1000 ft) to 900 m (3000 ft):
  - ✓ Continue climb at not greater than VZF + 20 km/h (VZF + 10 kt).
- At 900 m (3000 ft):
  - ✓ Accelerate smoothly to enroute climb speed.

Noise Abatement Take-Off Climb - Procedure B



Note: For purposes of these procedures the heights given in metres and feet, and speeds given in kilometers/hour and knots are considered to be operationally acceptable equivalents.

2.2 BEFORE START PROCEDURE

CAPTAIN	FIRST OFFICER
Receive the latest ATIS or takeoff data	
Obtain a clearance from GND CREW. (1)(2) CALL "PRESSURIZE SYSTEM." (3)	Push the MFD SYS Switch. Pressurize fuel and hydraulic system, as per the FCOM Amplified procedures
Verify that the brake pressure is 2,800 psi minimum. Verify that the system A(4) and B pressures are 2,800 psi minimum.	
Trim ..... Set Check each trim for freedom of movement. Stabilizer trim..... ____UNITS Set the trim for takeoff. Verify that the trim is in the green band. Aileron trim ..... 0 units Rudder trim ..... 0 units	
Have all gear pins removed and check visually (5) Verify all doors are closed.	
CALL "PUSHBACK/START-UP CLEARANCE."	Set Transponder mode to ALT ON/OFF or XPNDR. (6) PUSHBACK/START-UP clearance .....Obtain When Clearance is Obtained: ANTI COLLISION light switch ..... ON
Call "BEFORE START CHECKLIST."	Do the BEFORE START checklist.

(1) Get clearance to Pressurize HYD Systems. If push back is required - Confirm Tow Bar is Connected and nose gear Steering Lock Out Pin is Inserted.
(2) For OTP, it is acceptable to ask for clearance to pressurize at any appropriate time.
(3) If pushback is needed and the nose gear steering lockout pin is not installed, depressurize hydraulic system A during the hydraulic panel set step
(4) If hydraulic System A is not pressurized at this stage, pressurize System A after pushback or towing is complete and verify the pressure.
(5) Nose Gear Steering lockout Pin will be removed after pushback.
(6) Set transponder mode selector to ALT ON or XPNDR (with assigned transponder code if already received) at all airports before the request for push back or taxi whichever is earlier.

(7) It is mandatory to report any delay of more than 5 min with the reason, in addition to the Captain Report, whether it is before or after door closure through ACARS. If ACARS is not available report as follows:

- *At home base through IOCC or HOLLOWAY radio.*
- *At outstation through handling frequency and HOLLOWAY radio.*

END OF BEFORE START PROCEDURE

2.3 ENGINE START AND BEFORE TAXI PROCEDURE

The Engine Start procedure may be done during pushback or towing. If a single ENG taxi out (SETO) will be implemented, follow the procedures outlined in 2.3.2.

AFTEER OBTAINING START AND/OR PUSH BACK CLEARANCE

CAPTAIN	FIRST OFFICER
Inform ground crew of ATC's approval and/or instructions for pushback and/or start. (1) (2)	
Set or release the parking brake as directed by ground handling personnel Note block out time.	Check transponder mode selector is on ALT ON/OFF or XPNDR Record block out time
When ready to start engines: Call "PACKS OFF."	Air conditioning PACK switches..... OFF Select secondary ENG indication.
Check Duct pressure	
Call for engine start in sequence	
Perform engine start per FCOM ENG start amplified procedure.	
When pushback or towing is complete: Set the parking brake as directed by ground crew.	Confirm parking brake is set.
After the engines are stable at idle: Call ..... "FLAPS ___" (3)	Takeoff Flaps .....Set (3) MFD SYS switch .....Push GENERATOR 1 and 2 switches ..... ON PROBE HEAT switches ..... ON WING ANTI-ICE switch .....As needed ENGINE ANTI-ICE switches .....As needed PACK switches .....AUTO

	ISOLATION VALVE switch .....AUTO APU BLEED air switch ..... OFF ENGINE START switches .....CONT Recall ..... Check Verify that the LE FLAPS EXT green light is illuminated. Call "FLAPS ___ GREEN LIGHT." APU .....OFF (unless needed) after confirming with Captain
After the removal of ground equipment is confirmed release ground crew for hand signal. Check the steering lockout pin visually.	Confirm ground equipment removed and area is clear.
Flight controls .....Check (as per the FCOM Amplified procedures)	Follow rudder movement without applying pressure
Call "BEFORE TAXI CHECKLIST."	Blank the lower display unit. Do the BEFORE TAXI checklist.
Command for taxi clearance.	Obtain taxi clearance.
Before taxiing out get hand signals from the ground crew. Check left side is clear of moving vehicles or any obstruction and call "LEFT IS CLEAR" get "RIGHT IS CLEAR" confirmation from the First Officer.	Check right side is clear of moving vehicles or any obstruction. Then advise the Captain "RIGHT IS CLEAR."

*(1) **Caution:** Do not hold or turn the nose wheel steering wheel during pushback or towing. This can damage the nose gear or the tow bar.*

*(2) **Caution:** Do not use airplane brakes to stop the airplane during pushback or towing. This can damage the nose gear or the tow bar.*

(3) If taxi is through slush or standing water, in low temperatures, taxi with flaps up (refer FCOM for more information).

END OF STANDARD PROCEDURE FOR DUAL ENGINE TAXI-OUT

2.3.1 STANDARD PROCEDURE FOR SINGLE ENGINE TAXI-OUT (FOR NG ONLY)

Single engine taxi out (SETO) saves fuel and reduces carbon footprint of the airline. The procedure requires to taxi-out with single engine and start the other engine close to the departure time.

The flight crew should use the conditions listed in the SETO briefing checklist to implement single engine taxi out procedure. Whenever SETO is implemented, the SETO briefing and checklist will be used to supplement the standard takeoff briefing and normal checklist.

To avoid unnecessary confusion, the captain should inform his intention for a single engine taxi-out while communicating the push back and/or engine start clearance with the ground crew.

CAPTAIN	FIRST OFFICER
Accomplish number 2 engine start using normal procedures.	
After number 2 ENG is started: Call "FLAPS ___ SETO BEFORE TAXI PROCEDURE."	SETO BEFORE TAXI PROCEDURE Accomplish (2) (3)
After the removal of ground equipment is confirmed release ground crew for hand signal. Check the steering lockout pin visually.	Confirm ground equipment removed and area is clear.
Flight controls .....Check (as per the FCOM Amplified procedures)	Follow rudder movement without applying pressure
Call "SETO BEFORE TAXI CHECKLIST."	Blank the lower display unit. Do the SETO BEFORE TAXI checklist.
Command for taxi clearance.	Obtain taxi clearance.
Before taxiing out get hand signals from the ground crew. Check left side is clear of moving vehicles or any obstruction and get "RIGHT IS CLEAR" confirmation from the First Officer.	Check right side is clear of moving vehicles or any obstruction. Then advise the Captain "RIGHT IS CLEAR."

(1) Refer SETO BRIEIFING CHECKLIST (ATTACHEMENT F).
(2) When checking master caution recall verify that all system annunciator panel lights illuminate and then extinguish except ELEC, HYD, and AIR COND. Illuminated amber lights are DRIVE 1, ENG 1 HYD pump LOW PRESSURE and DUAL BLEED. For any other system light illuminated, accomplish appropriate non-normal checklist.

Engine no. 1 will be started while the airplane is taxiing for departure or holding at the holding point with consideration of a minimum engine warm-up time of 3 minutes before setting takeoff thrust.

F/O's are responsible to start the second engine (Engine no. 1) at the direction of the Captain if the start is done while the aircraft is taxing. If the start will be done while the aircraft is stationary the Captain will normally do the engine start. However, the Captain can delegate the F/O to start Engine no. 1 even when the engine is started while the aircraft is stationary.

CAPTAIN	FIRST OFFICER
When ready to start engine number 1: Call ..... "START NUMBER 1 ENGINE"	LEFT PACK switch ..... OFF Select secondary ENG indication. Confirm ENG #1 thrust lever is closed. Start ENG #1 (1)
	After Engine no. 1 start is completed: Announce "ENG STABILIZED AT TIME __" Timer ..... Start (2)
Call "SETO BEFORE TAKEOFF PROCEDURE"	Blank the lower display unit. SETO BEFORE TAKEOFF PROCEDURE Accomplish (3)
Call "SETO BEFORE TAKEOFF CHECKLIST"	Do the SETO BEFORE TAKEOFF checklist

(1) Refer 2.3.1 for normal engine start procedure. While the first officer is starting ENG 1, he will handle the engine start switch and engine start lever. The FO is responsible for monitoring the ENG start. If any of the conditions as listed in the FCOM normal procedure for aborted start condition occurs, the FO should do the memory item from aborted ENG start non-normal checklist. Further decision and direction will be the responsibility of the pilot in command.
(2) Start timer to observe engine warm up requirement.
(3) Refer SETO BRIEIFING CHECKLIST (ATTACHEMENT C)

END OF STANDARD PROCEDURE FOR SINGLE ENGINE TAXI-OUT

2.4 RUNWAY AND TAXIWAY INCURSION

Refer to FOM 2.5 (7.23) for Runway/Taxiway Incursion/Excursion Prevention.

Note: Use appropriate exterior lights to avoid runway/taxiway incursions/excursions during night and low visibility operations.

2.5 TAXI PROCEDURE

CAPTAIN	FIRST OFFICER
[DAY] Call "TAXI LIGHT AUTO/ON." / **[NIGHT]** Call "TAXI and RUNWAY TURNOFF LIGHTS ON." (1)	TAXI light.....AUTO/ON RUNWAY TURNOFF lights ON as instructed
Lookout for moving vehicles and obstacles along the intended taxi route. Pay due vigilance to taxiway/RWY incursion. (2) (3) (4) Select WX radar/Terrain ON and evaluate the departure area weather.	
Monitor appropriate frequency and copy ATC clearance if not already obtained. (5)	
If there is a change or modification in departure clearance, review T/O Briefing accordingly. Check on LEGS page	Make the necessary changes on MCP, navigational aids, radio setups and FMC (preferably when the A/C is stationary or in straight taxi route), in confirmation with the Captain.
Review ATC clearance (if obtained during taxi)	
When cabin ready message is received from team leader cabin services and approaching holding point and closer to take off time. Call "ADVISE CABIN."	Announce "**CABIN CREW BE SEATED FOR DEPARTURE"** on the PA.

(1) For low visibility use exterior lights as appropriate.
(2) In case of unintentional taxiway/runway incursion/excursion ATC must be advised immediately and assistance shall be requested as appropriate.
(3) The F/O must refer to the taxi chart, monitor taxi progress and assist the Captain. Each flight crew must make use of separate set of charts for all phases of flight.
(4) PA announcement shall not be made while the aircraft is taxiing.
(5) All clearance and weather information must be recorded on the OFP.

END OF TAXI PROCEDURE

2.6 BEFORE TAKEOFF PROCEDURE

CAPTAIN	FIRST OFFICER
Engine warm up requirement before setting the takeoff thrust: **[NG]** Verify an increase in engine oil temperature before takeoff. **[737-8]** Verify that the engine oil temperature is at or above 31°C. (1) Engine warm up recommendations: **[NG]** Run the engines for at least 2 minutes. **[737-8]** Run the engines for at least 3 minutes. use a thrust setting normally used for taxi operations.	
Before entering the departure runway verify that the runway and the runway entry point are correct.	
Before entering the runway check both sides of the runway and call "LEFT CLEAR"	Check both sides of the runway and call "RIGHT CLEAR" Set position light STROBE & STEADY. Set transponder mode selector to TA/RA. Tune (or check) Departure frequency on VHF1 STBY. Guard VHF2(2).
Make sure WX/TERR is selected and optimum range is adjusted.(3)	
Check TFC is displayed on ND and check traffic display for approach and departure area	
Call "BEFORE TAKEOFF CHECKLIST."	Do the BEFORE TAKEOFF CHECKLIST.
Check proper runway heading and marking	
	Obtain T/O clearance when instructed by the captain.(4)

(1) Running the engines for approximately 3 minutes with an OAT 0°C or greater allows the engine oil temperature to rise to 31°C or above.
(2) Operations VHF frequency 131.5 must be monitored until lineup at home base.
(3) One pilot should select WX and the other should be on TERRAIN. At Captain's discretion PF should select WX and PM TERRAIN.
(4) Consider wake turbulence separation.

END OF BEFORE TAKEOFF PROCEDURE

2.7 TAKEOFF

As part of the final flight deck preparation, review the TAKEOFF REF page to ensure the entries are correct, including noise abatement where applicable. Ensure V2 and initial cleared ALT set on the MCP, V1 on PFD and LNAV/VNAV armed as required. The map display, map range, runway heading and LEGS page sequence shall be consistent with the departure procedure.

A rolling takeoff is recommended

PF	PM
Check the T/O runway and take off path is clear of traffic and weather (crosscheck with the WX radar). Verify that the airplane heading agrees with the assigned runway heading	
On lineup position and when cleared for takeoff: Call "LANDING LIGHTS ON".	Set the FIXED LANDING light switches to ON. **[NIGHT&NG]** Set the RETRACTABLE LANDING light switches ON.
Verify that the brakes are released. Advance the thrust levers to approximately 40% N1. Allow the engines to stabilize.	
Push the TO/GA switch.[(1)]	Call out flight mode annunciations (N1/TOGA)
Verify that the correct takeoff thrust is set.	
	Monitor the engine instruments during the takeoff. Call out any abnormal indications. Adjust takeoff thrust before 60 knots as needed. During strong headwinds, if the thrust levers do not advance to the planned takeoff thrust, manually advance the thrust levers before 60 knots. Call "THRUST SET".
Once TO/GA is selected, the Captain must hold the thrust levers and ensure takeoff thrust is set prior to 60 Knots. The captain's hand must be on the thrust levers until V1.[(2)]	

Monitor airspeed. Maintain light forward pressure on the control column.	Monitor airspeed and call out any abnormal indications.
Verify 80 knots and call "CHECK."	Call "80 KNOTS." Call "THROTTLE HOLD"
Verify V1 speed.	Verify the automatic V1 call out or call "V1."
At VR, rotate toward 15° pitch attitude. After liftoff, follow F/D commands.(3) (4)	At VR, call "ROTATE." Monitor airspeed and vertical speed.
Establish a positive rate of climb.	

(1) Engines must be stabilized before the activation of TO/GA switch to assure a smooth acceleration of the engines.
(2) The decision to stop (reject) is the sole responsibility of the Captain and must be made prior to V1.
(3) *The flight director pitch command is not used for rotation.*
(4) *After liftoff, use the flight director commands, in conjunction with indicated airspeed and other flight instruments to maintain the proper vertical flight path.*

END OF TAKEOFF

2.8 INITIAL CLIMB (THROUGH 10,000 ft. AAL)

PF	PM
Maintain target pitch attitude and airspeed at V2 + 15 to 25kts.(1)	Monitor instruments.
	Verify positive rate of climb on the altimeter and call "POSITIVE RATE."
Verify a positive rate of climb on the altimeter and call "GEAR UP."	
	Set the landing gear lever to UP.(2)
At 400 ft radio altitude, call for roll mode as needed	Select or verify roll mode. Verify VNAV engaged (if armed before takeoff).
At thrust reduction height, verify that climb thrust is set. (3)	
Verify acceleration at the acceleration height.(4) Call "Flaps__" according to the flap retraction schedule.	

	Set the FLAP lever as directed. Monitor flaps and slats retraction.
Engage the autopilot when above the minimum altitude for autopilot engagement. (5)	
	After flap retraction is complete: • Call "FLAPS UP NO LIGHT" • Set or verify engine bleeds and air conditioning packs are operating • Set the engine start switches as needed • **[NG/ NIGHT]** Retractable landing lights OFF • Set the AUTO BRAKE select switch to OFF • **[NG]** Set the landing gear lever to OFF after landing gear retraction is complete.
Call "AFTER TAKEOFF CHECKLIST." (6)	
	Do the AFTER TAKEOFF checklist.
Establish on the published airway as cleared by ATC or within 25 NM from the airport of departure. (7) (8)	

(1) Follow FD commands if available. The flight director commands pitch to maintain an airspeed of V2 + 20 Kts after liftoff until another pitch mode is engaged.
(2) Do not retract landing gear until positive rate of climb is indicated on the altimeter.
*(3) **[AYL]** At thrust reduction height, **PF** call "SET CLIMB THRUST." **PM** Push the N1 switch. And verify that climb thrust is set.*
(4) For airplanes with restrictions to use VNAV (FMC and FCC versions) at acceleration height PF calls for flaps up speed and PM sets flaps up speed. Engage VNAV after flaps and slats retraction is complete
(5) Pilots are encouraged to delay autopilot engagement after takeoff so that pilots will have good opportunity to exercise their manual flying skills during good weather and light traffic conditions
(6) After Takeoff Checklist should be completed before starting the climb and cruise procedure.
(7) For night departures or whenever visual contact with terrain could not be established and maintained, do not deviate from SID until above MSA unless under radar vectors.
(8) Do not fly off published airway unless approved by ATC.

END OF INITIAL CLIMB (THROUGH 10,000 ft. AAL)

2.9 CLIMB

PF	PM
Manage the operation using the AFDS and FMC CDU.	Monitor compliance with clearance. Check MCP altitude selection. Revise CDU entries as necessary announcing changes and obtain PF confirmation before executing.
If applicable maintain 250Kts to 10,000 ft. MSL or as specified by ATC. When flying out of high terrain airports at night and/or in IMC, use max angle climb to minimum safe altitude.	Monitor flight progress.
At transition altitude: (1) Set and crosscheck the altimeters to standard.	
	During climb, set the affected center tank fuel pump switch to OFF when a center tank fuel pump LOW PRESSURE light illuminates. Set both center tank fuel pump switches to OFF when a center tank fuel pump LOW PRESSURE light illuminates if the center tank is empty.
Passing 10,000 ft. AFE: Call "EXTERIOR LIGHTS OFF."	Passing 10,000 ft. AFE: Call "10,000 ft." FIXED LANDING light switches ..... OFF Runway Turnoff light switches ..... OFF Taxi light switch ..... OFF LOGO light switch..... OFF WHEEL WELL light switch ..... OFF
Climb at ECON or turbulence penetration speed.	At the captain's discretion and when conditions permit but not below 10,000 ft. AAL select Fasten Belts sign to AUTO. (2)

(1) Terrain selection on PM side is recommended below MSA.

*(2) **Caution:** Do not allow the shoulder harness straps to retract quickly. Buckles can pull or damage circuit breakers.*

***Note:** To avoid any TCAS TAs or RAs on climb profiles, flight crew shall limit the vertical speed of an A/C to not more than 1,000 feet per minute for the last 1,000 feet climbing to an assigned altitude or flight level unless otherwise instructed by ATC.*

END OF CLIMB

2.10 CRUISE

Optimum altitude is the altitude that gives the best fuel mileage for a given trip length, cost index, and gross weight. The selected Operational flight plan cruise altitude shall normally be the optimum altitude or if not possible within ±2000 ft. of the optimum altitude.

Flight crew should as much as possible stay close to the speed and altitude schedule set in the Operational flight plan. Any deviation may invalidate the economics of the operation. Crew should properly assess the best course of action should there be limited cruise altitude selection options.

When established in a level flight attitude, if the center tank contains usable fuel and a center tank fuel pump switch(es) is OFF, set the center tank fuel pump switch(es) to ON again.

Set the affected center tank fuel pump switch to OFF when a center tank fuel pump LOW PRESSURE light illuminates.

Set both center tank fuel pump switches to OFF when a center tank fuel pump LOW PRESSURE light illuminates if the center tank is empty.

During an ETOPS flight start the APU before the ETOPS segment. The APU must be ON for the entire ETOPS segment. During the last hour of cruise, check fuel crossfeed valve operation.

2.10.1 BOTH PILOTS

- Monitor level off.
- Record altimeters reading on OFP after first level off above FL 285. Crosscheck altimeter every hour for all flights in RVSM airspace.
- Review engine out drift down, MEA, MORA, MOCA, depressurization escape route, and diversion airfield along the route.

Whenever a flight is operated over high terrain, especially those above 10,000 ft., both pilots must be mentally prepared for a rapid descent. They must be aware of any escape route during the descent to avoid high ground as they undertake the diversion to a suitable airport in these mountainous regions. TERR may be displayed on ND as needed to enhance situational awareness.

- Check destination, en-route alternate, and destination alternate weather periodically.
- Use FIX pages to the fullest extent by entering appropriate major airfields along the route.
- Monitor FMC accuracy by tuning a VOR along the flight path and crosscheck with RMI needles.
- Check and adjust cabin temperature as required. Make passenger address regarding:
    - Flight progress and points of general interest

- Delay, expected length of delay, arrival time
- En-route and destination weather
- Anticipated rough air and seat belts requirement
- Alternative airport landing if diverted etc. per the attachment PA guideline

- Tune and monitor IATA in flight broadcast frequency 126.9, emergency frequency 121.5 and/or pilot air-to-air frequency 123.45 as appropriate.
- Make sure you have the right frequencies on HF windows to receive SELCAL.

Note: Headsets must be worn and boom microphones shall be used during the sterile flight deck period. During cruise, Captain's flight deck speaker may be used but the volume shall be kept at the minimum usable level to avoid interference with normal crew conversation within the flight deck. The F/O must wear the headset at all times.

2.10.2 CAPTAIN

- Monitor the progress of the flight.
- Shall record and monitor time over waypoints and fuel status on OFP at least every 30 minutes provided that the cruise time is an hour or more, for the purpose of identifying trends and for comparison to the OFP.

2.10.3 FIRST OFFICER

- Monitor the progress of the flight.
- Make all appropriate radio calls and complete all essential paper works.
- Check and record time over waypoints and fuel status on the Operational flight plan. For flights less than one hour, verbal discussion with the PIC suffices inflight fuel monitoring.
- Advise captain any change of planned fuel, ETA (if more than 3 minutes ATC must also be advised), etc. from the OFP.
- Advise block out/takeoff time, ETA destination, total persons onboard, and aircraft status if any through company radio.
- Call Holloway and report operation normal as required.
- Be responsible in giving early warning on the maintenance status of the aircraft that adversely affect next dispatch through company radio (either HF or VHF) or ACARS.

2.11 PRIOR TO TOP OF DESCENT

Prior to top of descent both pilots are required to get destination and alternate airport weather. Enter expected approach into the FMC (This may be done by either pilot).

In areas of known or suspected possible map shift, raw data must be monitored.

FMC accuracy must be confirmed prior to top of descent and terminal control area holding patterns.

In marginal destination weather conditions, pilots must prepare for a possible diversion. Special attention should be given to an airport they are not familiar with.

Note: 10 minutes before the commencement of descent the captain shall inform cabin crew top of descent time to alert the commencement of sterile flight deck status.

2.11.1 APPROACH BRIEFING

Approach briefing must be done in as much as possible before the top of descent. Waiting for the ATIS or aerodrome actual weather is the main reason why approach preparations are delayed and workload increase becomes a threat to the crew. Therefore crew should strategically plan to avoid last-minute briefing that can very well extend to below 10,000 ft. AAL. Pay special attention to aircraft variants within the same type.

The Captain will initiate the briefing using the approach briefing flow checklist by determining who shall be the PF and the type of approach to be flown. Before starting briefing, PF will hand over the aircraft by saying "YOU HAVE CONTROL" and PM will acknowledge "I HAVE CONTROL" stating the flight condition (LNAV, VNAV, HDG SEL, …etc.). All the items listed on the approach briefing flow checklist must be covered in the approach briefing and as much as practical should be completed before top of descent.

Note 1: Decision altitude for all 2D approaches will be MDA+50 ft.

Note 2: Recheck visibility minimums required against the actual aerodrome weather condition before passing the final approach fix (FAF) or the outer marker (OM).

Note 3: Guard control wheel and thrust lever at all times leaving the IAF or first flap extension in case of radar vectors whichever comes first.

Note 4: B737-700 is category "C"

B737-800 and B737-8 are category "D"

Note 5: Before the acceptance of clearance for any instrument approach the flight crew has to check flight crew qualification, required approach charts, the capability of the aircraft, serviceability of airborne and ground equipment, and the required minimum metrological conditions including ceiling and visibility.

Note 6: Use briefing strip for RNP AR APP

2.11.2 APPROACH PREPARATION

Before the acceptance of a clearance for any instrument approach the flight crew has to check and do the following;

- Check for the flight crew qualification if they are capable of and authorized to do the approach (example low visibility takeoff and approach)
- Check for availability and accessibility of the required approach charts,
- Check for the capability of the aircraft, serviceability of airborne and ground equipment,
- Check for the required minimum metrological/weather conditions including ceiling and visibility are satisfied as per the requirement.
- Check the weather minima with the Jeppesen approach chart minima or company minima before acceptance of the approach. (Example, ILS, RNAV, VOR, Circling Approach etc.).
- Discuss approach related duties of the PF and PM (TASK SHARING)
- Discuss the stabilized criteria and operating conditions that require a missed approach to be initiated. As per SOP 2.17 & 2.19.
- Follow the appropriate briefing flow strip found in the flight deck.

The flight Crew shall not continue an instrument approach beyond IAF and Approach Ban Point unless the reported meteorological conditions including visibility or controlling RVR are equal to or above specified for the approach in use. The flight crew shall follow the following procedures;

- Flight crew shall check the weather condition has not deteriorated below the required minima before passing the IAF and Approach Ban
- If appropriate weather condition is not met before passing IAF Flight Crew shall hold or divert depending on the weather trend and fuel onboard.
- If the weather condition deteriorated after passing the IAF, the Flight Crew may continue up to the Approach Ban. The approach can continue to the approach minima if the weather condition is equal to or above the specified for the approach in use. If the weather condition is below the required, the Flight Crew shall make a Go-Around.
- If the weather condition deteriorated after passing the Approach Ban, the Flight Crew may continue up to the Decision point.
- At decision point, if the Flight Crew has the required visual cue, they can continue the approach. Refer to FOM 2.2(7.18) for the required visual cues.
- At or below the decision point (DA(H) or MDA(H)), if sufficient visual cue as per above is not met or lost, a rejected landing is mandatory.

Note: Flight Crew shall not continue an instrument approach to land at any Airport beyond a point at which the limits of the operating minima specified for the approach in use would be infringed. The flight crew shall follow the above procedures.

Note: For Non-ILS Approach, Flight Crew are required to do a stabilized constant descent profile during the final approach segment. For approach procedure use VNAV or IAN approach as indicated below.

PF	PM
Destination Weather Info .....Obtain	
EFB/Approach Plates ..... Prepare	
System annunciator lights ..... Recall and review	
Landing Performance .....Compute	
FMC CDU preparation ..... Verify	FMC CDU ..... Prepare • DEP/ARR – Arrival and approach procedure • LEGS – Speeds and altitude constraints • FLAPS/VREF speeds
EFIS control panel .....Set MINIMUMS.....Set Set the RADIO/BARO minimums as needed for the approach. QNH .....Preset COURSE.....Set Set course to final approach course.	Audio Control Panel ..... Set VHF NAVIGATION radios ..... Set AUTO BRAKE Select switch..... Set Standby instruments ..... Set Set QNH and STBY ILS (as applicable) EFIS control panel ..... Set MINIMUMS ..... Set Set the RADIO/BARO minimums as needed for the approach. QNH.....Preset COURSE ..... Set Set course to final approach course. LANDING ALTITUDE indicator .....Set (1)
Do the approach briefing per the approach briefing flow.	

(1) If landing Altitude is above 6000 ft, set LAND ALT indicator to destination field elevation at initial descent.

END OF DESCENT PREPARATION

2.12 DESCENT AND APPROACH PROCEDURES

PF	PM
	During the descent, verify the RNP as needed.
[RNAV (RNP) AR APP] In the approach briefing include speed and altitude restrictions, missed approach, engine failure, and unable RNP procedures	**[RNAV (RNP) AR APP only]** • Select VOR UPDATE - OFF on the NAV OPTIONS page. • Inhibit other navaids as needed per NOTAM.
	[RNAV (RNP) AR APP] On the RNP PROGRESS page verify RNP for the approach
[RNAV (RNP) AR APP] Select VSD.	
	Set both center tank fuel pump switches to OFF when a center tank fuel pump LOW PRESSURE light illuminates if the center tank is empty.
	Verify that pressurization is set to landing altitude
Call "DESCENT CHECKLIST."	Do the DESCENT checklist.
	Obtain descent clearance.
When cleared to descend: Initiate descent by setting MCP altitude and selecting the appropriate pitch mode. Monitor the automatic flight in compliance with the profile.	Advise ATC when leaving assigned cruise level whenever required.
At Top of Descent: Call "Fasten belts ON."	Set Fasten Belts ON.
Passing 10,000 ft. AFE: Call "EXTERIOR LIGHTS ON."	FIXED LANDING light switches ..... ON Taxi light switch ..... ON If nighttime: Runway Turnoff light switches ..... ON LOGO light switch ..... ON WHEEL WELL light switch ..... ON FASTEN BELT switch ..... Recycle

At transition level: Set QNH and crosscheck the altimeters.	
Update changes to the arrival and approach, as needed. (1)	
Call "APPROACH CHECKLIST."	Do the APPROACH checklist.
At 3000 ft. AGL or initial FLAPS selection: Call "ADVISE CABIN."	Announce **"CABIN CREW BE SEATED FOR LANDING"** on the PA.

(1) In case of holding, start reducing to airspeed within three minutes or 12NM before the holding Fix. Report over the initial FIX entering the hold. If arrival time is affected and workload permits, advise company through operation frequency.

Note: *To avoid any TCAS TAs or RAs on descent profiles, flight crew shall limit the vertical speed of an A/C to not more than 1,000 feet per minute for the last 1,000 feet climbing or descending to an assigned altitude or flight level unless otherwise instructed by ATC.*

END OF DESCENT AND APPROACH PROCEDURES

2.13 LANDING PROCEDURE ILS

PF	PM
Call "FLAPS ___" according to the flap extension schedule.	Set the flap lever as directed. Monitor flaps and slats extension.
When on localizer intercept heading: • Verify that the ILS is tuned and identified • Verify that the LOC and G/S pointers are shown. • Verify correct sensing. • Confirm approach clearance has been received.	
Arm VOR/LOC.	Check FMA and call "VOR/LOC ARMED."
Verify that the localizer is captured. Verify the final approach course heading.	
Arm APP. (1) If a dual channel approach is desired engage the second autopilot. (not below 800' AGL)	Check FMA and call "G/S ARMED."
	Call "GLIDESLOPE ALIVE."
At glideslope alive: Call "GEAR DOWN." "FLAPS 15."	Set the landing gear lever to DN.

	Verify that the green landing gear indicator lights are illuminated. Set the flap lever to 15. Set the engine start switches to CONT. **[NG][NIGHT]** Retractable landing lights ON
Set the speed brake lever to ARM. Verify that the SPEED BRAKE ARMED light is illuminated.	
At glideslope capture: Call "FLAPS __" as needed for landing.	Set the flap lever as directed.
Set the missed approach altitude on the MCP.	
Call "LANDING CHECKLIST."	Do the LANDING checklist.
At the final approach fix (LOM, MKR, DME), verify the crossing altitude.	
If an autoland is planned, Verify the AFDS status at 500 feet AGL.	
For a single-channel approach, disengage the autopilot and autothrottle no later than the minimum use height for a single autopilot operation. For a dual channel approach, disengage the autopilot after touchdown.	
(1) When using LNAV to intercept the final approach course, LNAV might parallel the LOC without capturing it. *Warning: The airplane can then descend on the glideslope with the localizer not captured. Use HDG SEL to intercept the final approach course, if needed.*	

END OF LANDING PROCEDURE ILS

Note 1: For an ILS approach, use low drag approach whenever possible. The recommended Low Drag procedure is as follows:

- Select flaps 1 within 15 NM to the runway
- Intercept the glideslope with flaps 5.
- At 2000 ft. AAL lower the landing gear and set flaps to 15 and arm Speed brake.
- At 1500 ft AAL set flaps 30/40.

Note 2: Comply with speed requirements given by ATC or indicated by the airport procedure.

2.14 NON PRECISION APPROACH

INSTRUMENT APPROACH OPERATIONS

There are two methods for flying instrument approach operations, 2D and 3D. In a 2D approach operation, only lateral guidance will be displayed to the pilot, for example, in the form of a very high frequency omnidirectional radio range (VOR) needle or ILS lateral deviation scale. A 3D approach operation will also provide vertical guidance in the form of a vertical deviation scale. The nature of the instrument approach operation depends on both the instrument approach procedure and the technique used to fly the procedure. Operations using a CDFA technique may be considered to be 3D or 2D depending on how the vertical profile is determined and on the guidance provided to the pilot

3D APPROACH OPERATIONS

A 3D instrument approach operation uses lateral and vertical navigation guidance. Lateral and vertical navigation guidance refers to the guidance provided either by a ground-based radio navigation aid such as an ILS or microwave landing system (MLS) or computer-generated navigation data from ground-based, space-based or self-contained navigation aids, or a combination of these. Manually calculated rate/angle of descent is not considered vertical guidance, therefore this is not considered to be a 3D approach operation. 3D operations are conducted to a DA/H, which allows for height loss after the commencement of the missed approach. 3D approach operations can be either

a) Type A with a DH of 75 m (250 ft) or above; or

b) Type B with a DH less than 75 m (250 ft).

2D APPROACH OPERATIONS

2D instrument approach operation uses lateral navigation guidance only. 2D operations are conducted to an MDA/H, below which the aircraft should not descend without adequate visual references. 2D approach operations can only be Type A with an MDH of 75 m (250 ft) or above.

Decision altitude for all 2D approaches will be the MDA+50 ft.

RNAV, Localizer, and VOR are the types of non-precision approaches. The use of autopilot and autothrottle reduces crew workload and allows more time for management and monitoring of the approach. Automatic flight is the preferred method of flying non-precision approaches. Manually flying non-ILS approaches in IMC increases workload and does not take advantage of the significant increases in efficiency and protection provided by the automatic system.

It is recommended that whenever feasible, non-precision approaches shall be flown using A/P, F/D, A/T, IAN, LNAV, and VNAV or V/S modes.

For airplanes not equipped with IAN, VNAV is the preferred method for accomplishing non-ILS approaches that have an appropriate vertical path defined on the FMC LEGS page.

For a VANAV approach refer to the FCOM Landing Procedure - Instrument Approach using VNAV.

Airplanes with IAN are capable of using the MCP APP switch to fly non-ILS approaches that have an appropriate lateral and vertical path defined on the FMC LEGS page. An Approach using IAN provides the functions, indications, and alerting features similar to an ILS approach while following the FMC glide path. Although non-ILS approaches using LNAV and VNAV can still be executed, IAN is normally used in place of LNAV and VNAV because of improved approach displays, alerts, and standardized procedures.

Note: The VNAV PTH mode contains no path deviation alerting. For this reason, the autopilot should remain engaged until suitable visual reference has been established.

In order to facilitate the use of automation, the approach plan and go around profile shall primarily be entered into the FMC by selecting from the FMC database.

If the approach to be flown is not in the database, another approach having the same plan view may be selected for reference. In this case, waypoint altitudes must be checked and modified as required. When an approach is flown by this "overlay" method, raw data should be monitored throughout the approach to assure obstacle clearance.

The aircraft must be configured early and speed must be reduced to the maneuvering speed appropriate to the flaps and aircraft position in the approach. Normally the aircraft shall cross the IAF with no more than flaps 5. FAF shall be crossed in the landing configuration and the applicable final approach speed.

2.14.1 APPROACH REQUIREMENTS RELATING TO RNP

With appropriate operational approval, approaches requiring RNP alerting may be conducted in accordance with the following provisions:

- AFM indicates that the airplane has been demonstrated for selected RNP
- at least one GPS or one DME is operational
- any additional GPS or DME requirements specified by Operations specification or by the selected terminal area procedure must be satisfied
- No UNABLE REQ NAV PERF - RNP alert is displayed during the approach.
- when operating with the following RNP values, or smaller

APPROACH TYPE	RNP
VOR, VOR/DME	0.5 NM
RNAV	0.5 NM
RNAV (GPS)/(GNSS)	0.3 NM

2.14.2 RNP AR APPROACH

Prior to beginning the approach, the crew must brief the approach and complete needed preparations. Review FCOM SP: Instrument Approach - RNAV (RNP) AR (Automatic Flight). These include, but are not limited to, the following items, which is included in the approach review card to aid briefing:

- equipment that must be operational prior to starting the approach
- selection of the approach procedure, normally without modifications, from the navigation database
- for airplanes without NPS, one pilot should have the map display in the 10 NM or less range to monitor path tracking during the final approach segment for airplanes with NPS, the map display range may be set as the crew desires
- TERR display selected on at least one side
- RNP progress page displayed on the CDU, as needed. For airplanes equipped with NPS, selection of the CDU page is at the crew's discretion
- inhibiting of navigation radio updating, as needed. VOR/DME updating must always be OFF. DME-DME updating must be inhibited as needed (per NOTAM).

LANDING PROCEDURE

PF	PM

Initially,
• If on radar vectors
  ○ HDG SEL
  ○ Pitch mode (as needed)
• If enroute to a fix
  ○ LNAV or other roll mode
  ○ VNAV or other pitch mode	

Select TERR on map. Select CDU: ACT RTE X LEGS page.	Select TERR or WX radar on map. **[for airplanes without NPS]** Select CDU: RNP PROGRESS page.
Use LNAV and VNAV or other pitch mode for initial descent. VNAV is required from the FAF inbound. Some approach procedures can require use of VNAV from the IAF inbound. (1) On intercept heading, select or verify LNAV.	
Call "FLAPS ___" according to the flap extension schedule or approach speed constraint.	Set the flap lever as directed. Monitor flaps and slats extension.
Approximately 2 NM before the final approach fix and after ALT HOLD or VNAV PTH is annunciated:	Approximately 2 NM before the final approach fix: Call "APPROACHING GLIDE PATH."

Set DA(H) on the MCP. Select or verify VNAV. Select or verify speed intervention, (as installed/needed).	
Maximum Lateral Deviation (XTK ERROR): NPS amber indication or 1 x RNP Maximum Vertical Deviation - FAF to DA: 75 feet Monitor NPS **[for airplanes without NPS]** Monitor RNP PROGRESS page	
Call: • "GEAR DOWN" • "FLAPS 15."	Set the landing gear lever to DN. Verify that the green landing gear indicator lights are illuminated. Set the flap lever to 15. Set the engine start switches to CONT. **[NG][NIGHT]** Retractable landing lights ON
Set the speed brake lever to ARM. Verify that the SPEED BRAKE ARMED light is illuminated.	
Beginning the final approach descent: Call "FLAPS __" as needed for landing.	Set the flap lever as directed.
Call "LANDING CHECKLIST."	Do the LANDING checklist.
At the final approach fix: • Verify the crossing altitude • Crosscheck the altimeters. Verify they agree within 100 feet.	
Monitor the approach. **[for airplanes without NPS]** At least one pilot must have map display set to 10NM range.	
When at least 300 feet below the missed approach altitude & **passed FAF**, set the missed approach altitude on the MCP.	
If suitable visual reference is established at DA(H), or the missed approach point,	

disengage the autopilot and autothrottle and maintain the glide path to landing.	

(1) Direct To modifications are not permitted when the fix is the beginning of an RF leg

Note

- ***[For airplanes without NPS]*** *If a go-around/missed approach is needed, track the required course manually using the trend vector and map until LNAV is selected.*
- ***[For airplanes with NPS]*** *Verify that the vertical RNP is 125 feet. While there are no vertical RNP values published on the approach chart, the use of 125 feet will cause the NPS amber deviation exceedance alert to occur at 75 feet or slightly less deviation, since vertical ANP will be at least 50 feet at all times.*
- *RNP AR Operations with minima below AFM & AOC minima are prohibited.*

Warning: *If an UNABLE REQD NAV PERF-RNP is shown during the approach, whether the lateral or vertical RNP are exceeded or not, do the Go-Around and Missed Approach Procedure unless suitable visual reference is established and maintained.*

END OF RNAV (RNP) AR LANDING PROCEDURE

2.14.3 RAW DATA MONITORING REQUIREMENTS

During localizer-based approaches, applicable localizer raw data must be monitored throughout the approach.

During non-localizer-based approaches where the FMC is used for course or path tracking, monitoring raw data is recommended if available. When using the FMC's (LNAV) without GPS updating to conduct a terminal area procedure or an instrument approach, raw data must be monitored and checked to ensure correct navigation. This check should be done no later than the FAF.

Checking raw data for correct navigation may be accomplished by either of the following:

(1) Selecting POS and comparing the displayed raw data with the navaid symbols on the map.
(2) Displaying the VOR and/or ADF pointers on the map display and using them to verify your position relative to the map.

Constant descent profile shall be used on the final segment of non-precision approaches.

2.14.4 INSTRUMENT APPROACH USING IAN

RNAV, GPS, VOR, NDB and LOC are examples of approach types that can be flown by using IAN.
The applicable approach procedure shall be selected from the ARRIVALS page. Select the G/S prompt OFF if flying an ILS approach where the G/S is inoperative or when the G/S data is unreliable. Do not manually build the approach to add waypoints to the selected FMC procedure.

For an ILS or LOC approach, select the appropriate localizer frequency. For all other approaches, select a VOR frequency in both VHF control panels.

The final approach course must be selected on both course windows.

IAN should be used only for approaches that have one of the following features:

- A published GP angle on the LEGS page for the final approach segment
- An RWxx waypoint at the approach end of the runway
- A missed approach waypoint before the approach end of the runway, (for example, MXxx)

Note: approach using IAN are not authorized for RNAV (RNP) AR with FMC version U12 and earlier.

Flight mode annunciations for IAN approach are as follows:

APPROACH TYPE	FMA
ILS with G/S out or LOC	VOR/LOC and G/P
GPS, RNAV, VOR, NDB	FAC and G/P

LANDING PROCEDURE USING IAN

PF	PM
Call "FLAPS ___" according to the flap extension schedule.	Set the flap lever as directed. Monitor flaps and slats extension.
When on localizer/final approach course intercept heading: • Verify that the navigation radios are tuned and identified (as needed) • Verify that the deviation pointers are shown. • Verify correct sensing • Confirm approach clearance has been received	
Arm the APP mode. (1)	
Use LNAV or HDG SEL to intercept the final approach course as needed.	
Verify that the localizer/final approach course (FAC) is captured. Verify final approach course heading.	
	Approximately 2 NM before the final approach fix: Call "APPROACHING GLIDE PATH."
Approximately 2 NM before the final approach fix: • Call "GEAR DOWN." • "FLAPS 15."	Set the landing gear lever to DN. Verify that the green landing gear indicator lights are illuminated. Set the flap lever to 15. Set the engine start switches to CONT. **[NG][NIGHT]** Retractable landing lights ON
Set the speed brake lever to ARM. Verify that the SPEED BRAKE ARMED light is illuminated.	
At glide path capture: Call "FLAPS__" as needed for landing.	Set the flap lever as directed.
Set the missed approach altitude on the MCP.	
Call "LANDING CHECKLIST."	

	Do the LANDING checklist.
At the final approach fix: • Verify the crossing altitude • Crosscheck the altimeters. Verify they agree within 100 feet.	
Monitor the approach.	
If suitable visual reference is established at MDA(H), DA(H), or the missed approach point, disengage the autopilot and autothrottle and maintain the glide path to landing.	
(1) Warning: When using LNAV to intercept the localizer, LNAV might parallel the localizer without capturing it. The airplane can then descend on the glide path with the localizer not captured.	

END OF LANDING PROCEDURE USING IAN

2.14.5 INSTRUMENT APPROACH USING VNAV OR V/S

Use the autopilot during the approach to give:

- Autopilot alert and mode fail indications
- Accurate course and glide path tracking
- Lower RNP limits.

All Ethiopian flights shall be operated on QNH altimeter setting. Should the flight crew receive altimeter settings and clearance in QFE they shall request a QNH from the ATC service.

VNAV should be used only for approaches that have one of the following features:

- A published GP angle on the LEGS page for the final approach segment
- An RWxx waypoint at the approach end of the runway
- A missed approach waypoint before the approach end of the runway, (for example. MXxx).

The recommended roll modes for the final approach are:

- For RNAV or GPS approach use LNAV
- For VOR approach use LNAV
- For LOC approach use LNAV or VOR/LOC

LANDING PROCEDURE USING VNAV OR V/S

PF	PM
Call "FLAPS ___" according to the flap extension schedule.	Set the flap lever as directed. Monitor flaps and slats extension.
When on final approach course intercept heading for LOC approaches: • Verify that the localizer is tuned and identified • Verify that the anticipation cue or LOC pointer is shown. • Verify correct sensing	
Select LNAV or arm the VOR/LOC mode. (1)	Check FMA and call "LNAV ARMED." or "VOR/LOC ARMED."
Use LNAV or HDG SEL to intercept the final approach course as needed.	
Verify that LNAV is engaged or that VOR/LOC is captured.	
Approximately 2 NM before the final approach fix and after ALT HOLD or VNAV PTH is annunciated: Set DA(H) or MDA(H) on the MCP. **[using VNAV]** Select or verify VNAV. Select or verify speed intervention, (as installed).	Approximately 2 NM before the final approach fix: Call "APPROACHING GLIDE PATH."
Call: • "GEAR DOWN" • "FLAPS 15."	Set the landing gear lever to DN. Verify that the green landing gear indicator lights are illuminated. Set the flap lever to 15. Set the engine start switches to CONT. **[NG][NIGHT]** Retractable landing lights ON
Set the speed brake lever to ARM. Verify that the SPEED BRAKE ARMED light is illuminated.	
Beginning the final approach descent: Call "FLAPS __" as needed for landing.	Set the flap lever as directed.

[using V/S] Set V/S	
Call "LANDING CHECKLIST."	Do the LANDING checklist.
When at least 300 feet below the missed approach altitude & **passed FAF** for VNAV or 300 feet above MDA(H) for V/S: Set the missed approach altitude on the MCP.	
At the final approach fix: • Verify the crossing altitude • Crosscheck the altimeters. Verify they agree within 100 feet.	
Monitor the approach.	
If suitable visual reference is established at MDA(H), DA(H), or the missed approach point, disengage the autopilot and autothrottle and maintain the glide path to landing.	
[using V/S] Turn both F/Ds OFF, then place both F/Ds ON.	
(2) Warning: When using LNAV to intercept the localizer, LNAV might parallel the localizer without capturing it. The airplane can then descend on the VNAV path with the localizer not captured.	

END OF LANDING PROCEDURE USING VNAV OR V/S

2.15 CIRCLING APPROACH

The circling approach should be flown with landing gear down, flaps 15, and at flaps 15 maneuver speed. Use the weather minima associated with the anticipated circling speed.

The circling approach may be flown following any instrument approach procedure. During the instrument approach, use VNAV or V/S modes to descend to the circling MDA(H). Use of the APP mode for a descent to the circling MDA(H) is not recommended for several reasons:

- The AFDS does not level off at MCP altitude.
- Exiting the APP mode requires initiating a go-around or disengaging the autopilot and turning off the flight directors.

Maintain MCP altitude or MDA(H) using ALT HOLD or VNAV ALT (as installed) mode. Use HDG SEL for the maneuvering portion of the circling approach.

Note 1: If the MDA(H) does not end in "00", set the MCP altitude to the nearest 100 feet above the MDA(H) and circle at MCP altitude.

Note 2: If VNAV ALT (as installed) is allowed to remain as the pitch mode during the circling maneuver with a higher altitude (e.g., missed approach altitude) set in the MCP, the VNAV ALT pitch mode will revert to CWS P.

When in ALT HOLD at MCP altitude or the MDA(H) and before commencing the circling maneuver, set the missed approach altitude.

When runway insight and at MDA, turn 45° track left or right (refer to the published chart for prohibited area or direction). When you start the turn, time 45 seconds and turn to downwind TRACK. See note below.

Note 1: Adjust airplane heading and timing so that the airplane ground track does not exceed the obstruction clearance distance from the runway at any time during the circling approach. Use of the CDU Fix page to enhance situational awareness by inserting the circling area radius from the intended landing runway threshold is recommended.

Note 2: Obstruction clearance areas are determined as radius(r) from threshold by the maximum IAS during the circling approach.

MAXIMUM IAS	CIRCLING AREA RADIUS (R) FROM THRESHOLD
180 Kts	4.2 NM
205Kts	5.28 NM

When abeam landing runway threshold time approximately 3 s/100' above AFE. And turn to runway heading. See notes below

Note 1: E.g. If circling altitude is 1000' AAL time will be 30 seconds, if circling altitude is 1200 AAL time will be 36 seconds, etc..

Note 2: In addition, reduce 3 second for every 10 kts of tailwind.

Note 3: It may help to use Position Trend Vector for turn guidance.

Before starting the turn to base leg, select landing flaps and begin decelerating to the approach speed plus wind additive. To avoid overshooting the final approach course, adjust the turn to final to initially aim at the inside edge of the runway threshold. Timely speed reduction also reduces turning radius to the runway. Do the Landing checklist. Do not descend below MDA(H) until intercepting the visual descent profile to the landing runway.

When intercepting the landing profile disengage the autopilot, disconnect the autothrottle, and continue the approach manually. At this point in the approach, the pilot's attention should be focused on intercepting the visual descent profile rather than attempting to set the MCP or FMC to allow continued use of the autopilot. After intercepting the visual descent

profile, cycle both F/D to OFF, then to ON. This eliminates unwanted commands for both pilots and allows F/D guidance in the event of a go-around. Complete the landing.

Note: If a go-around is selected with either flight director switch in the OFF position, the flight director pitch or roll command bar on the corresponding side will disappear when the first pitch or roll mode is selected or engaged.

If a circling approach is anticipated with one engine inoperative, maintain gear up, flaps 10 and flaps 10 maneuvering speed from the final approach fix until just before starting the turn to base leg. As an option, use flaps 5, and flaps 5 maneuver speed as the approach flaps setting for the circling approach. Before starting the turn to base leg, select gear down and flaps 15 and begin reducing speed to VREF 15 plus wind additive. Do not descend below MDA(H) until intercepting the visual descent profile.

If a missed approach is required at any time while circling, make a climbing turn in the shortest direction toward the landing runway. This may result in a turn greater than 180° to intercept the missed approach course. Continue the turn until established on an intercept heading to the missed approach course corresponding to the instrument approach procedure just flown. Maintain the missed approach flap setting until close-in maneuvering is completed. This ensures the airplane remains within the circling and missed approach obstruction clearance areas.

Different patterns may be required to become established on the prescribed missed approach course. This depends on the airplane position at the time the missed approach is started.

In the event that a missed approach must be accomplished from below the MDA(H), consideration should be given to selecting a flight path that assures safe obstacle clearance until reaching an appropriate altitude on the specified missed approach path.

A go-around must be initiated for the following conditions:

- At MDA and after reaching the missed approach point (MAP), there is no visual cue
- If visual reference is lost any time during the circling maneuver
- If the approach is not stabilized.

2.16 VISUAL APPROACH

Enter downwind at an altitude of 1,500 feet above the runway elevation with flaps 5 and flaps 5 maneuver speed. Maintain a track approximately 2 NM parallel to the landing runway.

Abeam threshold, start time. At 15 seconds, extend the landing gear, select flaps 15, arm the speed brake, and slow to flaps 15 maneuver speed. 30 to 45 seconds past the abeam touchdown point take the landing flaps and slow down to the approach speed, turn to base leg, adjust thrust as required, and establish on a shallow descent.

Allow the speed to decrease to the proper final approach speed and trim the airplane. Do the Landing checklist.

Roll out of the turn to final on the extended runway centerline. Use of Position Trend Vector is a very good tool for turn guidance. Maintain the appropriate approach speed. If controlling thrust manually, attempt to keep thrust changes small to maintain speed and avoid large trim changes.

An altitude of approximately 300 feet AFE for each NM from the runway provides a normal approach profile. Stabilize the airplane on the selected approach airspeed with an approximate rate of descent between 700 and 900 feet per minute on the desired glide path and in trim. Stabilize on the profile by 1000 ft. above touchdown.

2.17 STABILIZED APPROACH CRITERIA

All approaches shall be stabilized by 1000 feet AFE. A Stabilized approach is when a stable speed, descent rate, and vertical/lateral flight path in landing configuration is maintained with the rate of descent not greater than 1000 fpm, or if applicable, the maximum rate of descent approved for the particular approach. Sustained engine thrust at or near Idle is not consistent with the stable approach criteria.

Note 1: Do not attempt to land from an unstable approach. A go-around must be executed following an unstable approach. The decision to execute a go-around is not an indication of poor performance.

Note 2: At 1000 ft. the pilot monitoring calls STABILIZED or UNSTABILIZED according to the parameters listed below.

Note 3: If any time after descent below DA/H or MDA/H, the PIC cannot maintain the required visual reference, a go-around shall be executed.

Note 4: Except for ATC assigned speed all approaches shall be stabilized by 1,000 ft AFE, for instrument meteorological conditions (IMC). All approaches shall be stabilized by 500ft in Visual metrological conditions (VMC). If the instrument approach is not stabilized by 1,000 ft and in VMC conditions, at “1,000ft” call out the pilot flying must respond “checked and correcting” and the approach shall be stabilized by 500 ft.

An approach is considered stabilized when all of the following criteria are met:

1. The airplane is on the correct flight path (both lateral & vertical).
2. Speed brake is not extended.
3. Only small changes in heading and pitch are required to maintain the correct flight path.

4. The airplane should be at approach speed. Deviations of +10 knots to 0 knots are acceptable if the airspeed is trending toward approach speed.
5. The airplane is in the correct landing configuration (flaps & landing gear position, not handle position).
6. Sink rate is not greater than 1,000 fpm. If an approach requires a sink rate greater than 1,000 FPM, a special briefing should be conducted.
7. Thrust setting is appropriate for the airplane configuration.
8. All briefings and checklists have been completed.
9. ILS should be flown within one dot of the glideslope and localizer, or within the expanded localizer scale.
10. IAN approaches should be flown within one dot of glide path and final approach course (FAC) within the expanded scale.
11. During a circling approach, wings should be level on final when the airplane reaches 300 feet AFE.
12. Unique approach procedures or abnormal conditions requiring a deviation from the above elements of a stabilized approach require a special briefing.

At 100 feet HAT for all visual approaches, the airplane should be positioned so the flight deck is within, and tracking to remain within, the lateral confines of the runway edges extended.

As the airplane crosses the runway threshold it should be:

- on a stabilized flight path using normal maneuvering
- positioned to make a normal landing in the touchdown zone (the first 3,000 feet or first third of the runway, whichever is less).
- Initiate a go-around if the above criteria cannot be maintained.

2.18 MANEUVERING (INCLUDING RUNWAY CHANGES AND CIRCLING)

When maneuvering below 500 feet, be cautious of the following:

- descent rate change to acquire glide path
- lateral displacement from the runway centerline
- tailwind or crosswind components runway length available
- runway length available.

2.19 MISSED APPROACH/GO-AROUND

2.19.1 ALL ENGINES

PF	PM
Push the TO/GA switch.	
Verify that the thrust increases.	
Call "FLAPS 15" or "FLAPS___" as needed.	Set the flap lever as directed and monitor flap retraction.
Verify the rotation to go-around attitude.	
	Verify that the thrust is sufficient for the go-around or adjust as needed.
Verify a positive rate of climb on the altimeter and call "GEAR UP."	Verify a positive rate of climb on the altimeter and call "POSITIVE RATE." Set the landing gear lever to UP.
If the airspeed is within the amber band, limit bank angle to 15°.	
Above 400 feet radio altitude: Verify/select a roll mode.	Verify that the missed approach altitude is set. Observe mode annunciation
Verify the missed approach route tracked.	
At acceleration height: Call "FLAPS __" according to the flap retraction schedule.	Set the flap lever as directed. Monitor flaps and slats retraction.
After flaps are set to the planned flap setting and at or above the flap maneuvering speed: Select LVL CHG or VNAV.	
Verify that climb thrust is set.	
Verify that the missed approach altitude is captured.	
Call "AFTER TAKEOFF CHECKLIST."	**[NG]** Set the landing gear lever to off after landing gear retraction is complete. Set the engine start switches as needed. Do the AFTER TAKEOFF checklist.

END OF MISSED APPROACH/GO-AROUND ALL ENGINES

Note: If a missed approach is required following a dual autopilot approach with FLARE arm annunciated, leave the autopilots engaged.

Following a manual instrument or visual approach use the same procedure as above except:

- Rotate manually to 15^0 Pitch attitude
- Select or verify go-around thrust
- Engage autopilot as needed
- Above 400 ft. RA, the PM verifies/selects a roll mode

2.19.2 MANDATORY MISSED APPROACH

An approach must be discontinued if a visual reference has not been attained or cannot be maintained per the stabilized approach criteria mentioned above or any one of the below happens before or during the approach.

1. A navigation radio or flight instrument failure occurs which affects the ability to safely complete the approach.
2. There is a malfunction which jeopardizes the safe completion of the approach or landing.
3. The navigation instruments show significant disagreement.
4. There is a loss or a doubt about situational awareness.
5. On radar approach, radio communication is lost.
6. On ILS final approach, either the localizer or the glideslope indicator shows full deflection.
7. On RNP / RNP AR based approach:
    - An alert message indicates the ANP exceeds the RNP.
    - On RNP approach, at any time the cross-track error exceeds 1.0 ×RNP and the flight crew is unable to change to Non-RNP procedure.
8. On RNP AR based approach:
    - If an UNABLE REQD NAV PERF-RNP is shown during the approach, whether the lateral or vertical RNP are exceeded or not, do the Go-Around and Missed Approach Procedure unless suitable visual reference is established and maintained.
9. If not visual, execute a missed approach for:
    - ILS Approach more than ± 1 dot G/S or LOC

- VNAV Approach VNAV PATH changes to VNAV SPD below FAF
- VOR Approach more than 1 dot deviation
- IAN Approach more than ±1 dot Glide Path or FAC

10. If an aircraft is not in a position from which to make a normal landing within the designated touchdown zone without the use of abnormal maneuvering.
11. Whenever instructed by ATC.
12. Landing clearance has not been received or is issued and subsequently canceled.
13. Minimum height by which the landing clearance is not received normally by threshold (50ft).
14. ATC changes the final approach clearance resulting in rushed action from the crew or potentially unstable approach, or the approach is unstable.
15. Any of the following alerts occur:
    - GPWS
    - TCAS
    - Wind shear

2.20 LANDING ROLL PROCEDURE

PF	PM
Verify that the thrust levers are closed. Verify that the SPEED BRAKE lever is UP. Without delay, fly the nose wheel smoothly onto the runway.	Verify that the SPEED BRAKE lever is UP Call "SPEED BRAKES UP. " If the SPEED BRAKE lever is not UP call "SPEED BRAKES NOT UP."
Monitor the roll out progress.	
Verify correct autobrake operation.	
Without delay, move the reverse thrust levers to the interlocks and hold light pressure until the interlocks release. (1) Then apply reverse thrust as needed.	Verify that the forward thrust levers are closed. When both REV indications are green, call "REVERSERS NORMAL." If there is no REV indication(s) or the indication(s) stay amber, call "NO REVERSER ENGINE NUMBER 1" or "NO REVERSER ENGINE NUMBER 2", or "NO REVERSERS".
By 60 knots, start movement of the reverse thrust levers to be at the reverse Idle detent before taxi speed.	Call "60 Kts"
After the engines are at reverse idle, move the reverse thrust levers full down.	
Before taxi speed, disarm the autobrake. Use manual braking as needed.	
If an autoland was accomplished, disengage the autopilot before turning off the runway.	

(1) ***Warning:*** *After the reverse thrust levers are moved, only a full stop landing can be made. If an engine stays in reverse, safe flight is not possible.*

END OF LANDING ROLL PROCEDURE

2.21 AFTER LANDING PROCEDURE

Start the After Landing procedure when clear of the active runway.

CAPTAIN	FIRST OFFICER
SPEED BRAKE lever ..... DOWN (1) left Weather Radar/Terrain ..... OFF Left F/D..... OFF	
	Rudder trim .....Zero units VHF NAV frequency .....VOR Transponder .....ALT ON/ XPNDR (2) Stab. Trim.....5 units Flaps.....UP Autobrakes ..... OFF STBY approach mode selector ..... OFF Right F/D switch ..... OFF Right WXR/ TRR ..... OFF ENG start switches .....OFF/AUTO Position switch .....OFF/ Steady (3) Exterior lights .....As needed (4) PROBE HEAT switches.....OFF/AUTO APU .....As needed (5)

(1) Stows the Speed brake lever below 30kts GS.
(2) Do not change code.
(3) Position light shall be STEADY between sunset and sunrise and whenever the visibility is 3 kms or less (applicable for ground operation). Otherwise OFF.
(4) At night, turn off the landing lights after confirming with the Captain.
(5) Starts APU when approaching the gate, after confirming with the Captain.

END OF AFTER LANDING PROCEDURE

Note: While accomplishing the after-landing procedure the FO should give prior attention to vacating the runway, communicating with ATC, monitoring taxi path, and scanning for traffic. At busy airports, first put transponder to ALT ON/ XPNDR and exterior lights as required then copy and comply with ATC instruction before accomplishing the rest of the procedures.

2.22 SINGLE ENGINE TAXI IN

It is more important to observe the engine cool-down period. Damage to the engine due to insufficient cool-down time outweighs the advantages gained from single-engine taxi-in.

Engine cool-down time shall not be less than 3 minutes unless there is an abnormal condition requiring immediate shutdown. Allow timing error to ensure a minimum of 3 minutes cool-down period at a thrust setting normally used for taxi operations. It is recommended to do single-engine taxi-in procedure at airports where the taxi-in time or distance is long and/or is possible. Preferably the engine towards the majority of the turns should be shut down to avoid excessive tire scrubbing.

- If reverse thrust higher than idle is used, then a full cool-down period starts after the engines decelerate to idle reverse.
- If no higher than idle reverse thrust has been used cool down period of the engines begins at touchdown. Other factors like surface condition, weather, traffic congestion, etc. must also be considered for this procedure. The procedure may be accomplished at the pilot's discretion with all due consideration to all safety aspects.

Consider the following:

- taxi time more than engine cool down time
- runway crossing
- taxiway and apron contamination
- taxiway/apron slopes
- sharp turns requiring more than 90 degrees.

2.23 PARKING, SHUTDOWN AND SECURE

Start the Shutdown Procedure after taxi is complete. External power should be "ON" before shutting down both engines if APU is not available.

CAPTAIN	FIRST OFFICER
Parking brake SET.	
Verify that the parking brake warning light is illuminated.	
	Electrical power SET: If APU power is needed: Verify that the APU GENERATOR OFF BUS light is illuminated. APU GENERATOR bus switches ..... ON Verify that the SOURCE OFF lights are extinguished. If external power is needed: Verify that the GRD POWER AVAILABLE light is illuminated. GRD POWER switch ..... ON Verify that the SOURCE OFF lights are extinguished.
Engine start levers CUT OFF. (1) Operate the engines at or near idle thrust for a minimum of three minutes before shutdown to thermally stabilize the engines and reduce under cowl soak-back temperatures. Taxi thrust can be considered idle thrust for this purpose. If idle reverse thrust or no reverse thrust is used during the landing roll out, the three-minute period can begin when thrust is reduced to idle for landing. Routine cool down times of less than three minutes before engine shutdown can cause engine degradation.	ANTI COLLISION light switch ..... OFF FASTEN BELTS switch ..... OFF FUEL PUMP switches .....OFF (2) (3) (4) WING ANTI–ICE switch ..... OFF ENGINE ANTI–ICE switches ..... OFF Set Hydraulic panel: ENG HYD PUMPS switches ..... ON ELEC HYD PUMPS switches ..... OFF Air conditioning PACK switches .....AUTO ISOLATION VALVE switch ..... OPEN Engine BLEED air switches ..... ON APU BLEED air switches..... As needed (5) Exterior lights switches .....As needed TRANSPONDER Set code to 2000 and mode selector STBY.

EFB CLOSE FLIGHT ..... Select (as applicable)	
After the wheel chocks are in place: Parking brake .....RELEASE	
APU switch as needed	
Call "SHUTDOWN CHECKLIST."	Do the SHUTDOWN checklist.
	On the last leg of the flight: IRS mode selectors ..... OFF EMERGENCY EXIT LIGHTS switch .... OFF WINDOW HEAT switches ..... OFF Air conditioning PACK switches ..... OFF
Call "SECURE CHECKLIST."	Do the SECURE checklist.
Check and sign the Aircraft Logbook and OFP.	Complete Aircraft Logbook and finalize the OFP

*(1) **[737-8]** Caution: Do not apply rotational force when moving the engine start lever.*

(2) If extended APU operation is needed on the ground, position an AC operated fuel pump on. If fuel is loaded in the center tank, position the left center tank fuel pump switch on to prevent a fuel imbalance.

(3) Caution: Do not operate the center tank fuel pumps with the flight deck unattended.

(4) Caution: Center tank fuel pump switches should be positioned on only if the fuel quantity in the center tank exceeds 453kgs.

(5) Run the APU for two full minutes before using it as a bleed air source.

END OF PARKING, SHUTDOWN AND SECURE

2.24 POST FLIGHT WALK AROUND

After each landing, a flight crew will complete the post-flight inspection. The crewmembers will be inspecting the aircraft for general condition and looking for any abnormalities which may have occurred during the flight and log into the maintenance logbook.

2.25 BEST FUEL PRACTICE

Fuel is a major expense of an airline. Thus, using systemic fuel saving by implementing different procedures and control mechanisms can compensate for the extra cost imposed on the operation without jeopardizing safety. The following are best practices/initiatives that can enhance fuel cost saving and reduce carbon dioxide emissions:

1. Single engine taxi out (SETO).
2. Single engine taxi-in.
3. Make low drag approach during landing.
4. Minimize pilot extra uplift of fuel and report any extra uplifted fuel per FOM 2.12.
5. Use GPU or start APU as late as possible, 90 seconds before reaching the gate.
6. Climb optimization.
7. Use of idle reverse when conditions allow.

CHAPTER 3 NON-NORMAL PROCEDURES

3.1 NON-NORMAL SITUATION GUIDELINES

When a non-normal situation occurs, in adition to FOM 3.1 the following guidelines apply:

- **NON-NORMAL RECOGNITION:** The crewmember recognizing the malfunction calls it out clearly and precisely

- **MAINTAIN AIRPLANE CONTROL:** It is mandatory that the Pilot Flying (PF) fly the airplane while the Pilot Monitoring (PM) maintains situational awareness and supports the Pilot Flying to ensure obstacle clearance and airplane control. Maximum use of the autoflight system is recommended to reduce crew workload

- **ANALYZE THE SITUATION:** NNCs should be accomplished only after the desired flight path and appropriate configuration are correctly established. Review all warning lights, caution lights and other alerts to positively identify the malfunctioning system(s). In the event of multiple malfunctions, the flight crew needs to analyze the situation, use good judgment and determine which NNC takes priority.

Note: Pilots should don oxygen masks and establish crew communications anytime oxygen deprivation or air contamination is suspected, even though an associated warning has not occurred.

- **TAKE THE PROPER ACTION:** Although some in-flight non-normal situations require immediate corrective action, difficulties can be compounded by the rate the PF issues commands and the speed of execution by the PM. Commands must be clear and concise, allowing time for acknowledgment of each command prior to issuing further commands. The PF must exercise positive control by allowing time for acknowledgment and execution. The other crewmembers must be certain their reports to the PF are clear and concise, neither exaggerating nor understating the nature of the non-normal situation. This eliminates confusion and ensures efficient, effective, and expeditious handling of the non-normal situation. In the event of multiple malfunctions, consider doing memory items first followed by reference steps. Upon completion of an NNC, review all warning lights, caution lights and other alerts to determine the need to do other NNCs

- **EVALUATE THE NEED TO LAND:** If the NNC directs the crew to plan to land at the nearest suitable airport, or if the situation is so identified in the QRH section CI.2, (Checklist Instructions, Non-Normal Checklists), diversion to the nearest airport where a safe landing can be accomplished is required. If the NNC or the Checklist Instructions do not direct landing at the nearest suitable airport, the captain must determine if continued flight to destination may compromise safety.

3.1.1 NON-NORMAL SITUATION GUIDELINES OVERVIEW

This is a general overview of how non-normal situations should be conducted.

Consider maximum use of autoflight system to reduce workload, if available and appropriate. If flight directors are selected on, ensure the proper flight director modes are selected.

PF	PM
Recognizes and announces the non-normal situation. Other pilot acknowledges.	
MAINTAIN AIRPLANE CONTROL Ensure that the flight path is under control.	Monitor the flight path.
	ANALYZE THE SITUATION Review all warning lights, caution lights and other alert lights to identify the non-normal situation. Prioritize alerts and recommend course of action.
Acknowledges and confirms the Pilot Monitoring's identification or recommendation.	Cancels the FIRE WARN and/or MASTER CAUTION lights after the Pilot Flying has acknowledged it.
TAKE THE PROPER ACTION Do the NNC memory items based on each crewmember's area of responsibility.	
Call for the appropriate NNC.	
	Completes the NNC.
Review all warning lights, caution lights and other alerts, and do other NNCs as needed.	
EVALUATE THE NEED TO LAND Review options for diversion or continued flight.	

3.1.2 CREW COORDINATION AND HANDLING OF THE THRUST LEVER, THE ENGINE START LEVER AND THE ENGINE FIRE SWITCH DURING NNC

PF handles the Thrust Lever at all times. PF retards the affected engine Thrust Lever after the PM confirms that the PF has identified the correct engine.

The Engine Start Lever and Engine Fire Switch of the non-operating engine will be operated by the PM in confirmation with the PF. Coordinate activation of the Engine Start Lever as follows:

- PM places a hand on and verbally identifies the Engine Start Lever for the engine that will be shutdown
- PF verbally confirms that the PM has identified the correct Engine Start Lever
- PM moves the Engine Start Lever to cutoff.

If the NNC requires activation of the Engine Fire Switch, coordinate as follows:

- PM places a hand on and verbally identifies the Engine Fire Switch for the engine that is shutdown
- PF verbally confirms that the PM has identified the correct Engine Fire Switch
- PM pulls the Engine Fire Switch.

PF	PM
Engine Thrust Lever	
"THRUST LEVER ____ ENGINE CONFIRM" Slowly reduce the Thrust lever to idle and state: "CLOSE"	"CONFIRMED"
Engine Start Lever	
"CONFIRMED"	"ENGINE START LEVER ____ ENGINE CONFIRM CUTOFF" Move the engine start lever to cutoff and state: "CUTOFF"
Engine Fire Switch	
"CONFIRMED"	"ENGINE FIRE SWITCH ____ ENGINE CONFIRM PULL" Pull Engine Fire Switch and state "PULL"

*Note Before stating the memory items, **PF/PM** will place hand on appropriate lever/switch.*

3.2 PRIORITY REQUEST IN TRAFFIC SEQUENCE FROM ATC

Appropriate reporting is the responsibility of the crew during conditions that present a safety concern for the flight. Ethiopian airlines standard communication procedure must be used by the crew as follows:

1. To declare an emergency with anticipated imminent danger precede distress message with "**MAYDAY**" three times. Examples would be: for any fire, a smoke with an unidentified source, an uncontrollable smoke or when the calculated usable fuel predicted to be available upon landing at the nearest airport where a safe landing can be made is less than the planned final reserve fuel, loss of thrust on both engines, rapid decompression, etc.
2. To declare urgency, precede the message with "**PAN PAN**" three times. E.g. an engine failure, engine surge, or when a person gets seriously injured or sick in flight, etc.

Note: There should be no reluctance to declare an emergency. Request for assistance can be rescinded at any time and the emergency canceled later if appropriate. For a fire that is extinguished, the emergency may be canceled and urgency will be declared.

There are some situations where the flight crew must land at the nearest suitable airport. These situations include, but are not limited to, conditions where:

- The non–normal checklist includes the item "Plan to land at the nearest suitable airport."
- Fire or smoke continues
- Only one AC power source remains (engine or APU generator)
- Only one hydraulic system remains (the standby system is considered a hydraulic system)
- Any other situation determined by the flight crew to have a significant adverse effect on safety if the flight is continued.

It must be stressed that for smoke that continues or a fire that cannot be positively confirmed to be completely extinguished, the earliest possible descent, landing, and evacuation must be done.

If a smoke, fire or fumes situation becomes uncontrollable, the flight crew should consider an immediate landing. Immediate landing implies immediate diversion to a runway. However, in a severe situation, the flight crew should consider an overweight landing, a tailwind landing, an off-airport landing, or a ditching.

Checklists directing an engine shutdown must be evaluated by the captain to determine whether an actual shutdown or operation at reduced thrust is the safest course of action. Consideration must be given to the probable effects of running the engine at reduced thrust.

3.3 ENGINE FAILURE AFTER V1

If an engine fails immediately after takeoff perform the appropriate engine failure procedure and land at the nearest suitable airport. Do not delay landing in an attempt to restart the engine. If an engine fails during climb, cruise, or descent and if inflight engine restart is successful and all parameters are normal, consider continuing the flight.

PF	PM
Control the aircraft.	Announce "ENGINE FAILURE."
Rotate and adjust attitude to maintain V2 to V2 + 20 (Follow F/D command).	Verify positive rate of climb on the altimeter and call "POSITIVE RATE."
Call "GEAR UP."	Select landing gear lever to Up.
Verify one of these occurs: An engine failure, An ENG FAIL alert shows, An engine flameout. If it is engine fire, engine severe damage or engine separation: Do the "ENGINE FIRE OR ENGINE SEVERE DAMAGE OR SEPARATION AFTER V1."	
At 400 ft. AGL: Call for roll mode.	Select roll mode.
Follow engine out escape route (with maximum bank angle of 15 degrees) or maintain extended runway track if no especial engine out procedure is available for the departure runway.	
Call "START APU." Disengage autothrottle.	Start APU. Verify autothrottle is disengaged.
Engage autopilot when above minimum altitude for autopilot engagement (1)	Declare Urgency (**PAN PAN** three times). ATC shall be notified of engine out escape route.
At engine out acceleration height: Verify acceleration or call "SET FLAPS UP SPEED" (2)	Verify VNAV acceleration to flaps up speed or select flaps up speed.
Accelerate and Call "Flaps__" according to the flap retraction schedule.	Set the FLAP lever as directed. Monitor flaps and slats retraction.
After flaps are retracted and at or above the flaps up maneuvering speed: If VNAV is not available select LVL CHG.	

Call "SELECT CON THRUST." And manually adjust thrust lever to set CON thrust. Set MCP ALT selector to MSA, engine out altitude, or as directed by ATC if under radar control.	Select CON on the FMC. Verify: Thrust is manually adjusted to CON thrust. MCP altitude selector is set to MSA, engine out altitude, or as directed by ATC if under radar control.
Call "ENGINE FAILURE OR SHUTDOWN CHECKLIST."	Do the ENGINE FAILURE OR SHUTDOWN NNC.
Call "AFTER TAKEOFF CHECKLIST."	**[NG]** Set the landing gear lever to off after landing gear retraction is complete. Set the engine start switches as needed. Do the AFTER TAKEOFF Checklist.
After reaching MCP altitude: Adjust thrust manually on the live engine to maintain the selected speed.	Verify thrust setting on live engine to maintain selected speed.
Determine next course of action.	

(1) For airplane with FMC U10.7 or earlier, engage autopilot after flaps are up and LVL CHG is selected.

(2) If VNAV was armed for takeoff and an engine failure is detected below the ENG out acceleration height set on the CDU, VNAV will issue a target speed equal to the greater of current airspeed or V2, limited to be less than or equal to V2+20 Kts and will start acceleration to flaps up maneuvering speed at the ENG OUT acceleration height set on TAKE OFF ref page 2/2 of the CDU.

END OF ENGINE FAILURE AFTER V1

3.3.1 ENGINE LIMIT OR SURGE OR STALL AFTER V1

PF	PM
Control the aircraft.	Announce "ENGINE LIMIT." or "ENGINE SURGE OR STALL."
Rotate and adjust attitude to maintain V2 to V2 + 20. Follow F/D command.	Verify positive rate of climb on the altimeter and call "POSITIVE RATE."
Call "GEAR UP."	Select landing gear lever to Up.
Verify Engine Limit or Surge or Stall.	
At 400 ft. AGL: Call for roll mode.	Select roll mode.
Follow engine out escape route (with maximum bank angle of 15 degrees) or maintain extended runway track if no especial engine out procedure is available for the departure runway.	
Call "START APU." Disengage autothrottle.	Start APU. Verify autothrottle is disengaged.
Engage autopilot when above minimum altitude for autopilot engagement [(1)]	Declare Urgency (**PAN PAN** three times). ATC shall be notified of engine out escape route.
At engine out acceleration height Verify acceleration or call "SET FLAPS UP SPEED"[(2)]	Verify VNAV acceleration to flaps up speed or Select flaps up speed.
Perform memory items from the applicable non-normal checklist. [(3)]	
Accelerate and Call "Flaps__" according to the flap retraction schedule.	Set the FLAP lever as directed. Monitor flaps and slats retraction.
After flaps are retracted and at or above the flaps up maneuvering speed: Select LVL CHG. Call "SELECT CON THRUST." And manually adjust thrust lever to set CON thrust on the operating engine.	Select CON on the FMC. Verify: Thrust is manually adjusted to CON thrust.

Set MCP ALT selector to MSA, engine out altitude, or as directed by ATC if under radar control.	MCP altitude selector is set to MSA, engine out altitude, or ATC cleared level.
Call "ENGINE LIMIT OR SURGE OR STALL CHECKLIST."	Do the ENGINE LIMIT OR SURGE OR STALL NNC.
Call "AFTER TAKEOFF CHECKLIST."	**[NG]** Set the landing gear lever to off after landing gear retraction is complete. Set the engine start switches as needed. Do the AFTER TAKEOFF Checklist.
After reaching MCP altitude: Adjust thrust manually on the live engine to maintain the selected speed.	Verify thrust setting on live engine to maintain selected speed.
Determine next course of action	

(1) For airplane with FMC U10.7 or earlier, engage autopilot after flaps are up and LVL CHG is selected.

(2) If VNAV was armed for takeoff and an engine failure is detected below the ENG out acceleration height set on the CDU, VNAV will issue a target speed equal to the greater of current airspeed or V2, limited to be less than or equal to V2+20 Kts and will start acceleration to flaps up maneuvering speed at the ENG OUT acceleration height set on TAKE OFF ref page 2/2 of the CDU.

(3) Memory items from the applicable NNC can be initiated anytime above 400 ft. when the airplane and flight path is under control. While performing memory items continue with the next set of procedures simultaneously without delay.

END OF ENGINE LIMIT OR SURGE OR STALL AFTER V1

3.4 ENGINE FIRE OR ENGINE SEVERE DAMAGE OR SEPARATION AFTER V1

PF	PM
Control the aircraft.	Announce "ENGINE FIRE." or "ENGINE SEVERE DAMAGE." or "ENGINE SEPARATION."
Rotate and adjust attitude to maintain V2 to V2 + 20. Follow F/D command.	Verify positive rate of climb on the altimeter and call "POSITIVE RATE."
Call "GEAR UP."	Select landing gear lever to Up. Announce "ENGINE ___ FIRE" and Push FIRE WARN Light. (as applicable)
Verify Engine Fire or Engine Severe Damage or Separation.	
At 400 ft. AGL: Call for roll mode.	Select roll mode.
Follow engine out escape route (with maximum bank angle of 15 degrees) or maintain extended runway track if no especial engine out procedure is available for the departure runway.	
Call "Start APU." Disengage autothrottle.	Start APU Verify autothrottle is disengaged.
Engage autopilot when above minimum altitude for autopilot engagement (1)(2)	Declare Emergency (**MAYDAY** three times) ATC shall be notified of engine out escape route.
At engine out acceleration height Verify acceleration or call "SET FLAPS UP SPEED" (3)	Verify VNAV acceleration to flaps up speed or Select flaps up speed.
Perform memory items from the applicable non-normal checklist. (4)	
Accelerate and Call "Flaps__" according to the flap retraction schedule.	Set the FLAP lever as directed. Monitor flaps and slats retraction.
After flaps are retracted and at or above the flaps up maneuvering speed: Select LVL CHG.	

Call "SELECT CON THRUST." And manually adjust thrust lever to set CON thrust on the operating engine. Set MCP ALT selector to MSA, engine out altitude, or as directed by ATC if under radar control.	Select CON on the FMC. Verify: Thrust is manually adjusted to CON thrust. MCP altitude selector is set to MSA, engine out altitude, or ATC cleared level.
Call "ENGINE FIRE OR SEVERE DAMAGE OR SEPARATION CHECKLIST."	Do the ENGINE FIRE OR SEVER DAMAGE OR SEPARATION NNC.
Call "AFTER TAKEOFF CHECKLIST."	**[NG]** Set the landing gear lever to off after landing gear retraction is complete. Set the engine start switches as needed. Do the AFTER TAKEOFF Checklist.
After reaching MCP altitude: Adjust thrust manually on the live engine to maintain the selected speed.	Verify thrust setting on live engine to maintain selected speed.
Determine next course of action	

(1) The airplane shall be in trim and the flight director commands shall be satisfied prior to autopilot engagement.
(2) For airplane with FMC U10.7 or earlier, engage autopilot after flaps are up and LVL CHG is selected.
(3) If VNAV was armed for takeoff and an engine failure is detected below the ENG out acceleration height set on the CDU, VNAV will issue a target speed equal to the greater of current airspeed or V2, limited to be less than or equal to V2+20 Kts and will start acceleration to flaps up maneuvering speed at the ENG OUT acceleration height set on TAKE OFF ref page 2/2 of the CDU
(4) Memory items from the applicable NNC can be initiated anytime above 400 ft. when the airplane and flight path is under control. While performing memory items continue with the next set of procedures simultaneously without delay

END OF ENGINE FIRE OR ENGINE SEVERE DAMAGE OR SEPARATION AFTER V1

Note: In case of ENGINE FAILURE/FIRE OR SEVERE DAMAGE OR SEPARATION and the First Officer is the PF it will be the Captain's discretion to let the F/O continue flying. If the captain decides to take over, he/she calls out "I HAVE CONTROL." The first officer releases the control slowly and calls out "YOU HAVE CONTROL."

3.5 MISSED APPROACH/GO-AROUND ONE ENGINE INOPERATIVE

PF	PM
Push the TO/GA switch. Verify that the thrust increases or Advance the Thrust Lever to TO/GA. (1)(2)	Verify that the thrust increases.
Call "FLAPS 1." **[Fail operational airplanes]** If landing with flaps 30: Call "FLAPS 15"	Set the flap lever as directed and monitor flap retraction.
Verify the rotation to go-around attitude or rotate smoothly to go-around attitude. (3)	
	Verify that the thrust is sufficient for the go-around or adjust as needed.
Verify a positive rate of climb on the altimeter and call "GEAR UP."	Verify a positive rate of climb on the altimeter and call "POSITIVE RATE." Set the landing gear lever to UP.
If the airspeed is below Vref15+15 knots (white bug) limit bank angle to 15°.	
Above 400 feet radio altitude: Verify/Select/Command a roll mode. (4)	Select a roll mode as directed. Verify that the missed approach altitude is set. Observe mode annunciation
Verify the missed approach route tracked. (5)	
At acceleration height: Call "SET FLAPS UP SPEED" Accelerate and Call "Flaps__" according to the flap retraction schedule.	Select flaps up speed. Set the FLAP lever as directed. Monitor flaps and slats retraction.
After flaps are retracted and at or above the flaps up maneuvering speed: Select LVL CHG. Call "SET CON THRUST." And manually adjust thrust lever to set CON thrust. Set MCP ALT selector to MSA or as directed by ATC if under radar control.	Select CON thrust. Verify:

	Thrust is manually adjusted to CON thrust. MCP altitude selector is set to MSA or ATC cleared level.
Verify that the missed approach altitude is captured.	
After reaching missed approach altitude: Adjust thrust manually on the live engine to maintain the selected speed.	Verify thrust setting on live engine to maintain selected speed.
Call "AFTER TAKEOFF CHECKLIST."	**[NG]** Set the landing gear lever to off after landing gear retraction is complete. Set the engine start switches as needed. Do the AFTER TAKEOFF Checklist.
Determine next course of action	

(1) The pilot must control yaw with rudder and trim. Some rudder pedal pressure may be required even with full rudder trim.

(2) If the missed approach is initiated after LAND 3 or LAND 2 is annunciated, yaw is initially controlled by the autopilot.

(3) After TO/GA is engaged, the AFDS initially commands a go-around attitude, then transitions to maintain command speed as the rate of climb increases.

(4) If the missed approach is initiated after LAND 3 or LAND 2 is annunciated, yaw is initially controlled by the autopilot. Be prepared to immediately apply rudder input when selecting another roll mode, pitch mode, or when altitude capture occurs above 400 feet AGL because the autopilot reverts to single autopilot operation and automatic control of the rudder is discontinued.

Note: After the system reverts to normal (single) autopilot operation, the autothrottle should be disconnected.

(5) If the ENG out escape route, a visual pattern or any other route will be followed instead of the published missed approach routing, the crew should include the intended missed approach route during briefing and ATC should also be notified of such intention.

Note: Consider following an Engine-Out escape route in case of executing a missed approach at a high altitude airports if planning an overweight landing.

END OF MISSED APPROACH/GO-AROUND ONE ENGINES INOPERATIVE

3.6 ENGINE INOPERATIVE CRUISE/DRIFTDOWN

PF	PM
Control the aircraft.	Announce the failure: - "ENGINE FAILURE." "ENGINE LIMIT." or "ENGINE SURGE OR STALL." or "ENGINE FIRE." etc.
Disengage autothrottle.	Verify autothrottle is disengaged.
Call "SELECT CON THRUST" and manually adjust Thrust Lever to set CON thrust on the operating engine.	Select CON on the FMC.
Call "START THE APU."	Start APU.
Perform memory items from the applicable non-normal checklist (as required).	
	Select ENG OUT on FMC CRZ page and check ENG OUT Speed and Cruise altitude
	Declare Urgency (**PAN PAN** three times) or Emergency (**MAYDAY**) and request the drift down altitude
As applicable to the airspace turn 90 degrees to clear the airway and fly parallel until you determine on the diversion airport and receive clearance. (1)	
	Advise ATC of the Drift down altitude and enter diversion route on the FMC.
Set the engine out cruise altitude and speed on the MCP.	Check correct ENG OUT cruise altitude and speed from QRH or FPPM for ETOPS flights.
Descend in LVL CHG using the ENG OUT speed.	Monitor the descent to target altitude.
Call for the appropriate NNC.	Do appropriate NNC.

(1) The flight crew should follow procedure applicable to the airspace they are flying.

END OF ENGINE INOPERATIVE CRUISE/DRIFTDOWN

3.7 CABIN DEPRESSURIZATION AND RAPID DESCENT

This maneuver is designed to bring the airplane down smoothly to a safe altitude, in the minimum time, with the least possible passenger discomfort.

The crewmember observing the rapid loss of pressurization immediately call it out as clearly as possible. E.g., "CABIN CLIMBING."

3.7.1 BOTH PILOTS IN THE FLIGHT DECK

At first indication of rapid loss of cabin pressure:

CAPTAIN	FIRST OFFICER
Don oxygen masks and set regulators to 100%. (1)	
Silence the warning horn.	
Select Boom/Mask switch to mask (as installed). Establish crew communications. ("Captain Speaking"/ "First Officer Speaking")	
Call "CONTROL THE CABIN" Check if F/O can control the cabin pressure.	Try to control the cabin pressure: Pressurization mode selector ..... MAN Outflow VALVE Switch ..... Hold in CLOSE until the outflow VALVE indication shows fully closed If unable to control the cabin pressure (2): Announce "CABIN UNCONTROLLABLE."
Declare "RAPID DESCENT" when: • Rapid decompression occurs (i.e. differential pressure is zero in 10 seconds.) • Cabin alttitude is 14000' or above • Cabin is uncontrollable(see footnote 2)	
PASS OXYGEN switch ..... ON and hold for one second	
Set a lower altitude in the MCP altitude window. As required, remain on track or HDG SEL approximately 90° to the PF side If conditions permit. Consider high terrain, prohibited area, traffic, weather, escape route, etc (3) Select LVL CHG Select IAS in the IAS/MACH display by pushing the C/O switch. Confirm annunciations on the PFD.	Declare Emergency (**MAYDAY** three times) and advise ATC that you are making rapid descent. Transponder code .....Set 7700 Fasten Belts ..... ON Landing lights ..... ON Engine Start switches .....CONT Watch conflicting traffic both visually and by reference to TCAS. And monitor icing conditions.

Close Thrust Levers. Extend speed brake smoothly.	Select TERR as needed to enhance situational awareness.
Announce to the cabin "**ATTENTION, THIS IS YOUR CAPTAIN. PULL DOWN AND USE YOUR OXYGEN MASK, PULL DOWN AND USE YOUR OXYGEN MASK**."	Advise ATC that you are making rapid descent to MORA or 10,000 ft. whichever is higher. Call every 5,000 ft. in descent.
Refine altitude window to MORA or 10,000 ft. whichever is higher. Set target speed to MMO/VMO or If structural integrity is in doubt, maintain existing speed.	Silently read EMERGENCY DESCENT NNC. Advise the captain that checklist is complete by showing thumbs up or through intercom.
When clear of the airway: Fly parallel or follow the established escape route or ATC instructions.	
Set QNH as appropriate.	
Lookout for traffic and monitor icing conditions.	
At 2,000 ft. to level off: Set turbulence penetration speed or LRC. If structural integrity is in doubt maintain existing speed.(3)	Call "2,000 FT. TO LEVEL OFF."
At 1,000 ft. to level off: Speed Brake lever smoothly to down detent.	Call "1,000 FT. TO LEVEL OFF."

(1) When donning oxygen mask, put headset on the back of the neck for easy reach to put it back on over the air hose. Speakers may also be put on.

*(2) Determination of cabin controllability requires the use of multiple indications: cabin differential pressure, cabin altitude & cabin climb rate. If cabin differential pressure is 0 & cabin altitude is above 15,000 feet, cabin is uncontrollable. If closing the OFV switch results in a decrease in cabin altitude then the cabin can be considered controllabe as long as the cabin altitude is below 15,000 feet. Moving the OFV to CLOSE may likely have little or no **immediate** effect on the cabin altitude indicator but if the cabin rate of climb is **responsive** to pilot inputs, this does not necessarily mean the cabin is uncontrollable. Regardless of effects of closing the OFV switch, the cabin should be considered uncontrollable, if cabin altitude horn sounds or CABIN ALTITUDE WARNING light illuminates & cabin altitude is above 15,000 ft. (For detailed information refer Boeing FOTB 19-01)*

(3) Refer to the Jeppesen Airway Manual for contingency procedures in different airspaces.

(4) At level off altitude set LRC or reduce to turbulence penetration speed. If structural integrity is in doubt, maintain existing speed or lower speed as applicable and determine next course of action.

END OF CABIN DEPRESSURIZATION AND RAPID DESCENT – BOTH PILOTS IN THE FLIGHT DECK

Note 1: After level off and at or below 14,000 feet, advise passengers to remove the oxygen mask "**ATTENTION THIS IS YOUR CAPTAIN. WE ARE NOW AT A SAFE ALTITUDE. REMOVE YOUR OXYGEN MASK**".

Note 2: At level off, oxygen masks may be removed if cabin altitude is at or below 13,000 ft. For the first 30 minutes between 10,000 ft. and 13,000 ft., however at or below 10,000 ft. remove oxygen mask for the remainder of the flight. This should be done one at a time. F/O removes oxygen mask first then after a while the captain does.

Note 3: After removing the oxygen mask close the left oxygen mask panel door and push the RESET/TEST switch to disable the mask microphone and enable the boom microphone. Set the MASK/BOOM selector to boom (as installed). Failure to accomplish this step may lead to loss of communication with ATC through the head set boom mic. If further use of the oxygen mask is required opening the left oxygen mask panel door will restore oxygen flow to the mask and enable the oxygen mask microphone.

Note 4: The captain contacts the Team Leader Cabin Services through the interphone or calls her/him to the flight deck to get a preliminary briefing on the situation in the cabin. For more information refer FOM 3.5(7.2).

Note 5: Advise passengers of the existing situation and next course of action.

3.7.2 ONLY ONE PILOT IN THE FLIGHT DECK

At first indication of a rapid loss of cabin pressure:

CAPTAIN/ FIRST OFFICER
Don oxygen masks and set regulators to 100%. (1)
Silence the warning horn.
Select Boom/Mask switch to mask (as installed).
Try to control the cabin pressure: Pressurization mode selector ..... MAN Outflow VALVE Switch ..... Hold in CLOSE until the outflow VALVE indication shows fully closed
⬇ If cabin pressure is uncontrollable, cabin altitude is 14000' or higher or Rapid decompression occurs (if differential pressure is zero in 10 seconds) do the remaining procedures: ⬇
PASS OXYGEN switch ..... ON and hold for one second
Set a lower altitude in the MCP altitude window. As required, remain on track or HDG SEL approximately 90° to the PF side If conditions permit. Consider high terrain, prohibited area, traffic, weather, escape route etc... (2) Select LVL CHG Select IAS in the IAS/MACH display by pushing the C/O switch. Confirm annunciations on the PFD. Close Thrust Levers. Extend speed brake smoothly.
Announce to the cabin "**ATTENTION, THIS IS YOUR CAPTAIN/PILOT. PULL DOWN AND USE YOUR OXYGEN MASK, PULL DOWN AND USE YOUR OXYGEN MASK**."
Declare "MAYDAY" three times and Advise ATC that you are making rapid descent and set transponder code to 7700.
Refine altitude window to MORA or 10,000 ft. whichever is higher. Target speed MMO/VMO. If structural integrity is in doubt, maintain existing speed.
Fasten Belts ..... ON Landing lights ..... ON Engine Start switches ..... CONT
When clear of the airway: Fly parallel or follow the established escape route or ATC instructions.
Advise ATC that you are making rapid descent to MORA or 10,000 ft. whichever is higher.

Set QNH as appropriate.
At 2,000 ft. to level off: Set turbulence penetration speed or LRC. If structural integrity is in doubt maintain existing speed.
At 1,000 ft. to level off: Speed Brake lever smoothly to down detent.
After level off oxygen mask should not be removed until the other pilot comes to the flight deck.
After level off and at or below 14,000 ft.: Advise passengers to remove the oxygen mask "**ATTENTION, THIS IS YOUR CAPTAIN/PILOT. WE ARE NOW AT A SAFE ALTITUDE. REMOVE YOUR OXYGEN MASK**.", Then announce "**CAPTAIN/FIRST OFFICER TO THE FLIGHT DECK**."
After the captain/first officer is back to the flight deck: BOTH PILOTS IN THE FLIGHT DECK procedures apply.
(1) When donning oxygen mask, put headset on the back of the neck for easy reach to put it back on over the air hose. Speakers may also be put on. *(2) Refer to the Jeppesen Airway Manual for contingency procedures in different airspaces.*

END OF CABIN DEPRESSURIZATION AND RAPID DESCENT – ONLY ONE PILOT IN THE FLIGHT DECK

3.8 REJECTED TAKEOFF

During takeoff, the crewmember recognizing the non-normal situation will immediately call it out as clearly as possible. e.g. Engine failure, Engine fire, Cargo fire etc.

The Captain has the sole responsibility for the decision to reject the takeoff. Make the decision in time to start the rejected takeoff maneuver by V1. Rejecting a takeoff near V1 has too often resulted in the airplane stopping beyond the end of the runway.

If the decision is to reject the takeoff, the Captain clearly announces "REJECT," immediately starts the rejected takeoff maneuver and assume control of the airplane. If the first officer is making the takeoff, the first officer maintains control of the airplane until the Captain makes a positive input on the controls.

At low speeds up to approximately 80kts, the airplane shall be stopped if an event occurs that would be considered undesirable for continuing the takeoff or flight.

Before 80 knots, reject the takeoff for any of the following:

- activation of the master caution system
- system failure(s)
- unusual noise or vibration
- tire failure
- abnormally slow acceleration
- takeoff configuration warning
- fire or fire warning
- engine failure
- predictive windshear warning
- if a side window opens
- if the airplane is unsafe or unable to fly.

Additionally for Cargo airplanes

- If CARGO DOOR warning Light(s) illuminates with or without takeoff configuration warning horn.

Above 80 knots and before V1, reject the takeoff for any of the following:

- Fire or fire warning
- Engine failure
- Predictive windshear warning

- if the airplane is unsafe or unable to fly.

Additionally for Cargo airplanes

- If CARGO DOOR warning light(s) illuminates with or without takeoff configuration warning horn.

After rejecting a takeoff and stopping the aircraft, accomplish memory items, if there is no fire or smoke and conditions permit clear the active runway and call for the appropriate checklist.

Keep in mind and advise the ground personnel not to approach the main gear wheels. Have chocks placed under the nose wheels as soon as possible. Unless required, release the parking brake.

Note: It takes up to 20 min for the energy to build to peak temperature. Check break cooling schedule if a subsequent takeoff is to be made soon.

CAPTAIN	FIRST OFFICER
Call "REJECT"	
Without delay: Simultaneously close the thrust levers, disengage the autothrottles and apply maximum manual wheel brakes or verify operation of RTO autobrake. If RTO autobrake is selected, monitor system performance and apply manual wheel brakes if the AUTO BRAKE DISARM light illuminates or deceleration is not adequate. Raise SPEED BRAKE lever. Apply reverse thrust up to the maximum amount consistent with conditions. Continue maximum braking until certain the airplane can stop on the runway.	Verify actions as follows: Thrust levers closed. Autothrottles disengaged. Monitor RTO autobrake operation or monitor if maximum brake is applied, If SPEED BRAKE lever UP and call "SPEEDBRAKE UP." If SPEED BRAKE lever not UP, call "SPEEDBRAKE NOT UP." Reverse thrust applied. When both REV indications are green, call "REVERSERS NORMAL." If there is no REV indication(s) or the indication(s) stays amber, call "NO REVERSER ENGINE NUMBER 1", or "NO REVERSER ENGINE NUMBER 2", or "NO REVERSERS". Call out any omitted items.
When stopping is assured:	Decelerating through 60 knots: Call "60 KNOTS"

Start movement of the reverse thrust levers to reach the reverse idle detent before taxi speed. After the engines are at reverse idle, move the reverse thrust levers to full down.	
After parking brake is set: Announce to cabin "**CABIN CREW AT YOUR STATIONS. THIS IS YOUR CAPTAIN. I HAVE CANCELED THE TAKEOFF, REMAIN SEATED, REMAIN SEATED**."	Advise ATC the situation after the aircraft has come to a complete stop. "**ET___ STOPPING ON THE RUNWAY. STANDBY"**
Assess the problem.	
Accomplish appropriate memory items or NNC as required.	
Announce to cabin the reason for RTO and next course of action. If no action is required by the cabin crew advise "**RESUME NORMAL OPERATION."**	

END OF REJECTED TAKEOFF

Note 1: After a rejected takeoff the aircraft shall remain on the centerline.

Note 2: Flight crew shall request tower observations for possible fire or smoke around the wheel well area prior to vacating the runway.

Note 3: After a rejected takeoff comply with brake cooling requirements before attempting a subsequent takeoff.

Note 4: Verify brake temperature and tire pressure from systems page (as installed)

Consider the following:

- Consider securing the engine before vacating the RWY
- The possibility of wheel fuse plugs melting
- The need to clear the runway
- The requirement for remote parking
- Tower observation for possible fire or smoke around the wheel well area
- Wind direction in case of fire
- Alerting fire equipment
- Advising the ground crew of the hot brake hazard
- Advising passengers of the need to remain seated or evacuate
- Completion of NNC (if appropriate) for the condition which caused the RTO

3.9 EMERGENCY EVACUATION

The Captain initiates evacuation or accomplish expeditious normal deplaning by assessing the situation. Complete the appropriate memory items and verify the steps using QRH before executing emergency evacuation.

CAPTAIN	FIRST OFFICER
Stop the aircraft. Set parking brake.	
Announce "**CABIN CREW AT YOUR STATIONS**."	
Call "EVACUATION CHECKLIST."	Do the EVACUATION checklist.
Announce passenger evacuation through PA, "**EVACUATE, EVACUATE, EVACUATE. LEAVE THE AIRCRAFT IMMEDIATELY**." Notify available exits.	Notify ATC nature of emergency, location of the aircraft and request assistance.
Verify evacuation progress by checking door lights.	

END OF EMERGENCY EVACUATION

Note 1: Duty positions during evacuation:

- Captain Forward Cabin
- First Officer outside of the aircraft.
- DH or Augmented crew could assist passengers outside the aircraft

Note 2: The Captain initiates evacuation or Accomplish expeditious normal deplaning by assessing the situation.

Captain

- Assess exit and evacuation conditions
- Assist passengers in exiting the aircraft.
- Exit only after ensuring everyone has exited, or own life is in danger.

First officer

- Open exits if still they are closed.
- Help direct passengers to an assembly point

3.10 ON GROUND EMERGENCY

- During on Ground Emergency, stop the Aircraft immediately, notify ATC & Cabin Crew then perform the required NON NORMAL CHECKLIST. If evacuation is required, follow evacuation procedure.
- For engine fire use opposite side doors for evacuation.
- For APU fire use all doors for evacuation
- For Cargo Fire consider remote parking stand and deplane the passengers as quickly as possible using all available stairs.
- For engine fire on the ground perform Engine Fire on the Ground NON NORMAL CHECKLIST

3.11 VOLCANIC ASH ENCOUNTER

Volcanic ash eruption covers a wide area due to wind drift and can ascent to higher altitudes.

Volcanic ash report is disseminated through Sig wx charts, VAAC or PIREPs to pilots and flight dispatchers plan the routing to avoid Volcanic Ash. Avoidance is the best option and it is vital to know the symptoms of volcanic ash while flying through unpredictable Volcanic Ash suspected areas. Volcanic ash primarily affects the engine and the control surfaces might be corroded with acids formed from the gases erupted in the atmosphere.

Indication of volcanic ash :

- St. ELMO's Fire is seen on the windshield
- Engine parameter fluctuations (Stalls, EGT rise, Rotor Speed or flameout)
- Unreliable airspeed
- Acrylic odor (sulfur or electrical smoke) through the air conditioning system
- Red glow in the Engine nacelle due to static discharge.
- Areas covered by Volcanic Ash is associated with lighting and thunder due to the silicate electrical discharge
- Poor radio communication / reception
- Haze or other colors is seen visually
- Poor visibility through wind shields
- Cargo fire due to the smoke.

when pilots suspect Volcanic Ash encounter:

- Aviate, Navigate and communicate
- Perform the VOLCANIC ASH checklist.
- Refer to FOM 2.19 (7.13)

3.12 ALTERNATE FLAP OPERATION

When extending the flaps using the alternate system, the recommended method for setting command speed differs from method used during normal flap extension. Since the flaps extend more slowly when using the alternate system, it is recommended that the crew delay setting the new command speed until the flaps reach the selected position. This method may prevent the crew from inadvertently getting into a low airspeed condition if attention to airspeed is diverted while accomplishing other duties.

PF	PM
Call for flap 10 on the initial alternate flap extension. (To prevent the gear unsafe warning).	Select the flap lever to 10 and extend flaps to 10 using the alternate flaps position switch.
Instruct the PM to monitor flaps position 1, 5 and 10	Call as flaps move through 1, 5 and to 10.
After landing gear is extended: Call for flap 15.	Set the flap lever to 15 and extend flaps to 15 using the alternate flaps position switch.

CHAPTER 4 NON-NORMAL MANEUVERS

4.1 GROUND PROXIMITY WARNING (TERRAIN AVOIDANCE)

Refer maneuver section of QRH

4.2 UPSET RECOVERY

Refer maneuver section of QRH

4.3 APPROACH TO STALL OR STALL RECOVERY

Reduce the angle of attack until buffet or stick shaker stops then **advance the thrust levers** as needed (**promptly** & **smoothly**) to accelerate and promptly *return* the airplane to the *desired energy state*.

Refer maneuver section of QRH & FCTM

4.4 WINDSHEAR

Refer maneuver section of QRH

4.5 TRAFFIC ALERT AND COLLISION AVOIDANCE SYSTEM (TCAS)

TCAS is designed to enhance crew awareness of nearby traffic and issue advisories for timely visual acquisition or appropriate vertical flight path maneuvers to avoid potential collisions.

RA maneuver is mild and does not require large or abrupt control movements. Passengers and Flight Attendants may not all be seated during this maneuver. Smoothly and expeditiously return to appropriate altitudes and speeds when clear of conflict by following the FD command.

TRAFFIC AVOIDANCE

Immediately accomplish the following by memory whenever a TCAS traffic advisory (TA) or resolution advisory (RA) occurs.

For TA:

PF	PM
	All exterior lights ON and Fasten belt sign ON.
Look out for traffic using traffic display as a guide. Call out any conflicting traffic.	
If traffic is sighted, maneuver if needed	

Note: Maneuvers based solely on a TA can result in reduced separation and are not recommended

For RA, Refer maneuvers Traffic Avoidance section of QRH

During RA notify ATC as follows **"ETH____TCAS RA"**

4.6 PILOT INDUCED OSCILLATION

Pilot induced Oscillations (PIO) are inadvertent, sustained oscillations of the airplane resulting from interactions between the aircraft and control inputs by the pilot. They are often associated with tasks where the pilot is attempting to precisely and quickly accomplish a flight maneuver (such as the final phase of landing). In a fully developed lateral PIO, pilot control wheel inputs will be out of phase with the airplane roll response.

Flight crew should be aware of the potential for pilot induced roll oscillation when using high rate, high magnitude and rapidly reversed control wheel inputs. This potential is increased when in landing configuration with gusty wind conditions. Pilot techniques that utilize abrupt and pulsing control inputs may also contribute to these events.

The following action should be accomplished immediately when either pilot recognizes that PIO exists:

PF	PM
Announce the situation	
Immediately stop lateral control wheel inputs until the airplane stabilizes	Verify appropriate response
Initiate go-around if oscillations do not diminish or if the aircraft is not in a position from which a safe landing can be made	Recommend go-around if airplane is not stabilized for landing

CHAPTER 5 ETOPS/EDTO

REFER FOM 2.16 ETOPS POLICY and PROCEDURES

MAX DIVERSION DISTANCE TABLE

AIRCRAFT TYPE	MAX DIVERSION DISTANCE (NM)			REFERENCE WT. (TONES)	COMPANY SPEED
	60 MINS	120 MINS	180 MINS		
B737	413	806	N/A	65	.79/290
B738 & B73F	408	800	N/A	73	.79/290
B38M	415	814	N/A	77	.79/310

CHAPTER 6 LOW VISIBILITY OPERATION

6.1 INTRODUCTION

The B737 has been designed, equipped and certified for low visibility operations (CAT II/IIIA/IIIB). Each aircraft has been placarded as applicable.

All runways approved for All Weather Operations (low visibility operations) will have a Low Visibility Approach Chart in the Jeppesen Airways Manual.

6.2 LOW VISIBILITY TAXI PROCEDURE

The taxiway centerline lighting with its reduced spacing (generally 30m, but reduced to 15m to 7m on bends) facilitates taxiing in low visibility. Do not exceed 10kts during taxiing and the First Officer should make full use of taxi charts and call out ground speeds when indicating more than 10Kts to avoid the risk of possible incursion. Remember that in poor visibility ground equipment, aircraft wingtips and tails may not be as readily seen as the taxiway lighting.

a) Be fully aware of other aircraft taxiing in your vicinity.

b) Observe CAT II/III holding point markings/lights.

c) If at any time you are unsure of your ground position or miss a turn, advise ATC immediately.

Minimum RVR for taxi with visual guidance only is 75 meters.

6.3 LOW VISIBILITY TAKEOFF

The captain shall carryout all low visibility takeoffs

1. Confirm that the aircraft is lined up on the runway centerline lights and not the edge lights.

2. Tune the localizer frequency of the departure runway to crosscheck correct line up on the runway.

3. Check that the number of centerline lights is consistent with the reported RVR.

3. Use centerline lights or markings for directional control during the takeoff roll and stopping after a rejected takeoff. As speed increases, the streamlining effect of these improves, and directional control becomes easier.

4. Takeoff shall be rejected if visual reference is lost below 80kts.

5. Use of assumed temperature or fixed derate is authorized when low visibility procedures are in effect.

6. Takeoff minima stated in Jeppesen chart (10-9x or 10-20) or AOC Visibility minima whichever is higher is applied for all low visibility takeoffs FOM 2.2(7.5.4).

7. When weather condition is below RVR 500m (1600ft) and where only two RVR sensors are available, both are required and controlling. Where three RVR sensors are installed on the takeoff runway, the TDZ, MIDPOINT & ROLLOUT RVR values are controlling when operational. However, the failure of any one RVR will not affect operations, provided the remaining two RVR sensors are reporting values at or above the appropriate minima.

8. Furthermore, the pilot assessment can replace RVR/Visibility value for the first part of the takeoff roll which is the TDZ RVR, provided that the TDZ is inoperative and the rest of RVR reports meet the minimum required 150m RVR value, or the TDZ RVR report can be a minimum of 125m RVR and lights spacing of centerline of the runway shall be 15m or less and HIRL shall be 60M or less.

9. Low visibility must be enforced when visibility drops below RVR 400m.

6.4 LOW VISIBILITY APPROACH

Low visibility approaches may only be carried out when both the captain and the first officer are qualified and current, aircraft equipment is capable of CAT II or III operation and the intended runway of landing is certified for this operation with the associated chart.

Low visibility needs to be enforced at the aerodrome before commencing the approach.

Recommended operational procedures and guidelines are given in FOM2.120, FCTM, FCOM and.

B737 Alert Height (AH) is 200ft and a Decision Height (DH) of 20ft shall be used whenever DH is required for CAT IIIB approaches.

Alert height is called from memory and shall not be set.

6.5 CAT II/III APPROACH BRIEFING CHECKLIST

Before commencing a CAT II/III approach the crew will check and brief the following.

1. Both pilots are CAT II/III qualified and current
2. PIC shall be the PF
3. Aircraft maintenance status (Approach Category Status Placard)
4. Landing weight must be at or below maximum certified landing weight
5. **[B38M/ B738]** Check recall & verify that the NO AUTOLAND status advisory message is not shown on EICAS
6. Destination and alternate weather
7. Use of CAT II/III approach chart.
8. Runway status, CAT II / III, surface condition, use of auto brake, taxi chart and exits
9. Fuel status and action after go around

10. Alternate airport on ROUTE 2
11. Radio altimeter – CAT II/III minima (For CATIIIB with DH set 20'RA)
12. Baro minimums – CAT I minima
13. Use of anti-ice and exterior lights
14. make sure low visibility procedure is in progress*
15. Start APU at 10,000ft AGL (FL100)
16. Flaps 40 landing is recommended, at all approaches set Vref + 5Kts only
17. Adjust seats

Note: B737 is authorized for auto land only with flaps 30 or 40. Flaps 40 is the recommended flap setting for a better flight deck view during low visibility approaches.

* Not required for practice approach

B737-700 Disengage the autopilot after touchdown. Control the airplane manually.

B737-800 and B737-8 Disengage the autopilot before turning off the runway.

AFDS roll out mode performance cannot be assured when used on contaminated runways. If an autoland is accomplished on contaminated runway, the pilot must be prepared to disengage the autopilot and take over manually should roll out directional control become inadequate.

6.5.1 DH AND BARO SETTINGS

In order to facilitate quick reversion to a different minima while on approach, the minimums selector may be set in accordance with the following guideline.

- **CAT II** (RVR 300 > 550 ceiling 100 ft. < 200 ft.)

  Set BARO minimums to landing runway CAT I minima and set RADIO minimums to CAT II minima as per approach chart and leave minimums selector set on radio.

- **CAT IIIA** (RVR 200 >300 ceiling 50 ft. < 100 ft.)

  Set BARO minimums to landing runway CAT I minima and set RADIO minimums to 50 ft. or approved CAT IIIA minima and leave minimums selector set on radio.

- **CAT IIIB** (RVR 75 > 200 ceiling NO DH or DH < 50 ft.)

  Set BARO minimums to landing runway CAT I minima and Set RADIO minimums to 50FT or approved CAT IIIA minima.

  Leave the Minimum selector switch to RADIO and reset/blank the display.

6.5.2 RVR REQUIERMENTS

CATEGORY II RVR Requirements

RVR 500m (1600ft) or less – Touchdown required and controlling. Midpoint or rollout is also required and advisory. If MID point is reported, it shall not be less than 125m (400ft). If ROLLOUT is reported, it shall not be less than 75m (300ft).

CATEGORY III RVR Requirements;

a) For CAT IIIA approach the minimum required RVR;

TDZ: 200m (700FT), MID: 125m (400ft), ROLLOUT: 75m (300ft)

b) For CAT. IIIB approach the minimum required RVR;

TDZ, MID, & ROLLOUT 75m (300ft)

c) For all CAT III operations the following shall be considered:

- All available RVR reports are required and controlling, where four RVR reporting systems are installed (i.e. TDZ, MID, ROLLOUT and FAR END SENSORS) the FAR END RVR report is advisory information to the pilots or may be substituted for the ROLLOUT RVR report if that is not available.
- During fail operational landing system with fail passive or fail operational rollout system, any one of RVR reporting system may be temporarily inoperative. The other two are required and controlling.
- During fail passive landing system with fail passive or fail operational rollout system, either MID or ROLLOUT reporting system may be temporarily inoperative. The minimum TDZ RVR required will be 200m (700ft), MID RVR report, if available, shall not be lower than 125m (400ft) and if ROLLOUT RVR reported it shall not be lower than 75m (300ft)
- During fail operational or fail passive landing system, if rollout system fails, TDZ and MID RVR reports shall not be lower than 200m (700ft).
- During fail operational landing system with fail passive rollout system, TDZ and MID RVR reports if available shall not be lower than 125m (400ft).

6.6 EQUIPMENT REQUIREMENT

REQUIERED EQUIOMENT	CAT II	CAT IIIA	CAT IIIB
Autopilot	2	2	2
ILS reciever	2	2	2
Radio Altimeter	2	2	2
Decision height indicator	2	2	2
IRS	2	2	2
Authothrottle	1	1	1
Independent source of electrical power	2	2	2
Hydraulic system	2	2	2
Windshield wipers	With	With	with
Control column stb trim switches	With	With	with
Display source	2	2	2
TR unit	2	2	2
Bus tie braker	Closed	Closed	Closed
ISFD	Not required	Not required	Required
Engines	Both required	Both required	Autoland (performance permitting) authorized for flaps 30 only.
Autoland status annunciation	2 (LAND2 or LAND3)	2 (LAND2 or LAND3)	2 (LAND3 only)
Approach minima display	Required for each	Required for each	Required for each

Note 1: Autoland can be completed with ab autothrottle failure for a CAT II and CAT IIIA failures.

Note 2: If autopilot disconnects during a CAT II approach and there is sufficient visual conditions the landing can be completed manually.

6.7 AUTOLAND LIMITATION

Maximum allowable wind speeds when landing weather minima are predicated on autoland operations:

Headwind 25 knots

Crosswind 20 knots*

Tailwind 10 knots

- **Note for all CATIII operations the cross wind limit is 15 knots**

Whenever operating on contaminated runway, the crosswind and tail wind limitation should be the maximum of the autoland limitation or the crosswind guideline for contaminated runway on FOM 2.17 (7.15)

The maximum and minimum glideslope angles are 3.25 degrees and 2.5 degrees respectively.

The autoland capability may only be used with flaps 30 and 40 with both engines operative.

For airplanes equipped with a fail operational autopilot using either LAND 3 or LAND 2, Engine inoperative autoland may be used with flaps 30 if airplane performance permits.

There should be no reported windshear condition.

6.8 APPROACH AND GA ALERTS

Faults can occur at any point during an AFDS approach. Many non-normal situations or scenarios are possible. The flight deck is designed so that a quick analysis can be made for virtually all non-normal or fault situations by using the Autopilot/Autothrottle indicators, flight mode annunciations, master cautions system and for fail operational airplanes, autoland status annunciations.

For fail operational airplanes, faults leading to non-normal operations can be divided into two categories:

- Those occurring above AH
- Those occurring at or below AH.

The flight crew should be aware of the airplane equipment requirement for the approach. The following can be used for any AFDS fault indication:

Above alert height

Immediately after recognizing the fault from the master caution system, instrument flags or engine indications, check autoland status annunciation.

- If the autoland status annunciation has not changed, and the equipment is not required for the approach, continue the approach.

- If the autoland status annunciation has changed or the equipment is required for the approach, consider adjusting to the appropriate higher minimums (if the weather permits) or go-around. However, if suitable visual reference is established, consider landing.

At or below alert height

For fail operational airplanes: Below 200 feet AGL a safe landing and rollout can be made with any probable internal failure conditions. Flight crew alerts (lights or aural) may occur at any time during the approach. If a master caution or aural occurs below alert height, do not disengage the autopilot unless the autopilot system is not controlling the airplane adequately. Below alert height, the AFDS Fail operational refers to an AFDS capable of completing an ILS approach, autoland, and rollout following the failure of any single system component after passing alert height. The pilot should not intervene below AH unless it is evident that pilot action is required. If fault affects the autothrottle or autobrakes, assume manual control of thrust and braking, accomplish related procedures for system faults after rollout is complete and manual control of airplane is resumed.

The following table serves as a guideline for autoland, autopilot and autothrottle alerts.

ALERT	CAT II		CAT IIIA		CAT IIIB	
	Above 200	Below 200	Above 200	Below 200	Above 200	Below 200
LAND 3	LAND	LAND	LAND	LAND	LAND	LAND
LAND 2	LAND	Inhibited	LAND	Inhibited	Revert to CAT IIIA/II or GA	Inhibited
NO AUTOLAND	GA	GA	GA	GA	GA	GA
A/P Disconnect	GA *	GA *	GA	GA	GA	GA
A/T disconnect	LAND (Vref+ additive)	LAND	LAND (Vref+ additive)	LAND	Revert to CAT IIIA/II or GA	GA
ILS deviation alert	GA	GA	GA	GA	GA	GA

Note 1: For all go around (GA) conditions consider continuing approach or landing if suitable visual reference can be obtained and maintained.

Note 2: If the decision is to revert to a higher minima, the reported RVR and ceiling should be enough for the type of approach reverted to.

* Manual CAT II operations are authorized according to the following table adopted from ICAO Doc 9365 AN/910

CAT II operations minima coupled to below DH[1]		
Decision height	RVR/aeroplane Category A, B and C	RVR/aeroplane Category D
100 ft–120 ft	300 m	300 m[2]/350 m
121 ft–140 ft	400 m	400 m
141 ft–199 ft	450 m	450 m

1- The reference to "Coupled to below DH" in this table means continued use of the automatic flight control system down to a height which is not greater than 80 per cent of the applicable DH. Thus airworthiness requirements may, through minimum engagement height for the automatic flight control system, affect the DH to be applied.

2- For a CAT D aeroplane conducting an autoland, 300 m may be used.

Note 1: If A/P disconnects below 400ft RA after stabilizer trim bias, forward control column force may be required to hold the desired pitch attitude to maintain the glide path to landing.

6.9 AUTOLAND PROCEDURE

The following calls and actions will be carried out in addition to normal procedures for a coupled approach:

FAIL PASSIVE

CAPTAIN	FIRST OFFICER
Prior to Glideslope Intercept Arm VOR/LOC	CALL: "VOR/LOC ARMED" and after localizer intercept "VOR/LOC" CAPTURED.
On the localizer, cleared for the approach, arm the MCP APP Mode. Then, engage second autopilot in CMD >800 ft. RA.	CALL "G/S CAPTURED SINGLE CHANNEL AND/OR CMD OR LAND 3 OR LAND 2" as applicable.
After LOC and G/S Capture	
Confirm FLARE is armed and reply "Check."	Below 1,500 ft. RA, FLARE is annunciated, call "FLARE ARMED."
At 500 ft. RA	
	Call "500ft and A/P STATUS ANUNCIATION."
Call radio altitude, verify and call "FLARE ARMED."	

At 400 ft. RA Confirm the stabilizer is automatically trimmed an additional amount nose up.	
At FLARE	
	Once FLARE transitions green, call "FLARE GREEN." If not, call "NO FLARE."
If unable to use Autoland at any time during the approach but no later than 100 ft. above the touchdown zone and the required visual ques are met. Call "Manual landing" and disengage the Autopilot and the Autothrottle. Land the aircraft normally.	Respond "Check."
The autothrottle will begin retarding thrust at approximately 27ft. RA so as to reach idle at touchdown and RETARD annunciates in the FMA. Approximately 2 seconds after touchdown, the autothrottle will automatically disengage.	
At Touch Down	
The autopilot must be disengaged after touchdown and complete normal roll out procedures.	Confirm autopilot has disengaged and complete normal roll-out procedures.

FAIL OPERATIONAL

CAPTAIN	FIRST OFFICER
Prior to Glideslope Intercept Arm VOR/LOC	CALL: "VOR/LOC ARMED" and after localizer intercept "VOR/LOC CAPTURED."
On the localizer, cleared for the approach, arm the MCP APP Mode. Then, engage second autopilot in CMD >800 ft. RA.	CALL "G/S CAPTURED SINGLE CHANNEL AND/OR CMD OR LAND 3 OR LAND 2" as applicable.
After LOC and G/S Capture	
Confirm FLARE and ROLL OUT are armed and reply "CHECK."	Below 1,500 ft. RA, FLARE and ROLL OUT are annunciated, call "FLARE AND ROLL OUT ARMED."
	At 200ft RA for ILS CAT III approach Call "ALERT HIGHT"
At 500 ft. RA	
Call radio altitude, verify and call "FLARE AND ROLLOUT ARMED."	Call "500 ft and A/P STATUS ANUNCIATION."

At 400 ft. RA Confirm the stabilizer is automatically trimmed an additional amount nose up.	
At FLARE	
	Once FLARE transitions green, call "FLARE GREEN." If not, call "NO FLARE."
If unable to use Autoland at any time during the approach but no later than 100 ft. above the touchdown zone and the required visual ques are met. Call "Manual landing" and disengage the Autopilot and the Autothrottle. Land the aircraft normally.	Respond "CHECK."
The autothrottle will begin retarding thrust at approximately 27ft. RA so as to reach idle at touchdown and RETARD annunciates in the FMA. Approximately 2 seconds after touchdown, the autothrottle will automatically disengage.	
At Touch Down	
The autopilot must be disengaged if rollout guidance is not satisfactory after touchdown and complete normal roll out procedures, otherwise disengage autopilot at taxi speed.	

ATTACHMENT A - CONTACTS

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Capt. Yoftahe Adamu	Chief Pilot B777/787	Tel +251 911 22 27 77
Capt. Behirudin Abdo	Mgr. Boeing Fleet Training	Tel +251 911 21 55 31
Capt. Aklilu Yitbarek	Mgr. Airbus & Q400 Training	Tel +251 911 42 23 80
Capt. Theodros Asfaw	Chief Pilot A350	Tel +251 911 12 16 22
Capt. Mulualem Habte	Chief Pilot Q400	Tel +251 911 79 30 22
Capt. Hassen Gessesse	Chief Pilot B737	Tel +251 911 78 63 58
Ato Fasil Admasu	Dir. IOCC MOCC	Tel +251 966 08 88 25 Tel +251 11517 82 95
Ato Yossef Wabbe	Mgr. Dispatch and Navigation	Tel +251 11 517 86 53
Duty Manager		Tel +251 11 166 1212 / +251 11 517 8107
Crew Scheduling		Tel +251 11 663 1211 / +251 11 517 8232
Motor pool		Tel +251 11 663 1234 / +251 11 517 8419
Holloway Radio		Tel +251 11 517 8223
Dispatchers		Tel +251 11 517 8205 / +251 11 517 8007
ADDIS (HAAA) ACC/FIC		Tel +251 11 665 05 17/ +251 11 665 05 19
Bole TWR		Tel +251 11 665 05 16
MET Office		Tel +251 11 875 6389

ATTACHMENT B – TAKEOFF AND APPROACH BRIEFING FLOW

DEPARTURE BRIEFING

Threat (PF, PM)

Plan:

- Crew Experience and seat share policy
- Taxi, Departure Runway, SID
- Single Engine departure procedure
- T/Off alternate consideration
- Depressurization escape route before TOC
- Supplemental procedure
- RTO and Engine failure after V_1 procedure (on first flight for the paired crew)

Considerations:

- Any specific PM duties
- Other considerations (Bulletins, Notices, Special loads, gear pins, etc.)

APPROACH BRIEFING

Threat (PF, PM)

Plan:

- Route (STAR, Approach, Approach Mode, M/A procedure, Alternate Fuel and route)
- FMC programming
- Landing R/W Assessment, Exit, Taxi to gate
- Auto brakes
- Use of Reverse Thrust

Considerations:

- Any specific PM duties
- Other considerations (actions after Go-around, visual approach, all non-precision app MDA +50', etc.)

WHAT ARE OUR THREATS?

AIRPORT/RUNWAY/ATC

Bird activity
Clearance/ Reroute
R/W Changes
Hotspots
Contamination
Construction
ARR/DEP changes

ADVERSE WX

Visibility
Deicing
Winds
Precipitation

AIRLINE/Ops/Dispatch

Schedule Pressure
Delays
Paperwork

GROUND/RAMP/MX

Handling
Congestion
Logbook

ENVIRONMENT/PHYSIOLOGY

Terrain
Night
Traffic

AIRCRAFT

Systems
MELs
Automation
Performance

Fatigue Onset
Stress
Hydration
Nutrition

CABIN

Passengers
Interruptions

DEBRIRFING

To improve performance:

1. ***How do you think the flight went?**
2. **What would we do differently if we could do it again?**
3. **Are there any reports to complete/submit?**

** Debrief both excellent performance and areas to improve*

ATTACHMENT C – PASSENGER ADDRESS

GENERAL

Communication between flight crew and passengers is of a paramount importance in the safe, efficent and courteous conduct of Airline Transport Operation. The protocol, content and phase of this requirement is incorporated in various chapters of the FOM and SOP manuals as well as the service recovery section of the Ground Handling Manual. In line with keeping passengers well informed of routine and non-regular progress of the flight it is intended to prevent the occurrence of anxiety, worry, doubts and negative speculation.

Needless to say, all communication should be loud and clear, concise and to the point. Special care should be taken in choosing words and phrases to avoid confusion, alarm and even panic. Tonality of voice should be calming, assertive or even entertaining as appropriate.

For the sake of clarity and system efficiency, it is recommended to use the PA telephonic set, hand microphone and the boom microphone as a last resort. Speaking directly into a microphone mixes breathing sound with the voice reducing readability over the cabin speakers and so holding it slightly on the side of the mouth and or leaving a small gap improves the sound quality.

All announcements should first be made in Amharic and then in English. Though all essential announcements should be made by the captain, non-Amharic speakers may delegate this assignment to the F/O.

All initial announcements should be preceded by the name and rank of the announcer and thereafter by rank only.

PASSENGER ADDRESS

The passenger address (PA) system is one of our most effective service tools. It enables us to provide passengers a more informative journey. Proper use of the PA system enhances Ethiopian's professional image and promotes confidence in our service.

The following general rules should be followed when using the PA system.

- Think before you speak (If need be jot down important points) Tell the truth
- Don't make promises you cannot keep
- Avoid technical terms and aviation jargon (IFR, ATC). Use layman terms.
- Avoid words such as turbulence, thunderstorm, or foggy instead use bumpy or cloudy. Be careful with humor.
- Emphasize safety. We want our passengers to know that their well-being is important to us.
- Be friendly. Think of passengers as guests and treat them accordingly.

A review of our customer comment cards and letters tells us that our passengers want to hear the following from our flight crew:

Major points of interest pointed out along the route including large cities and geographical highlights.

During mechanical delays or irregular operations passengers want an explanation of the situation and how it is going to be handled. They also want continual updates on the situation.

Our passengers want to be able to hear PA announcements clearly. Please ask cabin crew if flight deck PA announcements are loud and clear during in flight conditions. Write-up PA system that require maintenance action.

After all passengers have boarded on the first leg of each flight, make a short welcome address appropriately timed before taxi. Conditions and time permitting this may include expected traffic condition, takeoff run-way and direction of departure and expected scenery during the maneuver along with the expected weather condition of the climb phase. Subsequent to this initial announcement any significant deviation or delay affecting departure and arrival times should without fail be communicated.

Do's and Don'ts

The essence of such announcements is to deliver the required information concisely, clearly and unambiguously. In so doing the tone of the announcer (with the captain busy with priority items it could be the F/O) should be:-

- calm and relaxed
- reassuring
- assertive (*it may be preferred for the F/O to use the term "this is your pilot speaking" to alleviate unfounded fears as to the wellness of the captain)

The choice of words should be in line with not raising undue alarm, create confusion and or panic. Examples:

- Engine malfunction instead of engine failure
- System inconsistence instead of system failure
- Rain shower instead of thunderstorm
- Bumpy air instead of turbulence
- Restricted visibility instead of fog
- etc….

Information on return to the ramp, the departure airport and diversion to a suitable airport should be presented as a necessity in line with safety and convenience and not out of lack of choice and or incapability.

The following templates are for reference and guidance:

PARKING POSITION ANNOUNCEMENT / የቅድመ በረራ መረጃ

የተከበራችሁ መንገደኞቻችን እንደምን _____ከበረራው ክፍል አብራሪያችሁ _____ ነኝ። ከሥራ ባልደረቦቼ "ከዋና / ከምክትል" አብራሪው ___ እና የዚህ በረራ የመስተንግዶ ሃላፊ _____ ጋር በመሆን የስታር አላያንስ አገልግሎት አባል በሆነው የኢትዮጵያ አየር መንገድ የበረራ ቁጥር _____ ለመብረር እንኳን በደህና መጣችሁ። አሁን ለበረራ የምናደርገውን ዝግጅት በማጠናቀቅ ላይ ነን ዝግጅታችንን እንደጨረስን የአውሮፕላኑን በሮች ዘግተን ጉዞአችንን እንጀምራለን።

በአሁኑ ሰዓት በዚህ ጉዞ ላይ ("ምንም መዘግየት አንጠብቅም" ወይም "በ_____ምክንያት ለ_____ ደቂቃ/ሰዓት ዘግይተን ጉዞአችንን እንጀምራለን"። የበረራ ጉዞ ከጀመርን በኋላ ስለበረራው ተጨማሪ መረጃ እንነግራችኋለን። አሁን ለመነሳት በምናደርገው ዝግጅት ስለበረራው ደህንነት የሚሰጠውን መመሪያ በጥሞና እንድትከታተሉ በትህትና አሳስባለሁ።

መልካም በረራ።

Good_____ ladies and gentlemen. This is your (Captain/First Officer) speaking.

My name is _____. Your (Captain/First Officer) is _____ and _____ is the Team Leader Cabin Services. Welcome onboard Ethiopian Airlines flight to _____ with star alliance service.

Currently, we are in the process of completing departure preparation. As soon as it is completed and doors are closed we will be requesting clearance. We are at the moment expecting (no delay/slight delay due to _____)

I will give you more information about our flight during cruise. In the mean time please comply with cabin safety instruction in order to facilitate timely departure.

Have a nice flight.

DELAY ANNOUNCEMENT / ስለበረራ መዘግየት

የተከበራችሁ መንገደኞቻችን ከበረራው ክፍል አብራሪያችሁ ነኝ። በግምት ለ _____ ደቂቃ/ሰዓት ከመነሻ ጊዘአችን ዘግይተን ጉዞአችንን እንጀምራለን ወይም ላልተወሰነ ጊዜ እንደምንዘገይ ተነግሮናል።

የምንዘገይበት ምክንያት ——————————————ሲሆን፣

- በአጋጠመን የአውሮፕላን ብልሽት
- በአውሮፕላን ማረፊያው አካባቢ በተፈጠረ የትራፊክ መጨናነቅ
- ለበረራ የማያሙች የአየር ሁነታ
- ከሌላ መዳረሻዎች ዘግይተው በሚመጡ መንገደኞች
- የመንገደኞች ሻንጣዎችን ለመለየት/ለመጫን/ለማውረድ
- በሰነድ ማረጋገጫ/በመንገደኛ መቀበያ ቦታ በተፈጠረ መጨናነቅ
- በደህንነት ቢሮ ሰራተኞች መመሪያ/ጥያቄ
- አውሮፕላኑ ከበረዶ መጽዳት ስላለበት
- የህክምና እርዳታ የሚያስፈልገው መንገደኛ በመኖሩ

ሁኔታውን ተከታትለን የሚገኘውን ውጤት አሳውቃችኋለሁ። በቅድሚያ ሁኔታውን ተገንዝባችሁ ለምታሳዩን ትዕግስት ላመሰግናችሁ እወዳለሁ።

Ladies and gentlemen, this is your captain speaking from the flight deck, we have (an undetermined delay/a departure delay of approximately _____ (minutes or hour(s))),

Due to:

- A technical problem
- Heavy traffic in the airspace / at the airport
- Bad weather conditions at _____
- The late arrival from the previous flight
- Baggage loading, baggage identification
- Congestion at passport control check-in counter.
- Last minute handling of luggage
- Airport security procedure
- Aircraft de-icing procedure
- Medical assistance required for (one/few) of our passengers

I will keep you updated on any progress as soon as we have new information. Thank you for your understanding and patience.

RETURNING TO PARKING POSITION ANNOUNCEMENT / ወደ ማቆሚያ ስለመመለስ

የተከበራችሁ መንገደኞቻችን ከበረራው ክፍል አብራሪያችሁ ነኝ።

በአጋጠመን _____ ምክንያት ወደ ማቆሚያ ቦታ ለመመለስ ተገደናል፣

- በአጋጠመን የአውሮፕላን ብልሽት
- በደህንነት ቢሮ ሰራተኞች መመሪያ/ጥያቄ
- የህክምና እርዳታ የሚያስፈልገው መንገደኛ በመኖሩ
- በአየር መንገዱ ኃላፊዎች መመሪያ

ሁኔታውን ተከታትለን የሚገኘውን ውጤት አሳውቃችኋለሁ።በቅድሚያ ይህንኑ ተገንዝባችሁ ለምታሳዩን ትዕግስት ላመሰግናችሁ እወዳለሁ።

This is your Captain speaking from the flight deck.

Due to:

- A technical problem
- Passenger needing medical assistance
- Company operational reasons, security reasons

We will return to our parking position. I will share with you any further information as soon as we receive it.

Thank you for your understanding and patience.

ABORTED TAKEOFF ANNOUNCEMENT / ከመነሳት መስተጓገል

የተከበራችሁ መንገደኞቻችን ከበረራው ክፍል አብራሪያችሁ ነኝ። በአጋጠመን የአውሮፕላን ብልሽት ምክንያት ለመነሳት ባለመቻላችን ይህንን በረራ ለማቋረጥ ተገደናል፡ሆኖም አስፈላጊው የጥገና ሥራ እንደተጠናቀቀ ጉዞአችንን እንደምንጀምር ተስፋ እናደርጋለን። በቅድሚያ ሁኔታውን ተገንዝባችሁ ለምታሳዩን ትዕግስት ላመሰግናችሁ እወዳለሁ።

This is your Captain speaking from the flight deck.

We had to cancel our takeoff due to a technical problem.

We hope that we will be able to continue our flight.

We thank you for your understanding and patience.

DE-ICING ANNOUNCEMENT

የተከበራችሁ መንገደኞቻችን ከበረራው ክፍል አብራሪያችሁ ነኝ: በአሁኑ ሰዓት የአውሮፕላኑን የውጭ አካል ከበረዶ ማጽዳት አስፈላጊ በመሆኑ ይህን ስናደርግ በአውሮፕላኑ ዉስጥ የሚፈጠረው ያልተለመደ ጠረን የታወቀና ምንም አይነት ጉዳት የሌለው እና ጽዳቱ ከተጠናቀቀ ጥቂት ደቂቃዎች በኋላ የሚጠፋ መሆኑን እገልጻለሁ፡ የሚወስድብንም ጊዜ በግምት 10 ደቂቃዎች ያህል ይሆናል አመሰግናለሁ።

This is your Captain speaking from the flight deck.

We will accomplish mandatory de-icing and anti-icing on the aircraft external structure.

It will take approximately 10 minutes. You may experience a different odor which is normal for this procedure and is harmless. It will go away in a few minutes after the procedure is completed.

CRIUSE ANNOUNCEMENT / የጉዞ መረጃ

የተከበራችሁ መንገደኞቻችን ከበረራው ክፍል አብራሪያችሁ ነኝ። በኢትዮጵያ አየር መንገድ የበረራ ቁጥር ከ ___ ወደ __ - ከእኛ ጋር ለመብረር በድጋሚ እንኳን በደህና መጣችሁ ፡ የስታር አላያንስ አገልግሎት አባል የሆነውን የኢትዮጵያ አየር መንገድ መርጣችሁ በመምጣችሁ በጣም ደስተኞች ነን።

ስለ በረራችን አጠር ያለ አጠቃላይ መረጃ ለመስጠት፣ ከተነሳን ጀምሮ ደቂቃዎች ሆኖናል፡፡ በአሁኑ ሰዓት ወደተፈቀደልን ከፍታ ___ እየወጣን ነው፡ (የተፈቀደልንን ከፍታ ይዘናል።)

የዛሬው መንገዳችን በ __ በኩል በ___ ፣ ___ ፣ ___ አየር ክልል ይወስደናል። ለተጨማሪ እና ተከታታይ መረጃ ከፊትለፊታችሁ በሚገኙት የምስል ማሳያዎች መጠቀም ትችላላችሁ።

ወደመዳረሻችን ___ በአገሩ አቆጣጠር___ ሰዓት እንደርሳለን፡ የአየር ሁኔታው ባለን መረጃ መሰረት (ጥሩ/ፀሃያማ/ደመናማ/ዝናባማ) ነው። በመንገዳችንም ላይ ያለው ሁኔታ (ጥሩ/ከደመና የጸዳ/አልፎ አልፎ ደመናማ) ሲሆን በነፋስ አቅጣጫና ፍጥነት መቀያየር የተነሳ ንቅናቄ ሊኖር ስለሚችል በተቀመጣችሁ ጊዜ የደህንነት ቀበቶዎቻችሁን እንድትታጠቁ እንመክራለን።

በኢትዮጵያ አየር መንገድ እና በሥራ ባልደረቦቼ ስም አብራችሁን በመጓዛችሁ እያመሰገንኩ ጉዞአችሁ ያማረ እና የተሳካ እንዲሆንና በምንሰጣችሁ አገልግሎት እንድትደሰቱ እጋብዛለሁ። መልካም በረራ።

Ladies and gentlemen. This is your Captain/First Officer speaking from the Flight Deck, once again welcome onboard Ethiopian airlines flight number ___ from ___ to ____.

As always it is a pleasure to have you onboard our flight and we are very glad you made it your choice to fly Ethiopian and the Star Alliance.

I would like to give you some information concerning our flight.

____ minutes have elapsed since departure. We are currently climbing to/cruising at flight level ___. We have about ____ of cruising time before we come to our top of descent.

Toady our route to ___, after leaving ____ airspace, we will be via _____, _____, _____ and ____ airspace. Estimate arrival ___ at ____ local time.

And the reported weather at ____ looks rainy, stormy, sunny, cloudy, partly cloudy, clear.

On behalf of Ethiopian Airlines and entire crew, i would like to thank you for flying with us and we wish you nice and pleasant flight. Lean back, relax and enjoy your service.

Thank you for your attention.

DESCENT ANNOUNCEMENT

የተከበራችሁ መንገደኞቻችን ከበረራው ክፍል አብራሪያችሁ ነኝ: አሁን ወደ መዳረሻችን ___ ስለተቃረብን ከፍታችንን መቀነስ እንጀምራለን። ለመድረስ የሚቀረን ጊዜ ___ ደቂቃዎች ያህል ነው የሚያርፉ ወይም የሚነሱ አውሮፕላኖች ካላዘገዩን በቀር በአገሩ አቆጣጠር ___ ሰዓት እናርፋለን;; አሁን ያለው አየር ሁኔታ (ጥሩ/ ፀሃያማ/ደመናማ/ዝናባማ) ነው፡ የሙቀት መጠኑ ___ ዲግሪ ሴንቲግሬድ/ፋራናይት ነው። በጉዞአችሁ እንደተደሰታችሁ ተስፋ እናደርጋለን፡አመሰግናለሁ ።

This is your Captain/First Officer speaking from the flight deck.

We are commencing descent to ____ airport.

We will be landing in about ____ minutes.

The current weather at ___ is rainy, sunny, cloudy, partly cloudy, snowy, clear and the temperature ____ degrees celsius.

Have a nice landing. Thank you.

DIVERSION ANNOUNCEMENT / የመዳረሻ ለውጥ መረጃ

የተከበራችሁ መንገደኞቻችን ከበረራው ክፍል አብራሪያችሁ ነኝ: በአጋጠመን (በማረፊያ ጣቢያው ባለው አመቺ ያልሆነ የአየር ሁኔታ፣ የአውሮፕላን ብልሽት ወይም የጣቢያ መጨናነቅ) ማረፍ ስላልቻልን ወደ ተለዋጭ ማረፊያ ጣቢያ ___ እንሄዳለን እዛም በአገሩ አቆጣጠር __ ሰዓት እንደርሳለን። ተጨማሪ መረጃ በጣቢያው ካረፍን በኋላ እንገልጽላችኋለን። አመሰግናለሁ።

This is your Captain speaking from the flight deck.

Due to:

- Unfavorable weather conditions,
- Heavy traffic,
- Technical problem

It is not possible to land at ___. We will continue our flight to _____ airport.

Our arrival estimate is _____.

Further details on the continuation of your flight will be given upon arrival. We thank you for your understanding.

EMERGENCY LANDING ANNOUNCEMENT/ ለአልተጠበቀ ሁኔታ መረጃ

የተከበራችሁ መንገደኞቻችን ከበረራው ክፍል አብራሪያችሁ ነኝ። አንድ ጊዜ ሁላችሁም በጥሞና እንድታዳምጡኝ እጠይቃለሁ። በአጋጠመን _____ ምክንያት፣

- የአውሮፕላን ብልሽት ምክንያት
- በወፎች መመታት
- በድንገት በታመመ መንገደኛ
- በደህንነት ሥጋት
- በድንገተኛ እሳት ወይም ጭስ

ጉዞአችንን መቀጠል ባለመቻላችን ይህንን በረራ በማቋረጥ ወደ _____ ማረፊያ ጣቢያ ለመሄድ ተገደናል፡፡

ሁኔታውን ተገንዝባችሁ ለምታሳዩን ትዕግስት ላመሰግናችሁ እወዳለሁ።

Ladies and gentelmen, this is your Captain. May I have your attention please.

We regret to inform you that we have to proceed to ____ airport due to:

- Bird strike,
- Hydraulic loss,
- Pressurization problem,
- A passenger suffering heart attack,
- Security reasons, decompression, smoke in the cabin.

ATTACHMENT D - USE OF MEL/CDL

USE OF MINIMUM EQUIPMENT LIST (MEL) AND CONFIGURATION DEVIATION LIST (CDL)

GENERAL

The MEL is a joint operational and maintenance document that comes under the purview of Ethiopian Airlines Operations. The MEL:

- Identifies the minimum equipment and conditions for an aircraft to meet the operating rules for the intended operation;
- Defines operational procedures necessary to maintain the required level of safety and to deal with inoperative equipment ; and
- Defines maintenance procedures necessary to maintain the required level of safety procedures necessary to secure any inoperative equipment.

MEL is developed on the base of Boeing and Ethiopian master MEL (MMEL) for their respective aircrafts and customized by as a function of its own operational policies and ECAA operational requirements.

The provisions of MEL/CDL at both home base and out stations are applicable until the aircraft starts to move under its own power for the purpose of takeoff. Any decisions and continue a flight following a failure or un-serviceable item after brake release for taxi must be subject to the good judgment and airman ship of the PIC.

After the engines have been started, the system annunciator panel must be checked by Recall in order to ascertain if there are any new failures which were not previously present. This must be done before completing the “BEFORE TAXI” checklist. System failures which occur after completing the “BEFORE TAXI” checklist will be considered as “failures in flight.”

DISPATCH

The MEL is not intended to provide for continued operation of an aircraft for an unlimited period of time. Repairs should be made as soon as possible with in the time limit imposed by Rectification intervals.

Rectification Intervals (A, B, C and D) have been introduced in accordance with definitions of MMEL/MEL.

Dispatch of the aircraft is not allowed after expiry of the rectification interval specified in the MEL unless the rectification interval is extended in accordance with the following;

A one-time extension (TLO) of the applicable rectification Interval B, C of D may be permitted for the same duration as that specified in the MEL provided ECAA is notified within 10 days of any extension authorized.

Although the concept of Rectification Interval does not exist for the (CDL/DDG). All (CDL/DDG) items are not allowed to be left un-rectified for an unlimited period of time as stated in the Flight Manual.

ATTACHMENT E- MAINTENANCE STATUS OF THE AIRCRAFT

The flight crew will obtain a preliminary briefing regarding the maintenance status of the aircraft from flight control personnel to determine any significant effect upon the intended operations. Should this be the case, the appropriate limitation will be observed and the operation planned accordingly.

When the flight crew boards the aircraft, a final and more complete maintenance status briefing will be conducted by the duty Foreman or his delegate technician. During preflight and inflight the ATL shall be accessible to the flight crew.

Additionally, the flight crew shall:

a) Review the previous maintenance discrepancies and corrective actions in the technical log book, which contains a journey records section and an aircraft maintenance record section (ECARAS 9.3.2.9), together with the current list of differed deficiencies and MEL/CDL requirements to determine the airworthiness of the aircraft.

b) Review all items entered in the Aircraft Technical Log (ATL) and addressed for proper sign off. Review a reasonable number of days to familiarize and discuss with maintenance on repeated items.

c) eview the Acceptable Deferred Defect (ADD) page and check if all items are in accordance with MEL/CDL and that those items have not expired beyond the category limit as shown on the system definitions section of MEL.

d) In the event there is a Time Limit Overrun (TLO) on any item, ensure that a properly and duly signed document is inserted in the ATL.

e) A maintenance discrepancy that requires an airworthiness release should be communicated with Production Department of the Technical Service. Flight Control should facilitate the required communication. Under no circumstances should any aircraft depart with defects affecting airworthiness that has not been processed according to this procedure.

f) The provisions of MEL/CDL at both home base and out stations are applicable until the aircraft starts to move under its own power for the purpose of takeoff. Any decisions to continue a flight following a failure or un-serviceable item after brake release for taxi must be subject to the good judgment and airman ship of the PIC. The following guidelines may help the captain to better assess the situation for the current and next flight sectors;

    1. If possible confirm the availability of the maintenance support at destination to prevent operational disturbances

2. Consider the performance penalties as informed in the MEL (i.e. takeoff performance, fuel consumption increase, etc…)
3. Consider operational restrictions as informed in the MEL (e.g. degradation of approach, landing, RVSM/RNP or ETOPS capability, etc.) Consider the environmental restrictions as informed in the MEL (e.g. OAT limitations, icing conditions limitations)
4. Consider the possible inter-relationship between all deferred MEL items (cumulative dispatch conditions requirements, etc.)

g) The PIC has the authority to reject an aircraft if, prior to each flight, he is dissatisfied with any aspect of airworthiness and or maintenance status of the aircraft.

MINIMUM EQUIPMENT LIST (MEL) AND CONFIGURATION DEVIATION LIST (CDL)

a) MEL (approved by ECAA) shall be provided for the use of the flight crew members and maintenance personnel assigned on operational control functions during the performance of their duties. The MEL shall be specific to the aircraft type and variant which contains the circumstances, limitations and procedures for release or continuance of flight of the aircraft with inoperative components, equipment or instruments.

b) CDL (provided and approved by the State of Design) shall be provided for the use of flight crew members and maintenance personnel assigned on operational control functions during performance of their duties. The CDL shall be specific to the aircraft type. The AOC holder operations manual shall contain those procedures accessible to the Authority for operations in accordance with the CDL requirements.

c) An aircraft is not permitted to depart with a defect that has not been processed in accordance with the MEL/CDL.

MNT-No. MNT-1452

Ethiopian ኢትዮጵያ

AIRPLANE MAINTENANCE RECORD LOGBOOK (MLB)

Log number No. 593175

A/C Reg. | A/C Type | Flt No. | Date DD MM YY

Purpose of Flight: Revenue ☐ Test ☐ Training ☐ Tech. Ferry ☐ Positioning ☐ VIP ☐

	Name	Reg. No.	Sector From	Sector To
Captain				
1st Officer				

Engine	Oil quantity (Quarts) ARR	Added	Total
Eng 1			
Eng 2			
APU			
IDG1/VFSG1 LH			
IDG2/VFSG2 LH			
BGEN1/VFSG1-RH			
BGEN2/VFSG2-RH			

Hydraulic uplift (indicate unit of quantity as gallons, quarts or %) system	ARR	Added	Total
Left/ (1)/(A)/(G)			
Right/ (2)/(B) /(Y)			
Center / (3)			

FUEL SERVICE FLIGHT TO

FUEL _____ (KG) (A). Cap. Sign. _____

BEFORE RE-FUELING (KG)(B)		METERED UPLIFT (L/L) (E)	
REQUIRED UPLIFT (KG) (C= (A) (B))		CALCULATED UPLIFT (L/L) (F=C: D)	
SPECIFIC GRAVITY (SG)(D)		DISCREPANCY (L/L) (G=E-F)	
FUEL USED ON GROUND (KG)		FUEL SUPPLIER	

Precautionary procedures code (incident/Accident)** _____ Nature of condition Code (incident/Accident)** _____

AIRCRAFT FUEL QUANTITY (KGS)

ARRIVAL TOTAL (KG)		****SPECIFIC GRAVITY
DEPARTURE FUEL TOTAL (KG)		
****FROM PREVIOUS LOG SHEET	****DEPARTURE FUEL (KG)	
	****METERED FUEL UPLIFT (L/L)	

CALCULATED BY/ ID NO:_____ Sign:_____ Date DD MM YY

Certificate Release to Service (CRS)*

Date DD MM YY	Check Type	Name	License/Auth. No	MRO App. No.	Signature/Stamp	Station
Time (UTC)	Daily			ECAA:002/88		
	Transit					
	ETOPS			Other:		
	MO					
	Other*					

Note:
*Briefly describe the other work done: _____
** The list of codes found at inside of the cover page
*** "The Aircraft identified above was inspected, Overhauled, Modified Tested & all Panels secured in accordance with current Regulations of ECAA and other regulations specified and is approved for return to service with respect to the work accomplished." Other regulations:

Item | ATA F.R.M Defect Code | Found By: Pilot ☐ Maint ☐

Defect Description

Repetitive (REP) ☐ NIL ☐ Incident/ Accident ☐ Action For ADD ☐

APU In-flight start Request By Maint. ☐ Altitude ☐
APU In-flight start number of attempt: ☐
Pass ☐ Fail ☐ Continue to:-_____

ID No _____ Sign/Stamp:_____

Action

System ATA | Ref:

MEL/CDL Deferral ☐ Critical Task/RII ☐ CDCCL ☐ Category _____ Deferral Reference _____ ADD Log page No:_____ Continue to:_____

ID No:_____ Sign/Stamp: _____ Station _____

Removal/Installation Data

MFG P/N Off_____
S/N Off_____
MFG P/N On:_____
S/N On:_____
Position:_____

Double Inspection

Name:_____
License/Auth. No:
Date DD MM YY
Date DD MM YY

Item | ATA F.R.M Defect Code | Found By: Pilot ☐ Maint ☐

Defect Description

Repetitive (REP) ☐ NIL ☐ Incident/ Accident ☐ Action For ADD ☐

APU In-flight start Request By Maint. ☐ Altitude ☐
APU In-flight start number of attempt: ☐
Pass ☐ Fail ☐ Continue to:-_____

ID No _____ Sign/Stamp:_____

Action

System ATA | Ref:

MEL/CDL Deferral ☐ Critical Task/RII ☐ CDCCL ☐ Category _____ Deferral Reference _____ ADD Log page No:_____ Continue to:_____

ID No:_____ Sign/Stamp: _____ Station _____

Removal/Installation Data

MFG P/N Off_____
S/N Off:_____
MFG P/N On:_____
S/N On:_____
Position:_____

Double Inspection

Name:_____
License/Auth. No:
Date DD MM YY
Date DD MM YY

Check Performed

Daily ☐ Other ☐ ETOPS ☐ Transit ☐ MO ☐ Performed by: Name: _____ Signature/Stamp : _____ ID No. : _____ Station : _____ Date DD MM YY

Captain Accepting the A/C

Name : _____ ID: _____ Sign. : _____ Date DD MM YY

For Training Flight Only

LDG	STN	Date/Time	Touch & go	TDOC	O.Shoot	From Seq.	Trans code	Tot. Ldg
log checked								

DISTRIBUTION: ALL COMPLETED LOGBOOK PAD AND DETACHED COPIES MUST BE SENT TO A/C RECORDED SECTION AS SOON AS POSSIBLE
ORIGINAL WHITE: SHOULD NOT BE DETACH FROM THE LOGBOOK., PINK COPY: SHOULD BE DETACHED BY MAINTENANCE PERSONNEL AT BASE STATION, YELLOW COPY: OUT STATION COPY (IT SHOULD BE DETACHED AT OUT STATION WHERE THE A/C IS RELEASED TO SERVICE)

Log number No. 593175

A/C DATA BOX

1.Aircraft Registration Number
2. Aircraft type
3. Flight Number
4. Date

FLIGHT CREW INFORMATION

5. Name of the assigned captain and first officer in the specified flight number
6. Sector: The aircraft departure and arrival airport

ENGINE/APU OIL UPLIFT

7. This Block details:
- Engine/APU remaining oil quantity at arrival in quarts
- the uplift (added) quantity in quarts
- The total qty in qurats

HYDRAULIC UPLIFT

8. This block details:
- the serviced quantity of hydraulic in quarts

FUEL

9. Fuel service flight to :- exact total fuel reading during aircraft dispatch and fuel assigned for taxi (A)
10. BEFORE RE-FUELING (KG)(B) :- Record of fuel quantity before and after fueling
- REQUIRED UPLIFT (KG) (C= (A)-(B)):- It is the diffrence between the mass of Fuel Service flight to (kg)(A) & Before re-fueling (kg)(B)
- Fuel specific Gravity(D):- specific gravity (hydrometer reading)
- FUEL USED ON GROUND (KG):- The mass of fuel used (consumed) on ground before A/C taxi
11. METERED UPLIFT (Lit) (E) :- Total volume of fuel (in liter) uplifted into the Aircraft
- CALCULATED UPLIFT (Lit) (F= C: D)=It is the quotient of REQUIRED UPLIFT (KG) (C= (A)-(B)) and Fuel specific Gravity(D)
-DISCREPANCY (Lit) (G=E-F) :- It is difference between volume (in liter) of Metered uplifted (E) & Calculated uplift (F) fuel
12. AIRCRAFT FUEL QUANTITY (KGS)
- ARRIVAL TOTAL (KG):- The total mass (in kilogram) of remaining arrival fuel
-DEPARTURE FUEL TOTAL (KG):- The total mass ((in kilogram)) of departure fuel
-****FROM PREVIOUS LOG SHEET :- Fuel data transferred from previous flight leg
-****DEPARTURE FUEL (KG):- The mass (in kilogram) of fuel uplfited for departure of the previous flight
-****METERED FUEL UPLIFT (Lit):- The volume (in liter) fuel uplifted on the previous flight
13. Calculated by:- The person who computes the fuel data

INCIDENT INFORMATION ENTRY

14. Entry for Nature and Precautionary Procedures code of Incident

Nature of Condition codes

A	B	C	D	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	X	Y	Z
Flame	Smoke	FOD damage	In-flight separation	Flight control affected	Pump failure	Electric power source loss	Flight attitude instrument malfunction	Warning indication	Fluid loss	Test function inoperative	Over temperature	False warning	other	Warning system indication	Engine Over Speed/ surge	Partial Engine power loss	Other Systems affected	Engine Case puncture	Suspected unapproved part	Engine flame out	Engine Stoppage in flight	Significant Failure report

Precautionary Procedures codes

A	B	C	D	E	E2	F	G	H	I	J	K	L	M	N	O
Unscheduled Landing	Emergency descent	Aborted take-off	Return to block	Engine shutdown	In-flight Engine shutdown	Activate fire extinguisher	Oxygen mask deployed	Deactivate system	Depressurization	Fuel dumped	None	Aborted approach	Evasive action	Hard Landing	Other

CRS BOX

15. This block comprises the aircraft release to service
- Check type
- Name, authorization No. , signature/stamp the authorized personnel ,
- date & time when the CRS given
- MRO approval No.

DEFECT ENTRY BOX

16. Entry for defect
- Item : Discrepancy number that contains the logbook page No. and sequence number. Example: B308802-1 where "B308802" logbook page Number and "1" is sequence Number.
- F.R.M code : 8-digit fault code and help flight crew to report a fault
- NIL: Mark "X" if there is no defect
- Incident box: Mark "X" if there incident related technical problem

DEFECT ENTRY BOX

- Found by : If the discrepancy is reported by PIC the "Pilot" box shall be marked as "X" . If the discrepancy is reported by authorized maintenance personnel the "Maint." box shall be marked as "X".
- Repetitive (REP*) box: Mark "X" if the discepancy is repetitive

Note: If a flight defect reoccurs 3 or more times on the same airplane in the Log Book during three consecutive flights even though three separate corrective actions have been taken to correct the problem it shall be classified as a Repetitive (REP) Squawk by flight crew.

- Defect description: Details of the discrepancy encountered during aircraft operations reported either by the Pilot In Command (PIC) or by the authorized maintenance personnel and ID No. and Signature of PIC or Maintenance Personnel.
- Action For ADD box : Write the ADD page No. in the box when the deferred remark is corrected in the Logbook.

17. APU Box
The APU box contains the following
a. APU in-flight start request by maintenance
b. number of in-flight attempt by the flight crew
c. pass/fail check box (mark by the flight crew)
d. altitude that the APU in-flight start is attempted (entered by the flight crew)

ACTION BOX

18. This block contains:
- System ATA: 4 digit ATA code of the defect example E.g. 32-40
- AMM Ref.: The Aircraft maintenance manual reference used to correct the defect.
- detail action taken regarding the discrepancy reported. Continue to the next item/page if the space is not enough and write the item No. where you continue to in the "continue to" space.

19. RII box: Mark "X" if RII inspection or critical task is performed.
CDCCL: Mark "X" if CDCCL task is performed
Definition.
RII: Required inspection item
CDCCL: Critical Design Configuration Control limitation
20. MEL/CDL box: Mark "X" if the defect is deferred per approved MEL/CDL
21. This block details the ADD logbook number on which the deferred defect transferred, Category and deferral reference
22. ID No., Signature/stamp of the maintenance personnel who perform the action and station where the action is taken
23. Removal installation data: Part Number on, Serial Number on, Part Number off, Serial Number off and Position of the component needed for rectification of the defect.
24. Double Inspection entry: Name and License/Auth. No of the personnel who perform the double inspection.
Critical, RII & CDCCL tasks require double inspection.

CHECK PERFORMED BOX

25. Checked performed etntry: the type of check accomplished and the person who perform the task are specified.

CAPTAIN ACCEPTANCE BOX

26. Entry for Pilot acceptance of the aircraft
- Name and signature of Pilot in command (PIC)
- Date

27. Purpose of the flight
- Specify the purpose of the flight by marking on the box provided

Abbreviations

- BGEN 1 : Backup generator of Engine 1 - for B777
- BGEN 2 : Backup generator of Engine 2 - for B777
-VFSG1 : Variable frequency starter generator of engine 1 - For B787
-VFSG2 : Variable frequency starter generator of engine 2 - For B787

ATA CHAPTER AND SECTION LIST

ATA	SEC.	TITLE
21		AIR CONDITIONING
	0	GENERAL
	-10	COMPRESSION
	-20	DISTRIBUTION
	-30	PRESSURIZATION CONTROL
	-40	HEATING
	-50	COOLING
	-60	TEMPERATURE CONTROL
	-70	MOISTURE / AIR CONTAMINANT CONTROL
22		AUTO FLIGHT
	0	GENERAL
	-10	AUTOPILOT
	-20	SPEED - ATTITUDE CORRECTION
	-30	AUTO THROTTLE
	-40	SYSTEM MONITOR
	-50	AERODYNAMIC LOAD ALLEVIATING
23		COMMUNICATIONS
	0	GENERAL
	-10	SPEECH COMMUNICATIONS
	-15	SATCOM
	-20	DATA TRANSMISSION AND AUTOMATIC CALLING
	-30	PASSENGER ADDRESS, ENTERTAINMENT, AND COMFORT
	-40	INTERPHONE
	-50	AUDIO INTEGRATING
	-60	STATIC DISCHARGING
	-70	AUDIO & VIDEO MONITORING
	-80	INTEGRATED AUTOMATIC TUNING
24		ELECTRICAL POWER
	0	GENERAL
	-10	GENERATOR DRIVE
	-20	AC GENERATION
	-30	DC GENERATION
	-40	EXTERNAL POWER
	-50	AC ELECTRICAL LOAD DIS.
	-60	DC ELECTRICAL LOAD DIS.
25		EQUIPMENT / FURNISHINGS
	0	GENERAL
	-10	FLIGHT COMPARTMENT
	-20	PASSENGER COMPARTMENT
	-30	BUFFET / GALLEY
	-40	LAVATORIES
	-50	CARGO COMPARTMENTS
	-60	EMERGENCY
	-70	ACCESSORY COMPARTMENTS
	-80	INSULATION
26		FIRE PROTECTION
	0	GENERAL
	-10	DETECTION
	-20	EXTINGUISHING
	-30	EXPLOSION SUPPRESSION
27		FLIGHT CONTROLS
	0	GENERAL
	-10	AILERON AND TAB
	-20	RUDDER /TAB
	-30	ELEVATOR AND TAB
	-40	HORIZONTAL STABILIZER
	-50	FLAPS
	-60	SPOILER
	-70	GUST LOCK AND DAMPER
	-80	LIFT AUGMENTING
28		FUEL
	0	GENERAL
	-10	STORAGE
	-20	DISTRIBUTION - DRAIN VALVES
	-30	DUMP
	-40	INDICATING
29		HYDRAULIC POWER
	0	GENERAL
	-10	MAIN
	-20	AUXILIARY
	-30	INDICATING
30		ICE AND RAIN PROTECTION
	0	GENERAL
	-10	AIRFOIL
	-20	AIR INTAKES
	-30	PITOT AND STATIC
	-40	WINDOWS, WINDSHIELDS, AND DOORS
	-50	ANTENNAS AND RADOMES
	-60	PROPELLERS / ROTORS
	-70	WATER LINES
	-80	DETECTION
31		INDICATING / RECORDING SYS
	0	GENERAL
	-10	INSTRUMENT AND CONTROL PANELS
	-20	INDEPENDENT INSTRUMENTS
	-30	RECORDERS
	-40	CENTRAL COMPUTERS
	-50	CENTRAL WARNING SYSTEMS
	-60	CENTRAL DISPLAY SYSTEMS
	-70	AUTOMATIC DATA REPORTING SYSTEMS
32		LANDING GEAR
	0	GENERAL
	-10	MAIN GEAR AND DOORS
	-20	NOSE GEAR / TAIL GEAR AND DOORS
	-30	EXTENSION AND RETRACTION
	-40	WHEELS AND BRAKES
	-50	STEERING
	-60	POSITION, WARNING, AND GROUND SAFETY SWITCH
	-70	SUPPLEMENTARY GEAR - SKIS, FLOATS
33		LIGHTS
	0	GENERAL
	-10	FLIGHT COMPARTMENT AND ANNUNCIATOR PANEL
	-20	PASSENGER COMPARTMENT
	-30	CARGO AND SERVICE COMPARTMENTS
	-40	EXTERIOR LIGHTING
	-50	EMERGENCY LIGHTING
34		NAVIGATION
	0	GENERAL
	-10	FLIGHT ENVIRONMENT DATA
	-20	ATTITUDE AND DIRECTION
	-30	LANDING AND TAXIING AIDS
	-40	INDEPENDENT POSITION DETERMINING
	-50	DEPENDENT POSITION DETERMINING
	-60	FLIGHT MANAGEMENT COMPUTING
35		OXYGEN
	0	GENERAL
	-10	CREW
	-20	PASSENGER
	-30	PORTABLE
36		PNEUMATIC
	0	GENERAL
	-10	DISTRIBUTION
	-20	INDICATING
37		VACUUM
	0	GENERAL
	-10	DISTRIBUTION
	-20	INDICATING
38		WATER / WASTE
	0	GENERAL
	-10	POTABLE
	-20	WASH
	-30	WASTE DISPOSAL
	-40	AIR SUPPLY
42		INTEGRATED MODULAR AVIONICS
	0	General
	20	Core System
	30	Network Components
45		CENTRAL MAINT SYSTEM (CMS)
	0	GENERAL
	-5	CMS / AIRCRAFT GENERAL
	-20	CMS / AIRFRAME SYSTEMS
	-45	CENTRAL MAINTENANCE SYSTEM
	-46	CMS / AIRFRAME SYSTEMS
	-50	CMS / STRUCTURES
	-60	CMS / PROPELLERS
	-70	CMS / POWER PLANT
46		INFORMATION SYSTEMS
	0	GENERAL
	-10	AIRPLANE GENERAL INFORMATION SYSTEMS
	-20	FLIGHT DECK INFORMATION SYSTEMS
	-30	MAINTENANCE INFORMATION SYSTEMS
	-40	PASSENGER CABIN INFORMATION SYSTEMS
	-50	MISCELLANEOUS INF SYS
47		NITROGEN GENERATION SYSTEM
49		AIRBORNE AUXILIARY POWER
	0	GENERAL
	-10	POWER PLANT
	-20	ENGINE
	-30	ENGINE FUEL AND CONTROL
	-40	IGNITION / STARTING
	-50	AIR
	-60	ENGINE CONTROLS
	-70	INDICATING
	-80	EXHAUST
	-90	OIL
51		STANDARD PRACTICES AND STRUCTURES - GENERAL
	0	GENERAL
	-10	INVESTIGATION, CLEANUP AND AERODYNAMIC SMOOTHNESS
	-20	PROCESSES
	-30	MATERIALS
	-40	FASTENERS
	-50	SUPPORT OF AIRPLANE FOR REPAIR AND ALIGNMENT CHECK PROCEDURES
	-60	CONTROL-SURFACE BALANCING
	-70	REPAIRS
	-80	ELECTRICAL BONDING
52		DOORS
	0	GENERAL
	-10	PASSENGER / CREW
	-20	EMERGENCY EXIT
	-30	CARGO
	-40	SERVICE
	-50	FIXED INTERIOR
	40	ENTRANCE STAIRS
	-70	DOOR WARNING
53		FUSELAGE
	0	GENERAL
	-10	MAIN FRAME
	-20	AUXILIARY STRUCTURE
	-30	PLATES-SKIN
	-40	ATTACH FITTINGS
	-50	AERODYNAMIC FAIRINGS
54		NACELLES / PYLONS
	0	GENERAL
55		STABILIZERS
	0	GENERAL
	-10	HORIZONTAL STABILIZER / STABILATOR OR CANARD
	-20	ELEVATOR - ELEVON
	-30	VERTICAL STABILIZER
	-40	RUDDER - RUDDEVATOR
56		WINDOWS
	0	GENERAL
	-10	FLIGHT COMPARTMENT
	-20	PASSENGER COMPARTMENT
	-30	DOOR
	-40	INSPECTION AND OBSERVATION
57		WINGS
	0	GENERAL
	-10	CENTER
	-20	OUTER
	-30	WING TIP
	-40	LEADING EDGE AND LEADING EDGE DEVICES
	-50	TRAILING EDGE AND TRAILING EDGE DEVICES
	60	AILERONS AND ELEVONS
	-70	SPOILERS
61		PROPELLERS / PROPULSORS
	0	GENERAL
	-10	PROPELLER ASSEMBLY
	-20	CONTROLLING
	-30	BRAKING
	-40	INDICATING
	-50	PROPULSOR DUCT
70		STANDARD PRACTICES
71		POWER PLANT
	0	GENERAL
	-10	COWLING
	-20	MOUNTS
	-30	FIRESEALS AND SHROUDS
	-40	ATTACH FITTINGS
	-50	ELECTRICAL HARNESS
	60	ENGINE AIR INTAKES
	-70	ENGINE DRAINS
72		ENGINE - TURBINE / TURBOPROP, DUCTED FAN / UNDUCTED FAN
	0	GENERAL
	-10	REDUCTION GEAR AND SHAFT
	-20	AIR INLET SECTION
	-30	COMPRESSOR SECTION
	-40	COMBUSTION SECTION
	-50	TURBINE SECTION
	-60	ACCESSORY DRIVES
	-70	BY-PASS SECTION
	-80	PROPULSOR SECTION (REAR MOUNTED)
73		ENGINE - FUEL AND CONTROL
	0	GENERAL
	-10	DISTRIBUTION
	-20	CONTROLLING GOVERNING
	-30	INDICATING
74		IGNITION
	0	GENERAL
	-10	ELECTRICAL POWER SUPPLY
	-20	DISTRIBUTION
	-30	SWITCHING
	-30	SWITCHING
75		ENGINE AIR
	0	GENERAL
	-10	ENGINE ANTI-ICING
	-20	ENGINE COOLING
	-30	COMPRESSOR CONTROL
	-40	INDICATING
76		ENGINE CONTROLS
	0	GENERAL
	-10	POWER CONTROL
	-20	EMERGENCY SHUTDOWN
77		ENGINE INDICATING
	0	GENERAL
	-10	POWER
	-20	TEMPERATURE
	-30	ANALYZERS
	-40	INTEGRATED ENGINE INSTRUMENT SYSTEMS
	-10	ENGINE ANTI-ICING
	-20	ENGINE COOLING
	-30	COMPRESSOR CONTROL
	-40	INDICATING
78		EXHAUST
	0	GENERAL
	-10	COLLECTOR - NOZZLE
	-20	NOISE SUPPRESSOR
	-30	THRUST REVERSER
	-40	SUPPLEMENTAL AIR
79		OIL
	0	GENERAL
	-10	STORAGE (DRY SUMP)
	-20	DISTRIBUTION
	-30	INDICATING
80		STARTING
	0	GENERAL
	-10	CRANKING

ATTACHMENT F – SINGLE ENGINE TAXI OUT BRIEFING

Conditions to be met for SETO (Single Engine Taxi Out) Procedure

- Engines are not considered cold. (Not more than 4 hrs since last shutdown)
- Apply only on 737NG (-700,-800,-800F)
- GTOW less than 70,000 Kg for -800s & 62,000 Kg for -700s
- No icing conditions exist
- LVP is not enforced
- Engine No. 2 has to be started first
- APU is available
- Dry surface conditions for taxi and takeoff
- Experience more than 100 hrs forapt and 300 hrs for FO
- A minimum of estimated 5 minutes taxi time
- Respect a minimum of 3 minutes of warmup time before setting takeoff thrust
- No significant uphill taxiway slope
- During taxi no tight turns have to be made & energy management technique using Ground Speed should be applied while turning towards live Engine.
- Taxi does not require crossing of active runway
- At the discretion of the Captain, SETO procedures should be performaed by reviewing the procedure prior to accomplishment and by reference to the procedure during its accomplishment. The workload/task management should be considered during the departure briefing for starting of Engine no. 1
- During Engine no. 1 start, ATC permitting, aircraft in stationary position is preferable, or in straight taxiway with max taxi speed of 10 kts is recommended.

SETO BEFORE TAXI PROCEDURE

After Engine no. 2 has been started and stabilized:

"FLAPS ___ SETO BEFORE TAXI PROCEDURE" .....Call C
Flap lever .....Set Takeoff Flaps FO
MFD SYS switch .....Push FO
GENERATOR 2 switch .....ON FO
PROBE HEAT switch .....ON FO
RIGHT PACK switch .....AUTO FO
ISOLATION VALVE switch .....CLOSE FO
Left PACK switch .....As Required FO
ENGINE no. 2 START switch .....CONT FO
Recall .....Check FO

Verify that all system annunciator panel lights illuminate and then extinguish except ELEC, HYD, and AIR COND. Illuminated amber lights are DRIVE 1, ENG 1 HYD pump LOW PRESSURE and DUAL BLEED.

Flight Controls .....Check C
"SETO BEFORE TAXI CHECKLIST" .....Call C
SETO BEFORE TAXI CHECKLIST .....Read FO

SETO BEFORE TAXI CHECKLIST

GENERATOR 2 .....ON
PROBE HEAT .....ON
ENG no. 2 Anti-Ice .....OFF
Isolation Valve .....CLOSE
No. 2 Engine Start Switch ..... CONT
Recall .....Checked
AUTOBRAKE ..... RTO
ENGINE no. 2 START lever .....IDLE detent
Flight Controls .....Checked
Ground Equipment .....Clear

SETO ENGINE NO. 1 START PROCEDURE

Start Engine no. 1 when appropriate and respect a minimum of 3 minutes of warmup time before setting takeoff thrust.

L PACK switch .....OFF FO
Lower DU ..... ENG FO
Engine no. 1 thrust lever .....CLOSE FO

Engine no. 1 Start lever operation and monitoring should be accomplished by FO, while the Captain remain responsible for taxiing. TFC permitting, stop the aircraft to accomplish Engine start.

"Start Engine no. 1" .....Call C

SETO BEFORE TAKEOFF PROCEDURE

After Engine no. 1 has been started and stabilized:

Lower DU ..... Blank
GENERATOR 1 switch .....ON FO
Left PACK switch .....AUTO FO
ISOLATION VALVE switch .....AUTO FO
APU BLEED air switch .....OFF FO
Engine no. 1 START switch ..... CONT FO
APU switch .....As Required FO

"SETO BEFORE TAKEOFF CHECKLIST" .....Call C
SETO BEFORE TAKEOFF CHECKLST .....Read FO

SETO BEFORE TAKEOFF CHECKLIST

GENERATORS .....ON
Anti-ice .....OFF
Isolation Valve .....AUTO
Engine Start switches ..... CONT
Recall .....Checked
AUTOBRAKE ..... RTO
ENGINE START levers .....IDLE detent
Flaps ..... _____ Green light
Stabilizer Trim ..... ____ Units

Intentionally Left Blank